Association for Project Management

APM PMQ

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Association for Project Management

Administration

Trainer
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Facilities
Days/Times
Breaks
Phones and Mobile devices

Materials

Introductions

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17 years training across a range of accreditations – APM, PRINCE2, P3O.

Career in public sector project and programme management (Department for Work and Pensions)

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Association for Project Management

Introductions

Warm up Exercise – Identify the key skills needed by a Project Manager in running a successful project

Name and Role/Organisation

Experience of Project Management, Including any other PM Qualifications

Key Objective for the Course

Course Syllabus

Module 1: Life Cycles

Module 2: Governance Arrangements

Module 3: Sustainability

Module 4: Business Case

Module 5: Procurement

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Course Syllabus

Module 6: Reviews

Module 7: Assurance

Module 8: Transition Management

Module 9: Benefits Management

Module 10: Stakeholder Engagement and Communication Management

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Course Syllabus

Module 11: Conflict Resolution

Module 12: Leadership

Module 13: Team Management

Module 14: Diversity and Inclusion

Module 15: Ethics, Compliance and Professionalism

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Course Syllabus

Module 16: Requirements Management

Module 17: Solutions Development

Module 18: Quality Management

Module 19: Integrated Planning

Module 20: Schedule Management

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Course Syllabus

Module 21: Resource Management

Module 22: Budgeting and Cost Control

Module 23: Risk and Issue Management

Module 24: Change Control

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Life cycles

Learning objective:

1. Understand the distinct life cycle stages used to structure and organise a project.

1a) Understand the distinctive features of linear, iterative and hybrid life cycles (including why projects are structured as phases in linear life cycles) and know when each is applicable.

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Life cycle options and choices

There are several approaches that organisations can chose from to deliver beneficial change through project work. The choice will determine the way projects, programmes and portfolios operate and play a vital role in determining their success.

Potential approaches range from highly predictive models such as a linear approach, assuming well established knowledge in regards the context through to highly adaptive models such as iterative, which operate in more uncertain, volatile and dynamic environments.

A linear life cycle:

- Is sequential with clearly defined outputs.
- Applies to stable low-risk projects with greater structure, where expectations of each phase are known.
- Provides a clear framework for the team to follow, with early definition of requirements and allows maximum control and governance.
- Allows for controls and information to be passed on to the next phase when predefined milestones have been reached and accomplished.

Concept

Development of an initial idea, through initial studies and high-level requirements management, assessment of viability including an outline business case.

Definition

Development of a detailed definition, plans and statement of requirements that include a full justification for the work. Captured in the Project Management Plan (PMP)

Deployment

Implementation of plans and verification of performance through testing and assurance to realise intended outputs, outcomes and benefits.

Transition

Handover, commissioning and acceptance of outputs to the sponsor and wider users, culminating in formal closure.

Source: Adapted from *APM Body of Knowledge*, 7th edition, Figure 1.2.2 (page 19).

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Iterative life cycles

Life cycles vary according to their nature, purpose and expected use. An iterative life cycle is based on the idea of simultaneous engineering, where different development steps are allowed to be performed in parallel.

An iterative life cycle:

- Repeats one or more phases, typically multiple deployment timeboxes/sprints.
- Is beneficial for evolving objectives or solutions, used in agile development projects and allows iterative learning.
- Works if the project scope might be vague, or solutions unclear, requiring greater flexibility in governance procedures.
- Allows the project manager and team to observe the benefits that staged functionality delivers and tweak the next iteration accordingly.

Source: Adapted from The DSDM Agile Project Framework (2014 onwards) as referenced in *APM Body of Knowledge*, 7th edition, Figure 1.2.3 (page 21)

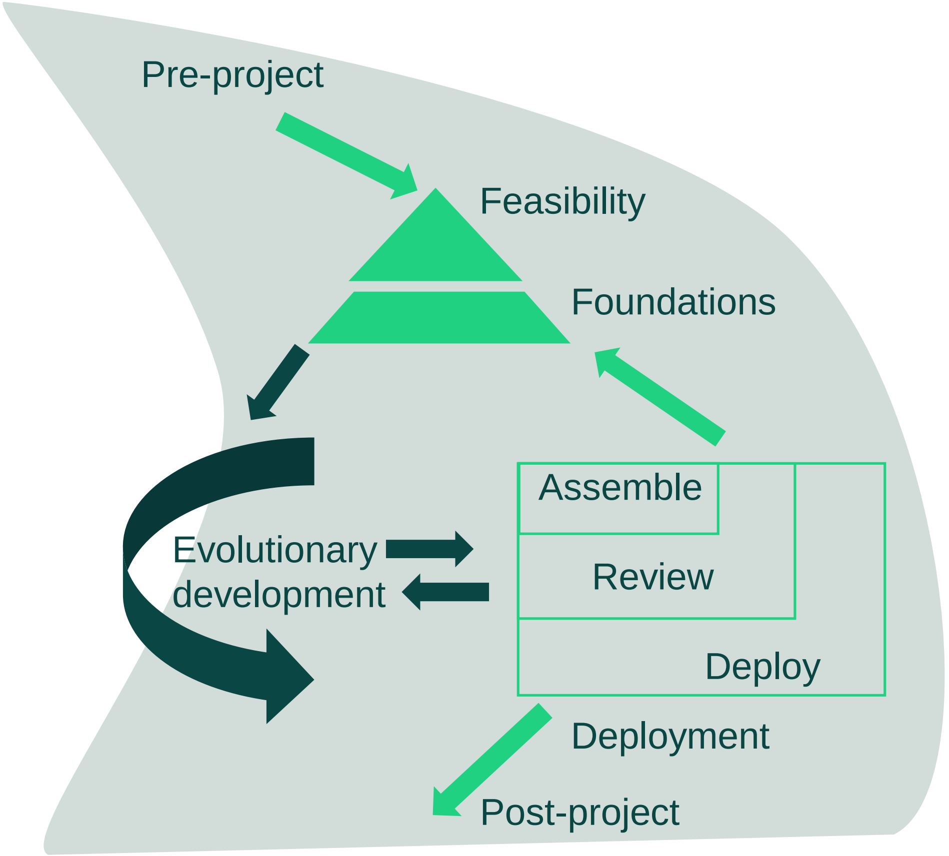
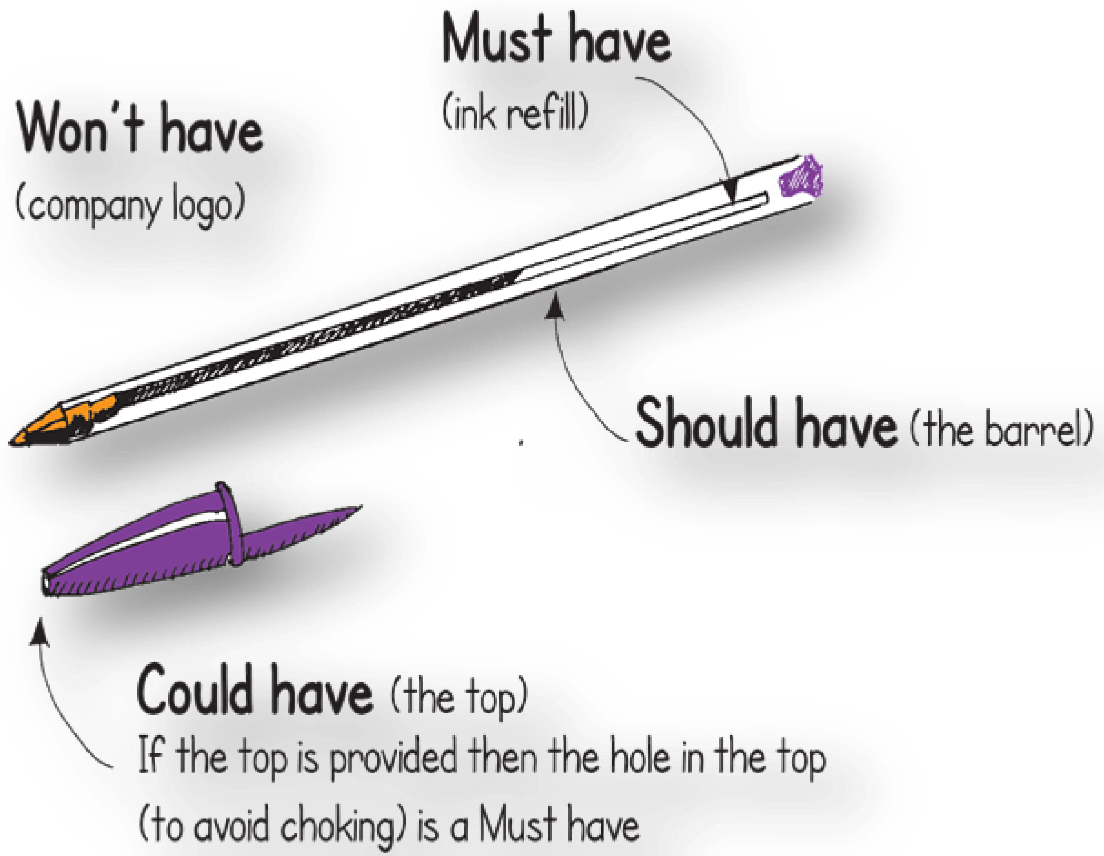


Figure 'Life Cycles' adapted from The DSDM Agile Project Framework (2014 Onwards) https://www.agilebusiness.org/dsdm-project-framework.htmls, Agile Business Consortium. Reproduced with permission; © Agile Business Consortium, 2014.

Prioritisation techniques

Must have some flexibility
What can be sacrificed if time and cost are threatened?

MUST
O
SHOULD
COULD
O
WON'T



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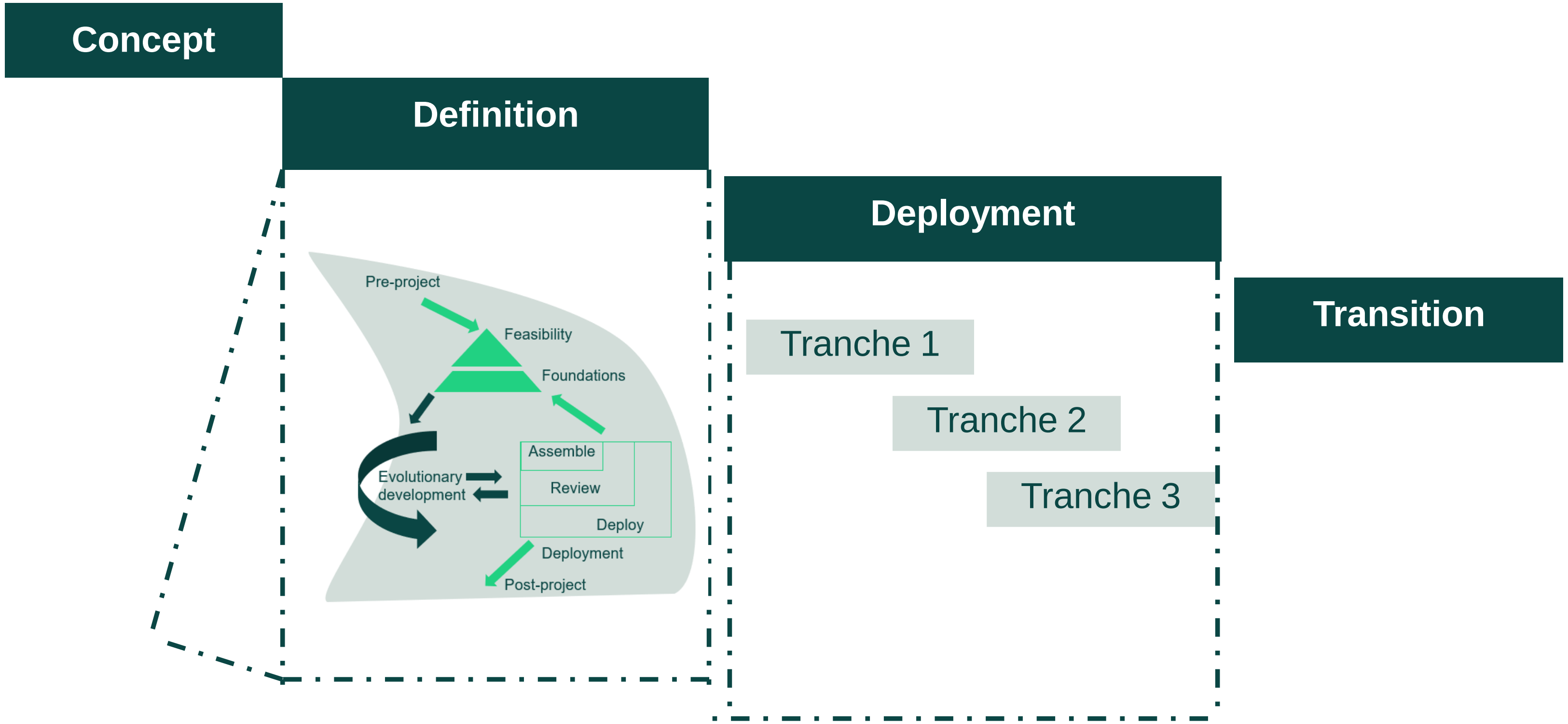
Hybrid life cycles

There is no one life cycle to suit all project contexts. Hybrid life cycles allow project managers to respond to complexities and uncertainties or the discovery of new information through iteration, by enabling a pragmatic mix of the predictive and adaptative elements of linear and iterative life cycles.

A hybrid life cycle:

- Adds iteration to a linear life cycle enabling a mix of approaches.
- Enables a project team to choose elements of both linear and iterative life cycles to best suit the project.
- Adapts – if a fixed requirement is set, the iterative style can be used during deployment to develop the output and add further functionality.



Source: Adapted from *APM Book of Knowledge*, 7th Edition, Figure 1.2.4 (page 23).
Figure 'Life Cycles' adapted from The DSDM Agile Project Framework (2014 Onwards) https://www.agilebusiness.org/dsdm-project-framework.htmls, Agile Business Consortium. Reproduced with permission; © Agile Business Consortium, 2014.

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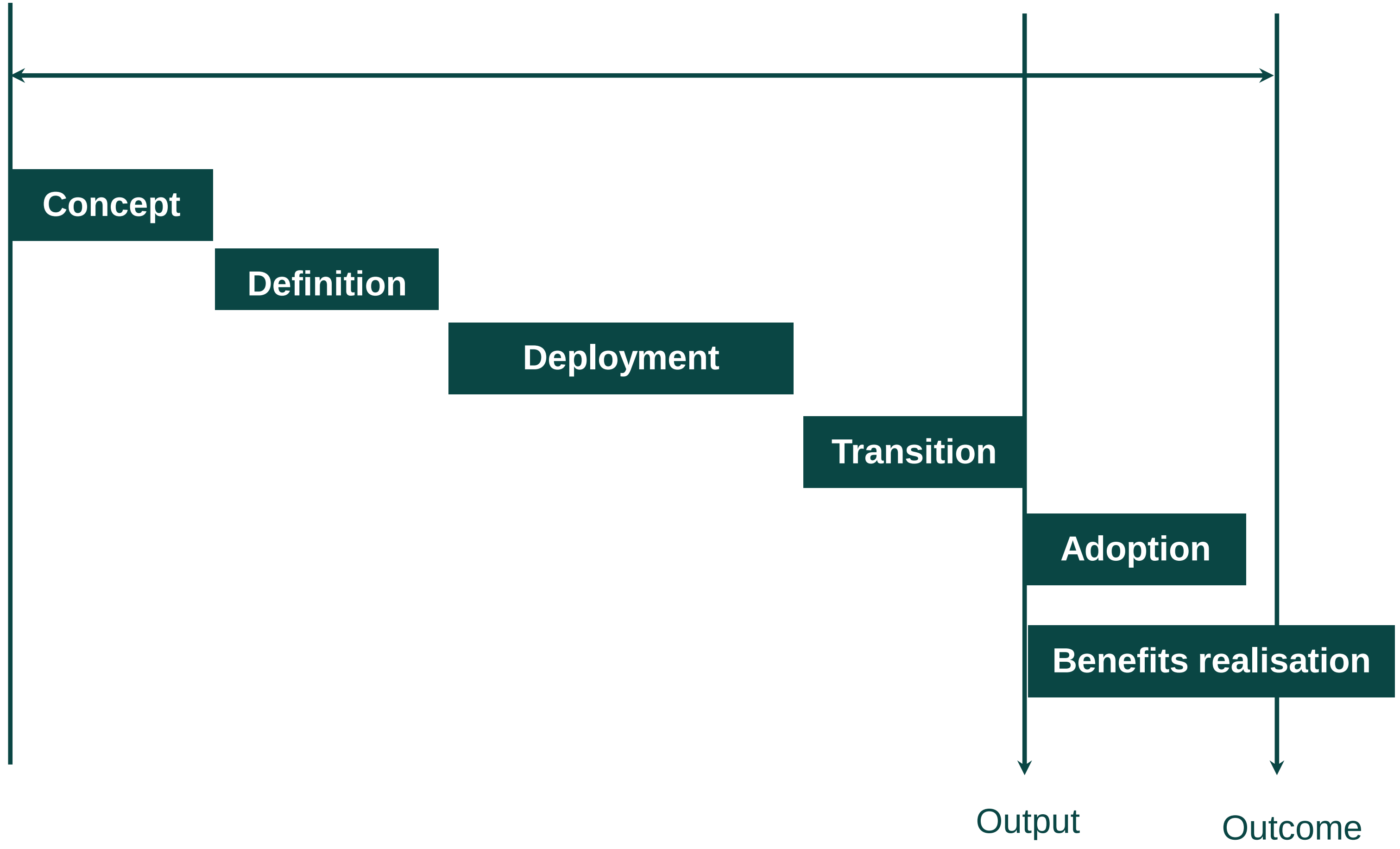
1b) Knowledge of the differences between a project life cycle and an extended life cycle.

Extended life cycle

An extended life cycle goes beyond the transition and closure phase, ensuring that accountability and governance of the investment stays with the change teams until the change is fully embedded by creating a connection with benefit realisation.

An extended life cycle will therefore include the following additional phases:

- **Adoption of outputs –** operations and sustainment required to utilise the new project and enable the acceptance and use of the benefits.
- **Benefits realisation –** realisation of the required business benefits or outcomes.



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Project versus extended life cycles

Before deciding on the most suitable life cycle approach, it should be decided whether a ‘project’ or an ’extended’ life cycle is suitable.

Project life cycle

- A project life cycle contains the phases up to transition where the project is closed and team disbanded.
- A business delivering the project on behalf of a customer may use the standard project life cycle, as they are likely to complete the contract at the handover of the project to the customer (end user).
- Accountability for the output is handed over, and achievement of outcomes and benefits is in the hands of the client

Extended life cycle

- An extended life cycle goes beyond the transition and closure phase to include the benefits realisation phase and adoption of outputs.
- This is suitable where the project is expected to incorporate management of change and benefit realisation.
- The extended life cycle is suited to projects that are funded within the organisation and are making a change to the business.
- The project team supports the embedding of the project.
- The team can determine the immediate benefits realised as the project goes operational and through immediate feedback to the organisation on the realisation of benefits.
- Governance and accountability for adoption of the output stays within the project until the change is fully embedded, thus preventing knowledge boundaries between project teams and operations.

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Project versus extended life cycles

Aspect	Project Life Cycle	Extended Life Cycle
Documentation Requirements	Primarily focuses on documentation required for project execution and closure, such as project plans, progress reports, and final reviews.	Also requires comprehensive documentation that extends into operational manuals, long-term maintenance schedules, and end-user support materials.
Quality Assurance	Quality assurance processes are designed to ensure that the project meets defined standards and specifications at closure.	Ongoing quality assurance is crucial to maintain the integrity, performance, and safety of the project outcomes throughout their lifecycle.
Skills and Expertise	Requires skills relevant to project management and technical execution specific to the project’s requirements.	Demands a broader range of skills, including operational management, technical support, and lifecycle management expertise.
Legal and Compliance	Compliance is targeted at meeting the legal requirements up to the project's delivery.	Extends compliance to include regulatory requirements that affect the ongoing use, updates, and eventual decommissioning of the project deliverables.

1c) Understand how the context and culture of an organisation, and the needs of a specific project, influence the choice of project life cycle and any adaptations that may be needed to the life cycle.

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Context, culture and project life cycles

Organisations favouring linear life cycles

- Where a highly structured, predictable and stable process is needed which offers maximum control and governance.
- Where there is availability of relatively perfect knowledge upfront.
- Which might be resistant to change and inflexible in terms of corrections and rework.
- Where knowledge / teams are divided into distinct phases, creating silos and knowledge barriers between the phases, particularly when different delivery agents will deliver different phases.
- Lower appetite for risk.

Organisations favouring iterative life cycles

- Where development projects are agile.
- Where departments are flexible, adaptable and open to change.
- They favour the idea of concurrency, or simultaneous engineering, where different development steps are performed in parallel.
- They can manage uncertainty regarding the scope by allowing the objectives to evolve throughout the life cycle as learning and discovery take place.
- Prototypes, timeboxes or parallel activities are utilised to acquire new insights, obtain feedback or explore high-risk options.

These two models alone would not suit all applications: hybrid life cycles enable a pragmatic mix of approaches to suit a specific situation.

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Context, culture and project life cycles

Organisational Context and Culture (In other words, how organisational factors influence delivery of change?)

Organisational context refers to the internal and external environment in which a project operates. This includes the organisation’s structure, existing processes, stakeholder expectations, and market conditions.

The following defines the influence on life cycle choice:

1. **Risk Tolerance:** Organisations with a high tolerance for risk may prefer iterative or hybrid life cycles that allow for flexibility and change. In contrast, organisations that prioritise stability and predictability might lean towards a linear life cycle.

2. **Decision-Making Style:** Hierarchical organisations may favour linear life cycles due to their clear structure and defined decision points. In contrast, organisations with a more collaborative or decentralised approach might find iterative methods more effective.

3. **Change Readiness:** The cultural readiness to embrace change influences whether an organisation can effectively implement iterative or hybrid life cycles, which require quick responses and adaptations.

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Context, culture and project life cycles

Project-Specific Needs

These are the unique requirements, goals, and constraints of a project, including scope, time, cost, quality, and stakeholder expectations.

The following defines the influence on life cycle choice:

1. **Project Complexity:** More complex projects might benefit from a hybrid life cycle that combines the structured approach of linear methods with the flexibility of iterative phases.

2. **Requirement Stability:** Projects with unstable or unclear requirements are better suited to iterative life cycles, allowing for regular reassessment and realignment with project goals.

3. **Stakeholder Engagement:** Projects requiring frequent stakeholder input and feedback are likely to succeed with iterative approaches, facilitating continuous improvement and alignment with user needs.

4. **Resource Availability:** This approach allows teams to plan and execute phases where sufficient resources are assured, using the linear part of the cycle, while leveraging the iterative phases to progress with available resources.

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Context, culture and project life cycles

Necessary Adaptations to Life Cycles

Purpose: Adaptations are adjustments made to the standard life cycle models to better fit the specific circumstances of the project and the organisational environment.

The following are the examples of adaptations:

1. **Phased Implementation:** In an organisation new to project management methodologies, introducing phases of a linear life cycle with checkpoints can prepare the team for more flexible approaches in the future.

2. **Custom Iterations:** For projects in dynamic sectors, customising the length and focus of iterations based on project progress and external feedback can optimise outcomes.

3. **Integration of Agile Techniques:** Incorporating agile techniques into traditional models (hybrid approach) can enhance responsiveness and stakeholder satisfaction in environments that value both innovation and control.

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1d) Knowledge of the strengths and limitations of different life cycles.

Strengths and limitations of different life cycles

Linear

Strengths

- Suitable for low-risk projects.
- Provides a clear framework for the team to follow.

Limitations

- Requires very early clarity on scope and governance.
- Longer process to ascertaining benefits realised.
- Less flexible to accommodating change.

Iterative

Strengths

- Beneficial for evolving objectives or solutions, and agile projects.
- Suitable where project scope might be vague.
- Allows iterative feedback and accommodates change more easily.

Limitations

- Lack of early certainty in terms of overall duration and cost.
- Could lead to complexities in managing resource.

Hybrid

Strengths

- Facilitates a mix of approaches to suit a specific development.
- Building agile working into a project or programme can offer increased efficiency and flexibility.

Limitations

- It requires great skill and clarity when using multiple different systems of working to be successful.

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 1.2 (Life cycle options and choices)

APM (2019) 'Hybrid project management: is it your next challenge?' apm.org.uk/blog/hybrid-project-management-is-it-your-next-challenge/

APM 'What is a life cycle?' apm.org.uk/resources/what-is-project-management/what-is-a-life-cycle/

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Sample Exam Questions

Learner Study Guide

Sample Exam Questions

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Governance arrangements

Learning objective:

2. Understand governance structures as a framework of authority and accountability for the delivery of a project, which align with organisational practice.

2a) Knowledge of different types of permanent and temporary organisation structures and their features (including functional, matrix, and project) and know that an organisation's governance approach will inform the approach used for a project.

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Permanent (BAU) and temporary (Project) organisational structures

- Permanent structures have stable teams for routine/operational work whereas specific initiatives, like projects, have temporary teams.

- Business as usual (BAU) or operations within an organisation are permanent structures that need specific resources to carry out routine tasks with a stable team environment.

- A fully **projectised structure** is temporary, where the core teams are brought together for the entirety of a project, then disbanded; therefore, building expertise, where all the staff are fully under the direction of the Project Manager.

- **The matrix structure** is one of the most common temporary structures used to manage projects, where authority is balanced between the functional line manager and the project manager. A matrix structure provides a mix of skilled resources on a temporary basis to the project, whereas a project structure provides skilled resources permanently to the project.

- In a **functional structure**, an individual will take instruction from a functional line manager only and therefore the line of communication is more straightforward. A project or matrix structure will provide more varied amount of work than a functional structure. Within a functional or matrix environment, the individual will be allocated tasks that match their capability, and these tasks could be repetitive.

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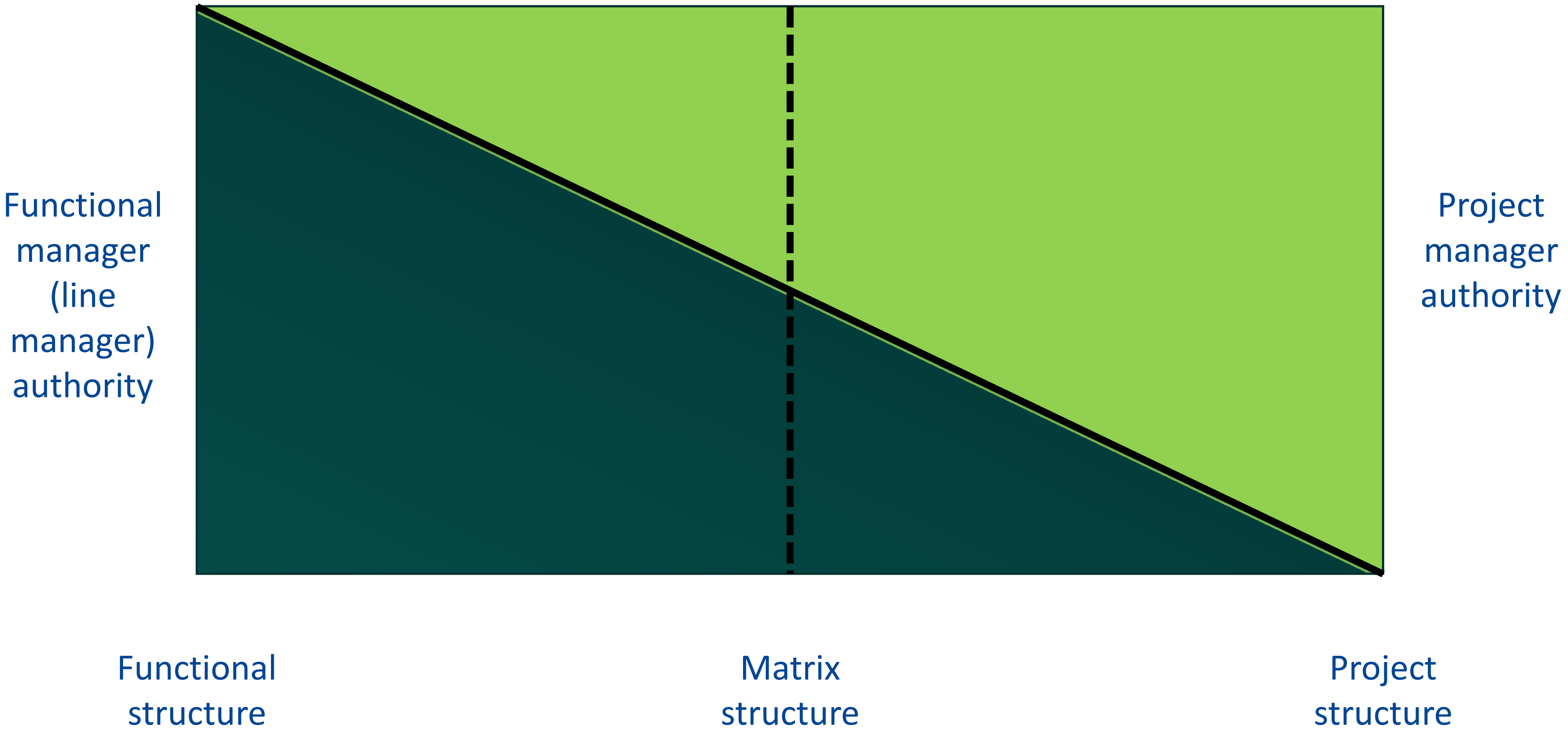
Organisational Continuum

Refer to the next slide, based on the information there, what type of organization structure does your organization have?



Permanent and temporary organisational structures

- The below table comparing permanent and temporary organisational structures:

Structure Type	Functional	Matrix	Project-based
Definition	Groups employees by specific skills and job functions in a hierarchical structure.	Combines functional and project-based structures, with dual reporting lines.	Forms teams specifically for projects, dissolving them after project completion.
Advantages	Clear career paths, specialised efficiency, easy management of tasks.	Resource flexibility, enhanced inter-department communication, good for complex projects.	High adaptability, focused decision-making, strong team motivation.
Disadvantages	Risk of silos, poor cross-department communication, less responsive to change	Potential for role confusion, conflict, and high administrative costs.	Job security concerns, frequent team changes, challenging resource management.
Typical Use	Stable operations like manufacturing or standardised services.	Industries with diverse, immediate projects such as IT and construction.	Project-centric fields like event management, construction, or consulting.

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2b) Understand why there are distinct roles within project management and know the responsibilities of each role (including users, project team members, the project manager, the project steering group/board and the product owner) and understand the differences in responsibilities of the project manager and project sponsor throughout the project.

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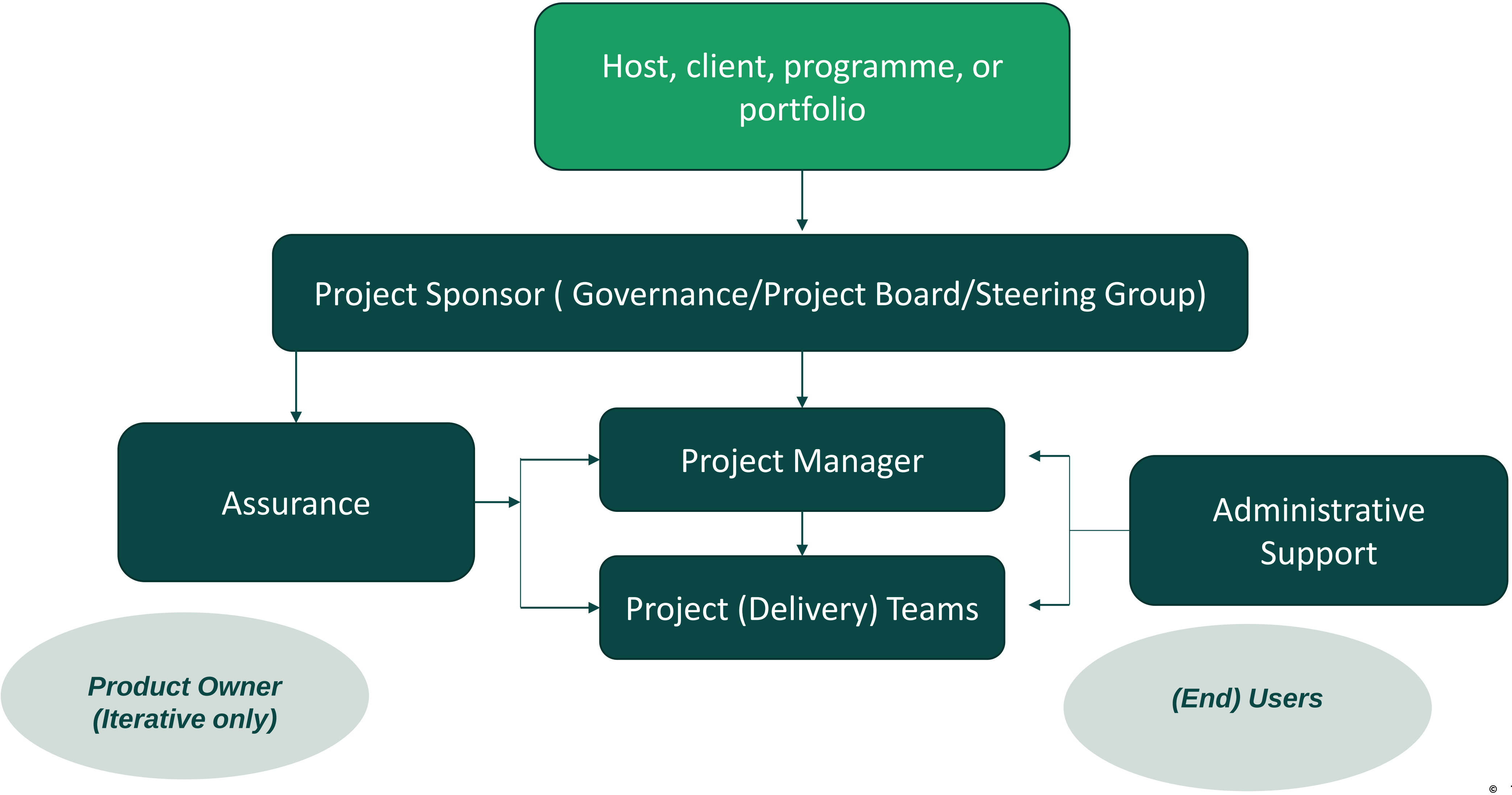
Example Project Team Structure



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Roles on a project

Project manager

- Manages the project day-to-day.
- Ensures successful delivery of outputs, identifying and managing risks, managing budgets and schedule.
- Communicates with and updates all project stakeholders.
- Organises, motivates, and leads the team, ensures knowledge and skills are aligned to assigned roles.
- Develops the project management plan, and monitors and controls the project's progress.

Project sponsor

- Communicates the vision and sets the high-level expectations, securing funding for the project.
- Owns and develops the business case during the definition phase, with support of the project manager.
- Identifies the business change and coordinates the justification of this change within the business case.
- Provides ongoing support to the project manager and overarching governance.
- Makes decisions on behalf of the business at phase reviews / decision gates.

Project steering group/board

- Guides the project in line with the strategic aims of the business.
- Responsible for the project's feasibility, business case and achievement of outcomes.
- Authorises the business case on behalf of the organisation, sanctioning a release of funds for the project to progress.
- Decides on any escalated issues from the project manager and supports higher-level decision making.

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Roles on a project

Product owner

- Typically, this role is more common within iterative or hybrid life cycles.
- Defines scope of work; leads the focus on product development.
- Acts as the intermediary between stakeholders and project team members.
- Creates vision and defines goals for the operability of the project's outputs.
- Accepts incremental delivery, performing a role to accept the function or scope as complete.

Project team members

- Perform project tasks as instructed by the project manager.
- Deliver project outputs.
- Report on progress of achievement against the project plan.
- Support the project manager in completing tasks within the project plan, such as identifying risks, communicating with stakeholders, and identifying and reporting changes.

Users

- Identify project requirements ensuring objective separation of musts and wants.
- Participate as members of the project team, identifying project constraints and dependencies.
- As subject matter experts, answer any questions the project team might have about the scope of work or the benefits required.
- Define acceptance criteria.

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2c) Understand why aspects of project management governance are required (such as the use of policies, regulations, functions, processes, procedures and delegated responsibilities) and understand the impact of a project’s life cycle on its governance framework and the limits of financial authority.

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Project management governance

Policies

- Provide guidelines as to what the **business** expects of projects e.g risk/reward balance
- Outline the boundaries for the business e.g.strategic objectives
- Influence how projects are conducted and demonstrate the culture of the organisation e.g. principles of governance
- Support governance of projects with various control points, such as decision gates, audits and evaluation reviews.

Regulations

- Ensure conformance to internal and external standards.
- Ensure that the work adheres to legal, regulatory, corporate, ethical and professional standards.
- May need a corporate level assurance function

Processes, procedures and functions

- Standardise methods of project working and provide a structured methodology for the delivery of projects.
- Ensure consistency of practice e.g some form of assurance to check
- Best practice to have a bank of processes, procedures, and standardised functions in place for a project community such as best practice principles for managing risk, controlling the budget, life cycle approach, etc.

Delegated responsibilities

- Clarify roles and increases efficiency; everyone has clarity on what is expected of them.
- Good governance expects that the roles and responsibilities of the team and wider stakeholders are clearly defined.
- Use of a responsibility assignment matrix (RAM or RACI) can support this activity.

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Impact of a project's life cycle on its governance framework

Photo credit: Denys Nevozhai

- Projects will follow a recognised project life cycle.
- Governance will be embedded into each phase of the project life cycle, with control points between each phase such as gate reviews, audits and evaluation.
- Governance ensures that all the requirements of a preceding phase of the chosen life cycle are met before work progresses to the next phase.
- The decision points between life cycle phases are commonly known as decision gates and rely on assurance of the work carried out so far.
- The life cycle approach adopted will impact on the level of flexibility.
  - For example, an iterative life cycle allows objectives to evolve and therefore gives greater flexibility to the project manager in terms of budget and schedule as the scope develops. A linear approach is likely to have a strict budget and schedule agreed early on, and any change requires escalation to the steering board for approval.

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2d) Understand the importance of linking projects to an organisation’s objectives.

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Linking projects to an organisation's objectives

- Projects, programmes and portfolios are introduced to enhance performance, bring about change and enable organisations to adapt, improve and grow.
- Organisational change is introduced through projects, programmes and portfolios to deliver business value.
- Delivering strategy is enabled through projects, programmes and portfolios. Portfolios structure investments in line with strategic objectives, while balancing, aligning and scrutinising capacity and resources. Programmes combine business as usual with projects and steady state activity dictated by strategic priorities.
- Projects are transient endeavours that bring about change and achieve planned objectives. They are intended to deliver the beneficial change required to implement, enable and satisfy the strategic intent of the organisation.

Photo credit: Gerrit Vermeulen

Linking projects to an organisation’s objectives

- Linking projects to an organisation's objectives involves aligning project goals with the strategic aims of the company. This ensures that every project contributes directly to the broader business agenda and enhances overall organisational success.
- ***The following defines the importance of strategic alignment:***

1. **Enhances Strategic Cohesion:**

- When projects are aligned with organisational objectives, they contribute to a cohesive strategy that integrates various initiatives across the company. This alignment helps prevent projects from operating in silos, ensuring that all efforts are synergistic and collectively drive the organisation toward its long-term goals.

2. **Improves Resource Utilisation:**

- Aligning projects with organisational objectives ensures that resources are allocated efficiently and effectively. This strategic alignment helps organisations avoid wasteful expenditures and optimise the use of their assets, whether it's time, money, or personnel, by focusing on projects that offer the most substantial strategic value.

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Linking projects to an organisation's objectives

3. Increases Stakeholder Engagement:

- Stakeholders are more likely to support projects when they understand how these initiatives fit into the larger business context. By linking projects to clear organisational goals, stakeholders, including investors, employees, and partners, can see the direct benefits and are thus more engaged and supportive.

4. Enhances Decision-Making:

- Strategic alignment simplifies decision-making processes. Project managers and team members have a clear framework within which to make choices about project directions, priorities, and tactics. This clarity comes from understanding how their decisions impact broader business objectives, leading to more effective and quicker decision-making.

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Linking projects to an organisation’s objectives

5. **Boosts Motivation and Morale:**

- Employees are more motivated when they see how their work contributes to the success of the organisation. Linking projects to organisational objectives helps team members understand their role in the bigger picture, which can enhance job satisfaction and morale, leading to higher productivity and better project outcomes.

6. **Facilitates Better Risk Management:**

- When projects are closely tied to the company's objectives, it becomes easier to identify, assess, and manage risks related to the strategic goals of the organisation. Project managers can prioritise risk management efforts based on how potential issues might impact the achievement of organisational objectives, thus ensuring more targeted and effective risk mitigation strategies.

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 1.3.1 (Governance principles)

APM (2020) ‘The golden rules of governance’ apm.org.uk/blog/the-golden-rules-of-governance/

APM ‘Governance, an enabler of project success’ apm.org.uk/resources/what-is-project-management/what-is-governance/

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Sample Exam Questions

Learner Study Guide

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Sustainability

Learning objective:

3. Understand sustainability as balancing the environmental, social, economic and administrative considerations that will impact a project.

3a) Understand why sustainability responsibilities, principles and priorities are considered within a project and the impact they may have.

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What is sustainability in project management?

- Sustainability in the project profession is an approach to business that balances the environmental, social and economic aspects of project-based working to meet the current needs of stakeholders without compromising or overburdening future generations.

- Building sustainability into the vision at project inception is key – if sustainability is planned in as a key requirement from the outset, it cannot then be taken out.

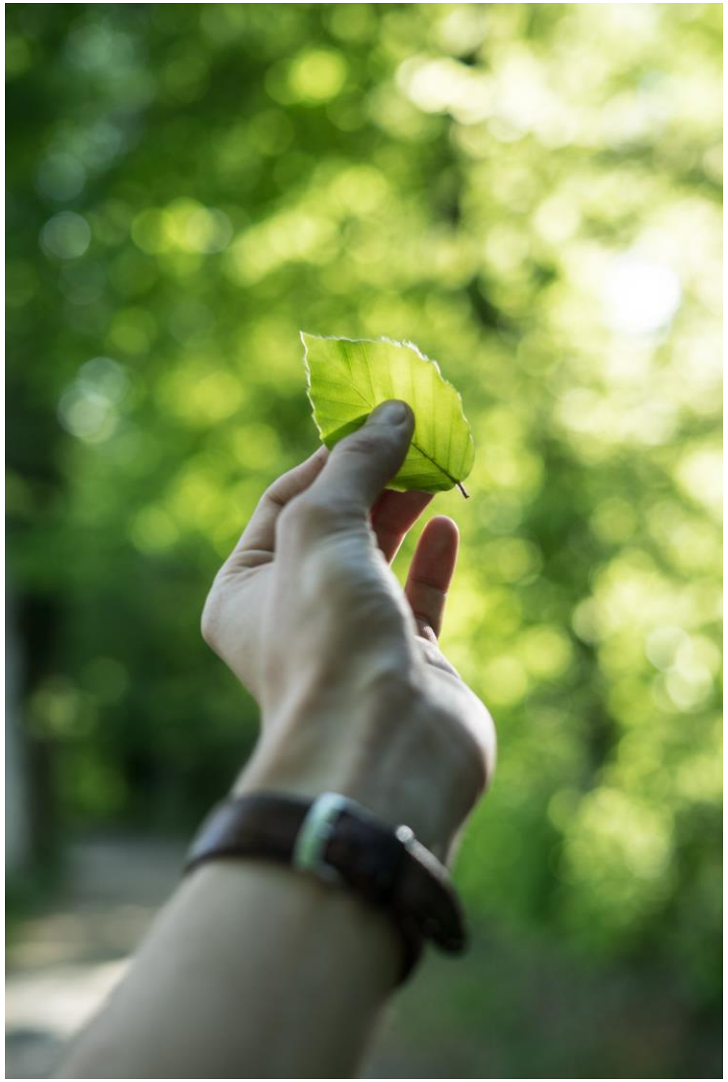

Photo credit: Tobias Weinhold

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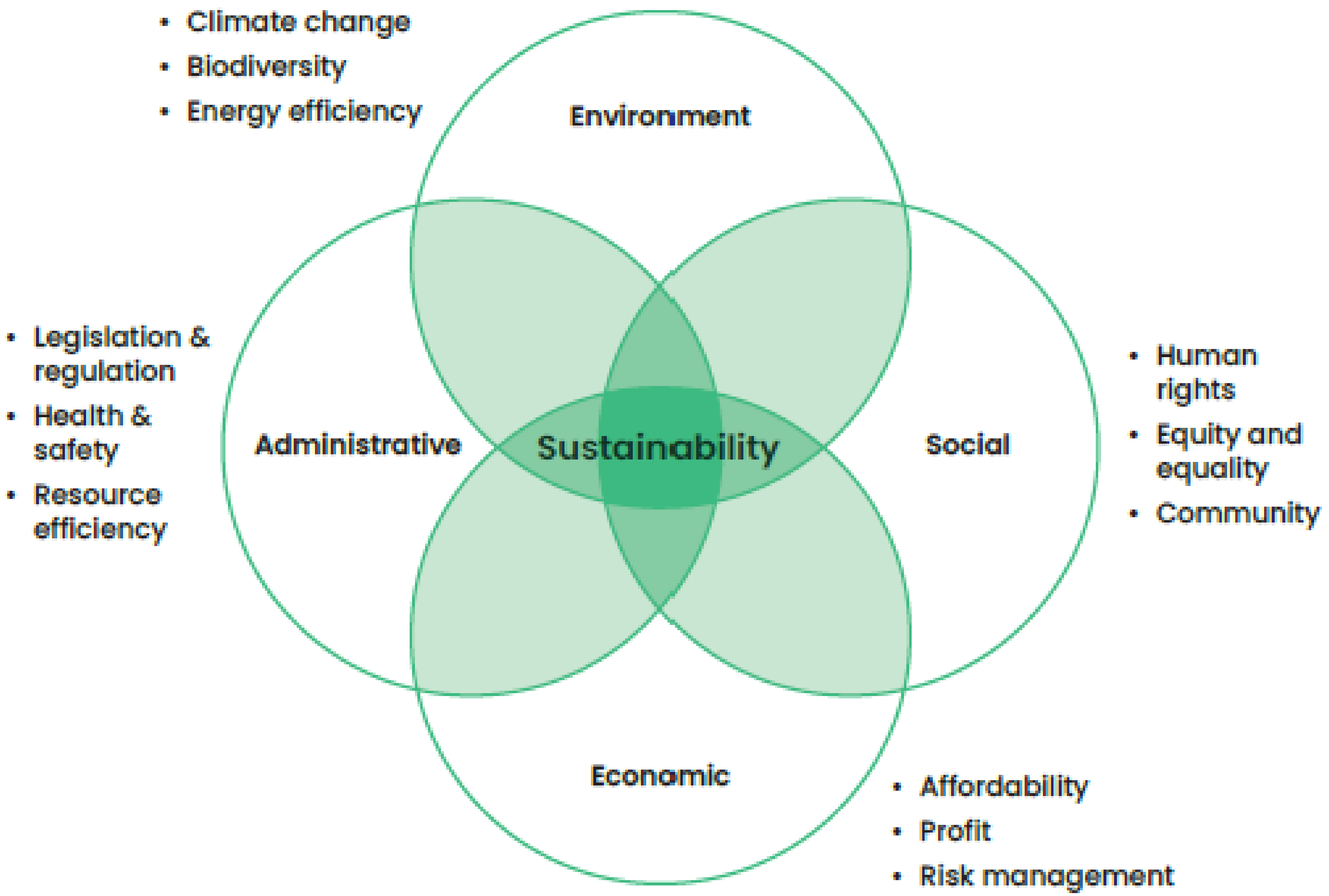
Balancing different concerns

Sustainability requires balancing environmental, social, economic and administrative considerations.



In your own organization which of the four 'broad' Principles is the main driver?

Source: *APM Body of Knowledge*, 7th edition.

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Sustainability vision and objectives

- Project sustainability involves both individual and organisational responsibility.
- The business case should be aligned to support the sustainability agenda.
- There should be alignment with sustainability frameworks, standards and best practices, such as the UN's Sustainable Development Goals (SDGs) or environmental, social and governance (ESG) criteria.
- Sustainability initiatives will influence product design, manufacturing processes, energy consumption, waste management and stakeholder relationships.

Photo credit: Zbynek Burival

3b) Knowledge of how sustainability measures are monitored and reported on.

Stakeholder perception

Explore stakeholder perception and actively manage sustainability expectations and outcomes, ensuring that stakeholders are engaged and responsive to sustainability practices.

Internal communities

- Organisations need the ability to support ongoing business-as-usual operations.

Adjusting the culture of the organisation, internally, to make sustainability a part of day to day thinking

"we want to do it" not "we are made to do it"

External communities

- Organisations should report on carbon emissions, energy consumption, waste management, social impact and ethical practices.
- Sustainability messages should be tailored to different target audiences and be effective for engaging stakeholders.

Environmental, social and economic impacts

- Transparency, credibility, and the importance of data and metrics are key in communicating sustainability performance.
- Aim for sustainable procurement.

Global Reporting Initiative (GRI)

- The GRI provides support relating to data collection methods, data analysis techniques, and continuous improvement processes for sustainability practices.

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 1.3.3 (Sustainability)

APM (2022) ‘Thinking beyond a project to bring sustainable value to society’ apm.org.uk/blog/thinking-beyond-a-project-to-bring-sustainable-value-to-society/

APM (2021) Make climate action part of your project apm.org.uk/blog/make-climate-action-part-of-your-project/

APM ‘Delivering sustainability in projects’ https://youtu.be/r9kka5Ge3io

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Sample Exam Questions

Learner Study Guide

Sample Exam Questions

Brainstorm of question 3.3

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Business case

Learning objective:

4. Understand a business case as the justification for the initiation, investment and/or continuation of a project in terms of benefits, costs and risks.

4a) Knowledge of the tools and techniques used to determine factors which influence and impact a project’s business case (including PESTLE, SWOT and VUCA).

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Five elements of the business case

Strategic context: The compelling case for change.

Economic analysis: Return on investment based on investment appraisal of options.

Commercial approach: Derived from the procurement strategy.

Financial case: Affordability to the organisation in the time frame.

Management approach: Roles, governance structure, life cycle choice, etc.

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Tools and techniques

PESTLE

- The Political, Economic, Sociological, Technical, Legal and Environmental influences that could impact the project.

SWOT

- The Strengths, Weaknesses, Opportunities and Threats relating to the project.

VUCA

- The Volatility, Uncertainty, Complexity and Ambiguity relating to the project.

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SWOT

Strengths	Weaknesses
Opportunities	Threats

SWOT

Strengths

Consider the strengths of the organisation, and their influence / impact on the project.

Weaknesses

Consider the weaknesses of the organisation, and their influence / impact on the project and the risk they might pose.

Opportunities

Consider the elements in the world outside the organisation that the project could exploit.

Threats

Consider the external elements outside the organisation that could be problematic for the project and the risk they might pose.

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PESTLE

Select two PESTLE factors and consider how these might impact your project

Political	• Might lobbying or a change of government impact the project? • What about local government policies? • Office politics?
Economic	• Is the project subject to market volatility, exchange rates, inflation, etc.? • Availability of funding vs Financial Benefits
Social	• Will the project impact on people’s health or living standards? • Will It require a change in behaviour, attitudes to things?
Technological	• How mature is the current technology? Will it support our proposed solution? • Is technology likely to change?
Legal	• Is new legislation likely? Is contract law or employment law a factor? • Internal policies which may constrain us?
Environmental	• Does the project have an environmental impact? • Sustainability issues

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VUCA

Factor	Description		Response	Description
Volatility -	the nature, dynamics and speed of change – things are happening fast and in unpredictable and non-repeatable ways eg Material prices due to market turmoil	V	**Vision**	Accept and embrace change, keep your end goal in mind
Uncertainty	the lack of predictability and likelihood of surprise – the past is not an accurate predictor of the future.	U	**Understanding**	Pause for thought, invest in analysis of the environment
Complexity	the multiple forces, chaos and confusion – there are so many factors involved it's difficult to know what affects what	C	**Clarity**	One person can't understand all, so develop teams & communicate well
Ambiguity	the fuzziness of reality, potential for misreads, and cause and effect confusion – who, what, when, where, how and why is unclear	A	**Agility**	Plan ahead in small chunks, be flexible, build in contingency

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4b) Understand the importance of regularly reviewing the impact of any changes in a project to the business case and know that the business case forms the baseline for the project.

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Baseline and reviews of the business case

- The business case justifies the initial and continued investment in the project and enables the project manager to agree success criteria with the project sponsor and other stakeholders.
- It provides a record of decisions made by governance and documents the options considered.
- As a document, it will be reviewed periodically and at decision gates throughout the life cycle to ensure the investment remains aligned to expected benefits.
- In the definition and deployment phases, the business case can be reviewed against progress made, as well as checking progress against the benefits outlined.
- The project manager will use the business case, which defines why the project is needed, as the starting point for developing the project management plan.
- If the benefits are no longer viable, then the senior management team may stop the project.

Photo credit: Dylan Gillis

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 1.3.7 (Business case)

APM (2018) *Starting Out in Project Management*, 3rd edition

APM (2019) ‘How to nail your business case’ apm.org.uk/blog/how-to-nail-your-business-case/

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Procurement

Learning objective:

5. Understand procurement as securing the provision of resources, choosing strategies for obtaining best value from supply chains.

5a) Understand the purpose and importance of a procurement strategy and know the typical contents of a procurement strategy.

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Procurement

- Process of acquiring goods and services.
- Includes the development of a procurement strategy, preparation of contracts, selection and acquisition of suppliers, and management of the contract.
- The procurement strategy is a project document that describes the mechanics of how the project will go about procuring and subsequently managing services and goods, and it should be formulated alongside the business case.
- It may also be a component of the project management plan as there will be numerous procurement implications for the project manager throughout the life of the project. It forms a communication tool between the project manager and project stakeholders.

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Make or buy decision

Should the goods be made in-house or could we procure them elsewhere at a lower cost or better quality?

- Understanding of the functional and technical specification required to understand the cost and time needed.
- Understanding of capacity, speed and availability as well as the benefits of specialist or technical support.
- Definition of the required quality aspirations.
- Engagement with relevant internal and external stakeholder groups.

Photo credit: Headway

5b) Know the stages of a supplier selection process, including how to plan the procurement process and conduct negotiations (for example ZOPA, BATNA and ‘Win Win’).

Supplier selection process

Many organisations have a specific process for selecting suppliers. There will be rules and guidelines to follow.

There may be guidelines for a competitive tender process where several suppliers are asked to submit prices for a specified piece of work in a commercially competitive manner. The objectives are to achieve best value for the price charged, and to ensure the process and selection are transparent and fair.

It is important that qualified resources are engaged in the process.

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Tender process

Research the market

Define requirements, and review number of potential suppliers

Pre-qualify suppliers

Check capability, financial stability, references

Issue an Invitation to Tender (ITT)

Instructions, questions, evaluation criteria and scoring matrix

Respond to queries from bidders

Opportunity to seek clarifications

Receive and evaluate the bids

Against selection criteria; consider compliance with ITT and added value

Award a contract to a successful bidder

Agree to the terms of a binding contract, understanding respective liabilities and responsibilities

Enter into contract and contract administration

In accordance with contract terms agreed

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Plan for negotiation

When trying to reach an agreement with a selected supplier, several negotiation techniques could be used.

BATNA

Best Alternative to a Negotiated Agreement

This is the best fallback position.

It is useful to understand this position for both parties.

ZOPA

Zone of Possible Agreement

This is the bargaining range where the lowest of the suppliers quoted price range and the highest of the customer's price range overlap.

There will be a successful outcome to the negotiation when the final agreement takes place anywhere within the ZOPA range.

'Win Win'

This is a collaborative approach.

It focuses on problem-solving and communication.

It depends on a strong trust-based relationship.

ZOPA example:
Software Suppliers Price range = £1m-£1.4m
Project Software Budget = £900k-£1.2m
ZOPA = £1m to £1.2m

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5c) Know the features of different contractual relationships and understand why different methods of supplier reimbursement are used and when it is appropriate to use them (including fixed price, cost plus fee, per unit quantity and target cost).

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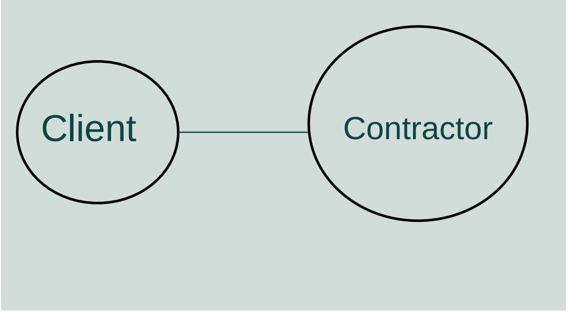
Contractual relationships

How many suppliers should we buy from? How long-term do we want the relationship to be?

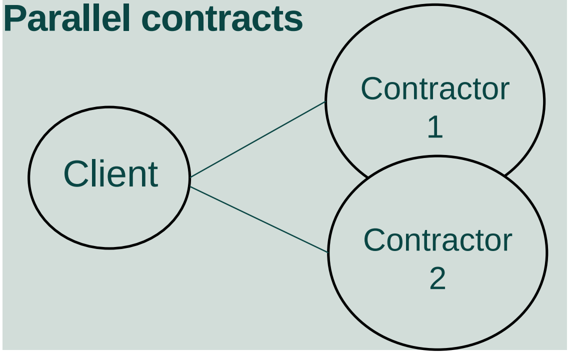
Single contract



Simplest form

Client buys services / goods from one supplier.

Parallel contracts

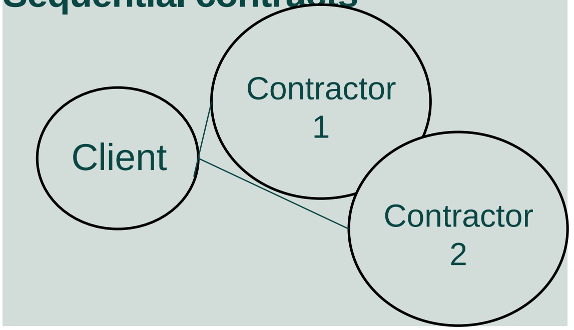


Duplication or replication of work

Several different contracts where suppliers are doing related work at the same time.

The client holds multiple contractual relationships.

Sequential contracts

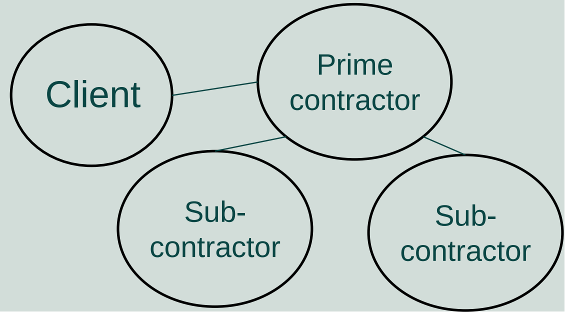


Dependency between activities or work packages

Dependency of completion of one activity prior to another activity commencing.

The client holds multiple contractual relationships.

Prime contracts



Management of sub-contracted work

The client holds one contractual relationship, but sub contractors deliver elements of the specialist work for the Prime contractor.

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Pro’s & Con’s of single, multiple or integrated suppliers

The supplier strategy chosen will affect the cost and time required for procurement and contract negotiation, the leverage on product pricing and the level of responsiveness of the supplier(s).

Single supplier

Preferred supplier with strong existing relationship used for all requirements: preferential rates / volume pricing.

Less time and cost attached to procurement.

Risk of over-reliance on one supplier: shortages, disruption.

Multiple suppliers

Price competition between suppliers.

Reduced risk of supply shortage / disruption.

Increased cost in contract negotiations and management resource.

Integrated supplier

Integration of supplier into the project team.

Optimum service delivery through immediacy of contact, easier communications, greater transparency and understanding.

Difficulties in feeding back issues in service or maintaining confidentiality of certain aspects of the project.

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Conditions and forms of contract

- A contract is a legally binding agreement between two or more parties setting out obligations of each party.
- There should be an ‘offer’ made by one party, and acceptance by the other party; intention to create legal relations between the parties; consideration passing from one party to the other in return for the provision of goods or services covered by the contract; definite terms, so that it is clear as to what conditions the parties are agreeing; and legality, with only properly incorporated firms or competent persons entering into the contracts.
- Typical contract conditions should include general information, provider’s responsibilities in terms of design and subcontracting, time, quality, payment, dealing with change, compensation or risk, property ownership, insurances and dispute resolution.
- Specialist advice should be sought by the project team in relation to amending, drafting and agreeing contracts so that level of risk exposure and legal ramifications are clearly understood.
- Bespoke forms of contract can be drafted to apply to specific projects or procurement requirements.
- Standard forms of contract in engineering and construction are the Joint Contracts Tribunal (JCT) and the New Engineering Contract (NEC), which have established best practice and are industry recognised.

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Supplier reimbursement terms

The decision about which reimbursement approach is selected is often based on level of risk and how this risk is apportioned between the parties. It can also depend on the level of maturity of scope development and understanding of what is required.

Time and materials / unit price	Cost plus	Contract target cost / price	Fixed price
£100 per hour / buy 10 and they cost £10 each, buy 100 and they will be £9 each	Agree all reasonable costs will be reimbursed plus agreed fee	Target cost of £20k in the contract, plus agreed fee. Final Cost less than target the supplier gets share of saving	Agree £20k in the contract, pay them £20k
• Client pays per unit. • Time overrun risk lies with client. • Suitable when specification is vague.	• Client pays actual cost of work plus an agreed amount for supplier overhead and profit. • Profit could be a percentage (lower risk for supplier) or a fixed fee (higher risk for supplier). • Suitable where works are urgent, or an earlier start is required.	• Target cost or price is agreed upfront; client and supplier work together to achieve it. • Risk is shared between the parties at an agreed rate. • Incentivised collaboration. • Suitable where contractual relationships are collaborative and where there are opportunities to incentivise collaborative ideas .	• Price agreed based on agreed scope of work. • Time overrun risk lies with contractor. • Price may be higher due to greater level of risk for supplier . • Suitable where scope is clearly defined, and a transactional relationship is preferred or tolerated.

Risk to client

Risk to supplier

Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Sections 2.1.4 (Procurement strategy), 4.2.1 (Contract award)

APM Contracts and Procurement Specific Interest Group (2017) *Guide to Contracts and Procurement* apm.org.uk/book-shop/apm-guide-to-contracts-and-procurement/

APM Contracts and Procurement Specific Interest Group (2020) *Procuring for agile projects* apm.org.uk/v2/media/2t5pppeb/procuring-for-agile-white-paper-akt12811-lr.pdf

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Reviews

Learning objective:

6. Understand reviews as a way of gathering information to provide an assessment on the status of a project and the ongoing viability of the work.

6a) Understand the benefits of conducting reviews throughout the life cycle (including decision gates, benefits reviews and audits).

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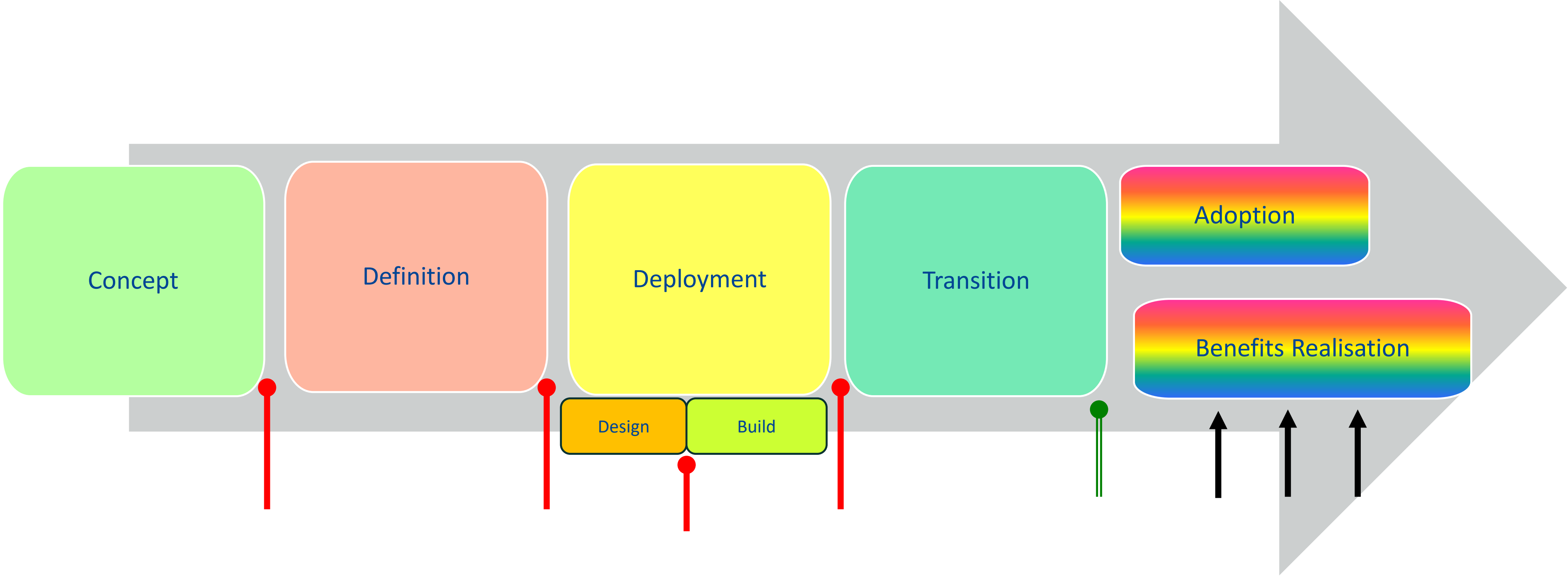
Reviews



Key:

Decision Gates

Post-project review

Benefit realisation review

Audits - throughout

Types of review

Decision gates

- If the life cycle is linear, these will be event driven. If iterative, these will be time-bound. They lead to increased senior stakeholder involvement.
- Held at the end of each stage or iteration primarily to gain agreement from the sponsor and wider governance board or changes to be incorporated.
- Check that the project is still on track to deliver the benefits as defined in the business case.

Benefits reviews

- Held during benefits realisation, typically 6–12 months after handover.
- Check that the project achieved the business benefits identified within the business case.
- Actions or improvements needed to support the realisation of the benefits can be determined.
- Typically undertaken by the project sponsor.

Audits

- Provide independent compliance assessment and increase stakeholder confidence.
- Held to check that the project is following the agreed governance and processes as defined within quality management plans.
- Conducted by personnel independent of the project.

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6b) Know the factors which would typically be reported on to help ensure successful project outcomes.

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Factors to be reported on

Progress against the schedule	• Understanding how the project's tasks are progressing against the schedule will help the project manager to assess progress on the overall project and to identify where controlling actions might be required if progress isn't as planned.
Actual spend against planned spend	• Understanding how the project is performing against a set spend curve and any corrective action needed where overspend is identified.
Earned value	• Tracking project spend and understanding the value of the work achieved.
Audit findings and number of non-conformances	• Understanding whether processes being followed are correct and generating the required outputs or alternatively considering any recovery actions needed.
Risks and issues	• Understanding and communicating transparently the risks and issues facing the project and how these need to be monitored and managed.

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6c) Understand the importance of producing information and collecting data to inform decision making and communicate actions and decisions to stakeholders.

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Importance of information and data

Status reporting	Estimating the project	Risk identification	Problem solving and option appraisals	Information Management	Knowledge management
• So that the project manager can better understand the status of the project in relation to cost, schedule and quality against the acceptance criteria in the project management plan.	• So that more realistic schedules and costs can be developed using past project data.	• So that the need for any subsequent corrective actions to be undertaken or the need for escalation can be determined.	• So that options and solutions to solve project problems and issues can be generated using past project experience.	• To inform the decision-making process at key points, such as gate reviews, by ensuring the relevant information is available in an appropriate format.	• To provide an understanding of when best to schedule quality assurance and quality control activities for the project. • By assessing what was done on past projects and by understanding the optimum times within a project life cycle, activities can be scheduled.

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6d) Understand why activities may be re-planned after a review.

Re-planning after a review

Progress monitoring leads to the production of meaningful reports. Such reports allow the assessment of whether any re-planning of activities is needed in the following areas:

- Where achievement of planned scope is not to the required quality.
- Where motivation and satisfaction of team members is low.
- Where performance of contractors and the health of the relationships in the supply chain is poor.
- Where committed costs and cash flow are above original forecasts.
- Where changes to the risk profile and impact on time buffers or cost contingency are needed because of newly identified or changed risks.
- Where effectiveness of communication with stakeholders is lacking.

Photo credit: Markus Winkler

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Sections 2.2.2 (Decision gate), 4.3.1 (Progress monitoring and reporting)

APM (2013) 'What makes a good project reviewer?' apm.org.uk/blog/what-makes-a-good-project-reviewer/

APM (2012) 'Project Reviews, Assurance and the PMO' apm.org.uk/blog/project-reviews-assurance-and-the-pmo/

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Assurance

Learning objective:

7) Understand assurance as the ability to provide confidence to the governance board that a project is on track to deliver objectives.

7a) Know the purpose of assurance within a project; including awareness of the scope, priorities and strategic aims of assurance activities.

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Assurance reviews

- Objective and independent approach agreed by sponsor and other stakeholders.
- Provide confidence to the governance board the project is on track.
- Support decision making and change control.
- Improve risk assessment.
- Increase probability of projects meeting their objectives.

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Assurance activities within a project and priorities

These work together: any weakness in one discipline damages all three.

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Assurance activities

1. Set Controls

- Application of a management system, comprising policies, procedures, processes, standards.
- Intended outcome is to control risks.

2. Ensure Compliance

- Management assurance, comprising monitoring, checks and audits through risk management, quality assurance, PMOs.
- Intended outcome is to confirm the control of risks.

3. Independent review

- Assurance through independent reviews by internal audit, external audit (e.g. NAO), independent peers or external scrutiny.
- Intended outcome is to provide strategic overview of system of control.

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A successful assurance plan will meet certain criteria

- Assurance is independent, objective and proportionate to the work.
- Assurance is targeted where the greatest risks exist.
- There are clear accountabilities for assurance arrangements, activities and outputs.
- Timings and cycles of assurance activities are planned and coordinated.
- The governance route for reporting outcomes and resolving issues is clear.

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 1.3.2 (Assurance principles)

APM (2012) ‘Project Reviews, Assurance and the PMO’ apm.org.uk/blog/project-reviews-assurance-and-the-pmo/

APM (2019) ‘What is the value of assurance?’ apm.org.uk/blog/what-is-the-value-of-assurance/

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Transition management

Learning objective:

8. Understand transition management as integration of the outputs of a project into business-as-usual.

8a) Know the basic requirements needed to support a successful transition, including considering business-as-usual throughout a project and planning for transition from the outset of the project.

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Actions for a smooth transition

- Managing **stakeholder** expectations from the outset to increase the likelihood of the project outputs being accepted.
- Identifying and engaging with end users and project sponsors as **key stakeholders** throughout the project.
- Aligning needs with outputs where project constraints allow and **communicating** any digression from these.
- Setting the path for handover and acceptance by familiarising key **stakeholders** with the deliverables from the outset.
- **Clear communication** to reduce the risk of the project manager misunderstanding the requirements or the end users of the deliverables.

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Preparing for new business-as-usual

- Preparing for the new business-as-usual should be considered early in the project life cycle.
- Business readiness should be considered as a continuous activity throughout the project to ensure that the intended users of the project outputs and those who will be expected to do things differently have bought into the outcomes.
- Business readiness activities can be carried out by the project team. These will require them to proactively engage with people to understand the processes and routines performed in business-as-usual, listen to and appreciate the sources of any resistance, and to identify, empower and motivate champions for the 'new normal'.
- Techniques, such as **change impact analysis**, will help understanding of:
  - physical or process impacts on the wider system
  - feelings about the change
  - knowledge or skills gaps that need to be addressed

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Change impact analysis

Change impact analysis is a method of analysing the scope and scale of the change to the new business-as-usual state.

It helps to build an understanding of the organisational impacts on people, performance, processes, systems and culture.

It supports the planning and communication of change effectively, identifying potential issues and highlighting the resources required.

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The change impact analysis process

1. Define the change, its purpose, objectives, scope and timeline.
2. Identify organisational areas such as people, performance, processes, systems, culture and individual stakeholders that will be impacted.
3. Evaluate the level of impact that the change will have on each area.
4. Develop an action plan to address the impacts of the change, which should be aligned with a common and shared vision, and communicated regularly throughout the project.

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8b) Understand the importance of knowledge transfer in the transition process; this includes learning from experience and continuous improvement.

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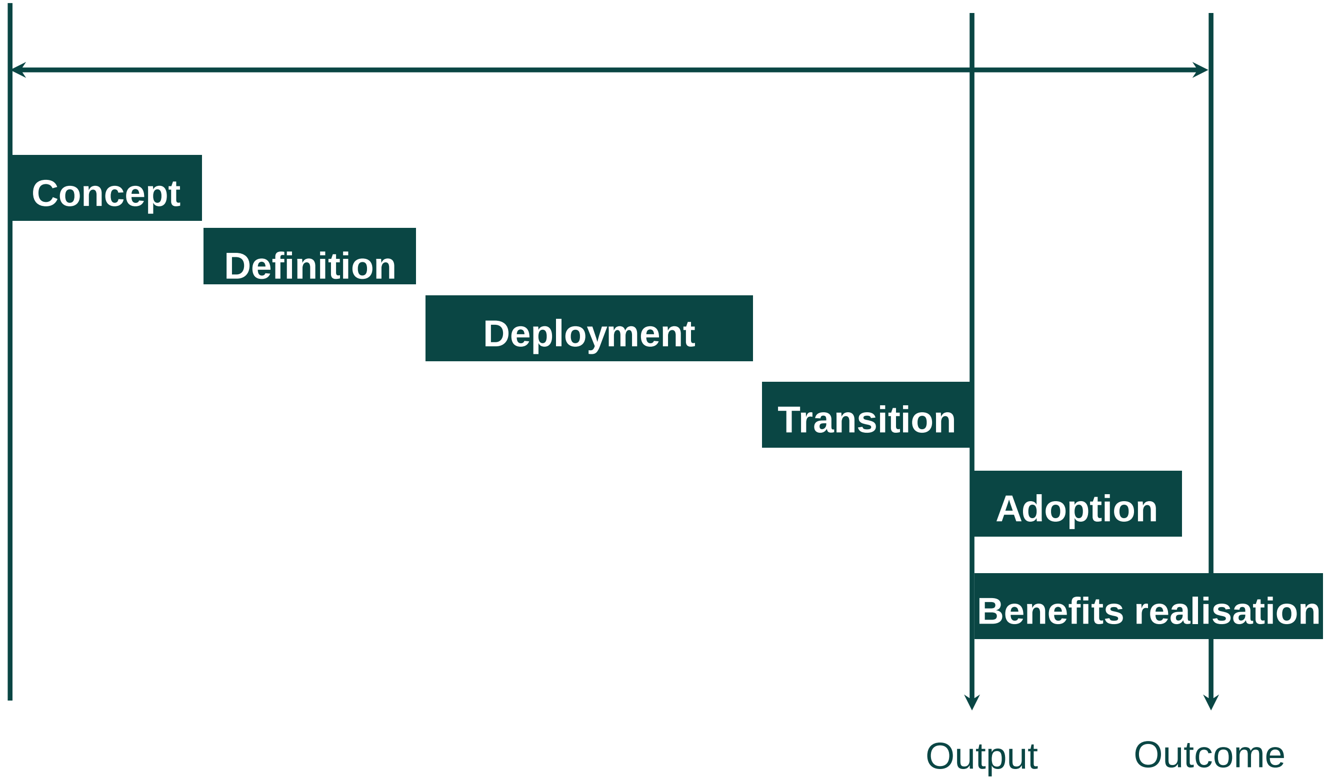
Progressive implementation

- The relationship between the people responsible for deployment of new capabilities and those responsible for business readiness and adoption in business-as-usual is carefully managed as part of governance.
- The sponsor is a key champion of this work to align the knowledge and experience of the temporary project team with the functional and operational teams to ensure the transfer of knowledge can take place.
- An **extended life cycle approach** would make the project team accountable for change management and therefore prevents the formation of knowledge boundaries between project teams and operational teams.
- It is important to provide effective learning and development from experience so that knowledge can be transferred to nurture future talent and users of the project outputs and to promote continuous improvement.



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Transition planning

Transition planning starts at the beginning of a project. Information provided by the investing organisation is used to shape the solutions and plans, including details of what needs to be handed back to the organisation at the point of transition.

The transition process may include:

- Acceptance of all pertinent documentation (containing all prescribed information related to the deliverables, including guarantees and warrantees.)
- Acceptance certificates signed by the sponsor.
- Transfer of responsibility for the deliverables from the project team to the sponsor or users.
- Formal transfer of ownership.

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Information handed over to business as usual

The information provided at handover for linear projects, or at relevant points of a project delivered under an iterative life cycle, can include:

- process maps
- project documentation
- ownership of supplier relationships
- ongoing licensing arrangements

As an example, for a building project, the handover information might include:

- user manuals
- operating procedures and maintenance manuals
- asset registers
- design information in the form of architectural and / or engineering drawings
- survey information
- health and safety files
- warranty information and commissioning certificates

Training on new systems, software and knowledge transfer should be organised by the project manager with the users at transition or at an appropriate point in the life cycle. For example, in an iterative or hybrid project, training and knowledge transfer may be required frequently, and before the end of a project.

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8c) Understand how to engage stakeholders to agree a transition plan, including transfer of risks.

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Engaging stakeholders with a transition plan

Early engagement

It is important to engage the relevant stakeholders early in the project. Stakeholder mapping will help to identify those that will be impacted by or interested in the transition process, such as senior leaders who will drive the transition, or departmental change agents providing personalised transition leadership to colleagues with existing or new roles.

Strategic alignment of goals and understanding of the future state

Collaborative planning with stakeholders throughout the project will help to develop a shared vision and a common understanding of the organisational and strategic impact, benefits and financial value goals expected.

Communicating benefits

Communicating benefits to stakeholders from an early stage in the project will support common understanding. Designating a benefits champion can support the communication of expected benefits with stakeholders throughout the project and help to prepare them for the transition process.

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Agreeing a transition plan with stakeholders

A benefits management plan

This will define how the benefits will be managed through transition into operational use and the roles and responsibilities involved; benefits should be documented in terms of priority interdependencies, value, timescales, and ownership. Techniques such as MoSCoW (M – Must have; S – Should have; C – Could have; W – Won't have) can be used.

Mitigating conflict

The possibility of conflict between the user and the project manager should be considered; this may occur if the project does not meet its acceptance criteria, or user expectations.

Transfer of risk

Any remaining risks at project closure should be communicated to those involved in the adoption phase.

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Sections 2.3.1 (Business readiness), 2.3.2 (Transition of project outputs)

APM ‘What is change management and organisational change?’ apm.org.uk/resources/what-is-project-management/what-is-change-management/

APM ‘Managing effective change’ https://youtu.be/WlPsgegZ5Mk

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Benefits management

Learning objective:

9. Understand benefits management as monitoring of benefits realisation throughout a project.

9a) Understand what is meant by benefits management (including identification, definition, planning, tracking and realisation).

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Benefits management life cycle: stages

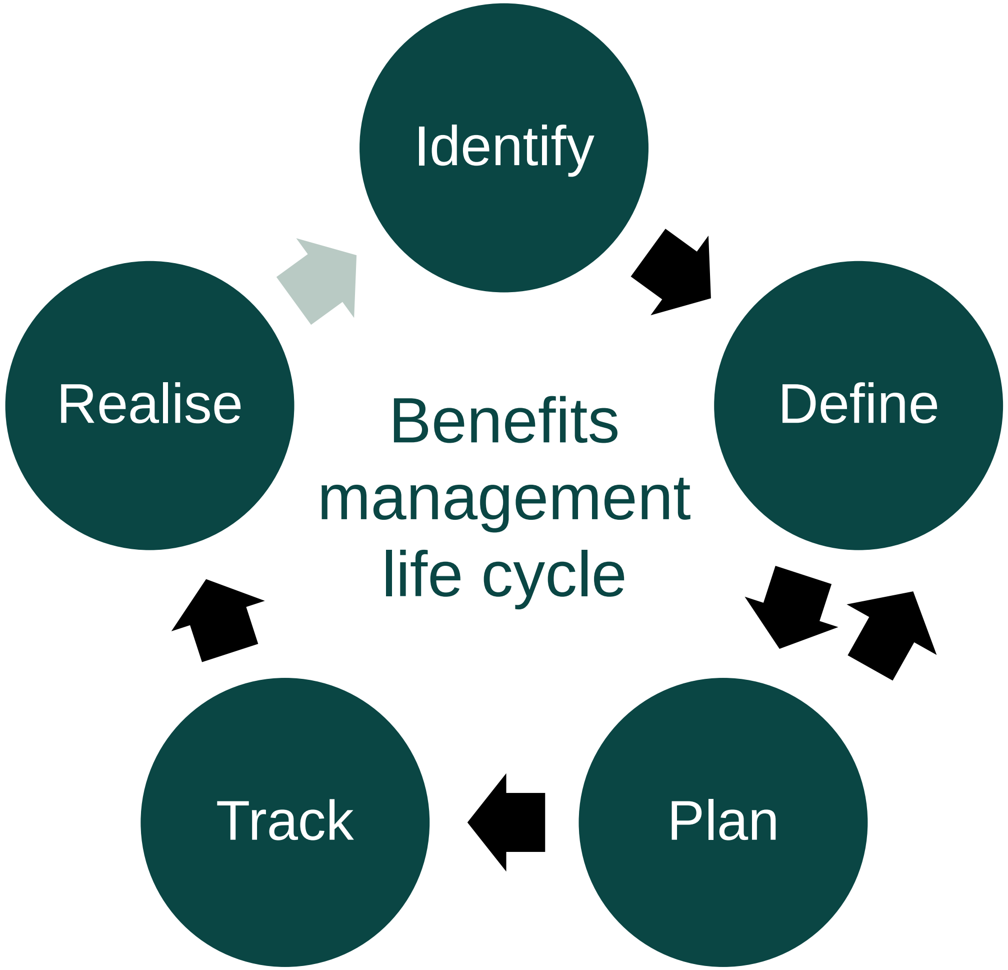

Figure ‘The Benefit Realisation Cycle’ based upon APMG International’s Managing Benefits by Stephen Jenner (TSO, 2012). Reproduced with permission of APMG International. All rights reserved; ©APMG International, 2012

Five stages of benefits management

Identification

- If benefits are not identified, then they will not be managed and return on investment could be missed. Requirements are captured from sources such as the business case, and stakeholders.

Definition

- A benefits management plan will define how the benefits will be managed through transition into operational use and the roles and responsibilities involved. It will also define how the benefits should be measured and the regularity of the management overview. Each benefit (and dis-benefit) should be documented in terms of priority, interdependencies, value, timescales and ownership.

Planning

- This involves capturing baseline measurements, agreeing targets, setting the timeline and milestones for realising benefits, and identifying their dependencies on project outputs. This allows the impact of any changes during the deployment phase to be assessed on the planned benefits.

Tracking

- Tracking benefits throughout the project deployment phase will inform when changes are needed to the scope or timescale and ensure outputs are still supporting said benefits. Agreement between the project manager and the sponsor on which benefits are to be tracked should be made early in the project life cycle, setting baseline measures against which to monitor progress.

Realisation

- Realisation of benefits comes when the project outputs are embedded and the operation changes for the better. A change manager or nominated representative may be put in place to track the realisation. The bulk of benefits may only be realised after the project is completed and therefore long-term actions and monitoring may be needed for continued realisation.

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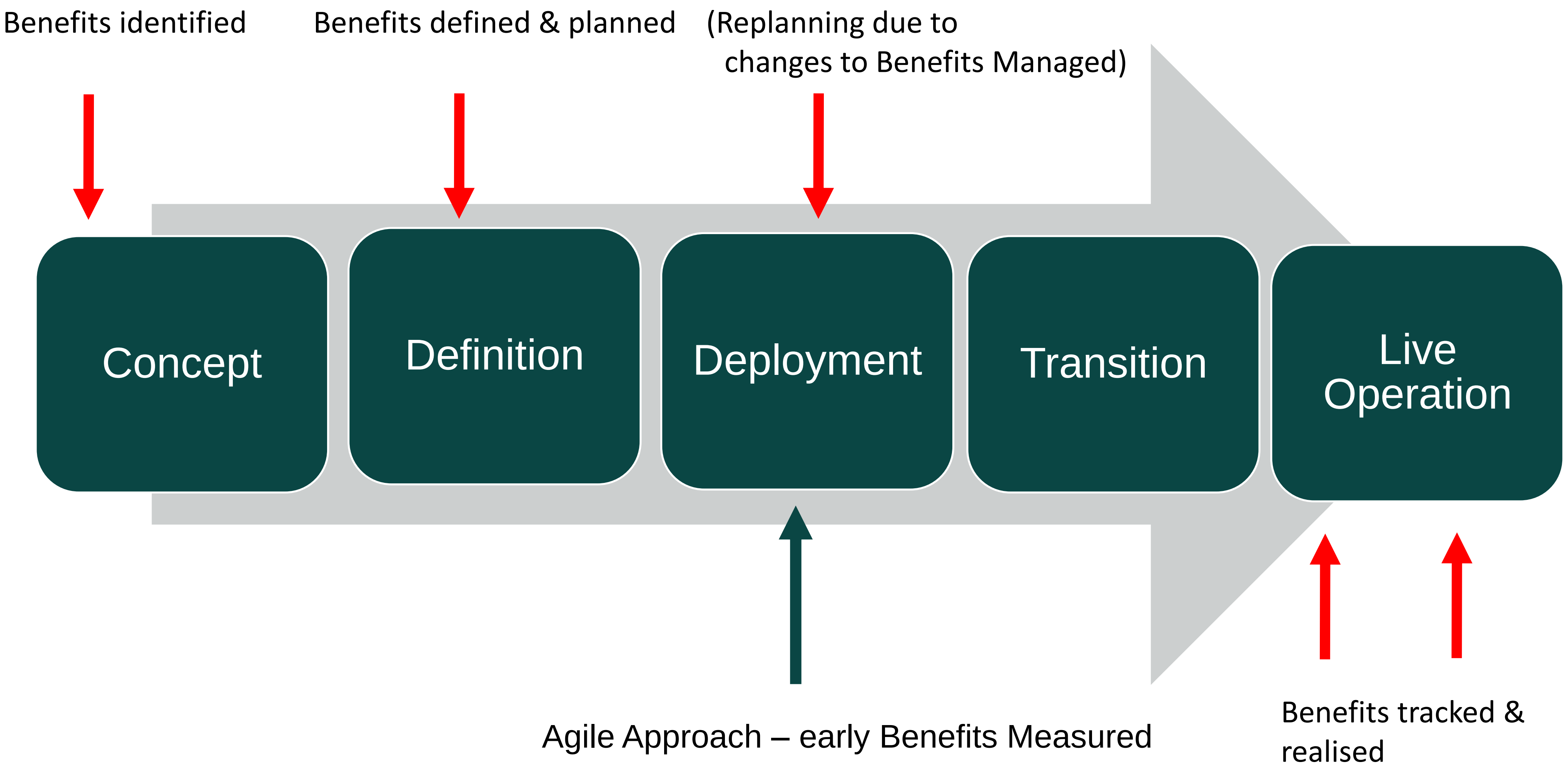
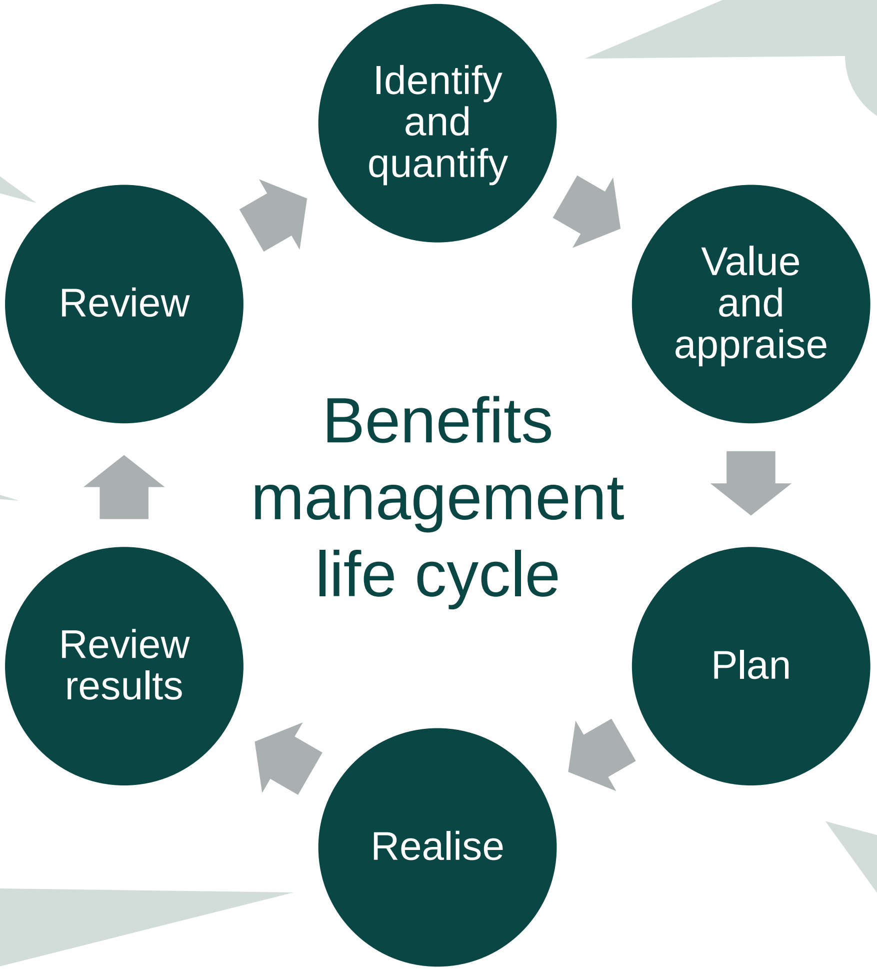
Timing of Benefits Management



Alternative Benefits management life cycle: stages



Identify and quantify
- Identify and involve stakeholders in the benefits and change process.
- Align benefits to strategic objectives.
- Use benefits maps to identify relationships between benefits.

Value and appraise
- Produce an outline business case.
- Establish baseline measurements.
- Establish realistic benefit targets.
- Undertake analysis to understand stakeholder interests and concerns.

Plan
- Describe benefits and relate changes.
- Establish responsibility for benefits realisation.
- Categorise and structure benefits in terms of the type of change needed.
- Establish change success criteria.

Realise
- Monitor, track and report benefits realisation.
- Optimise changes for maximum benefits realisation.
- Evaluate realised benefits.

Review results
- Identify additional benefits.
- Identify the benefits possible through business changes.
- Identify the benefits possible from further investment.

Review
- Assess benefits management capability development.
- Identify how to improve the benefits management process.

Figure 'The Benefit Realisation Cycle' based upon APMG International's Managing Benefits by Stephen Jenner (TSO, 2012). Reproduced with permission of APMG International. All rights reserved; ©APMG International, 2012

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Benefits management life cycle: other considerations

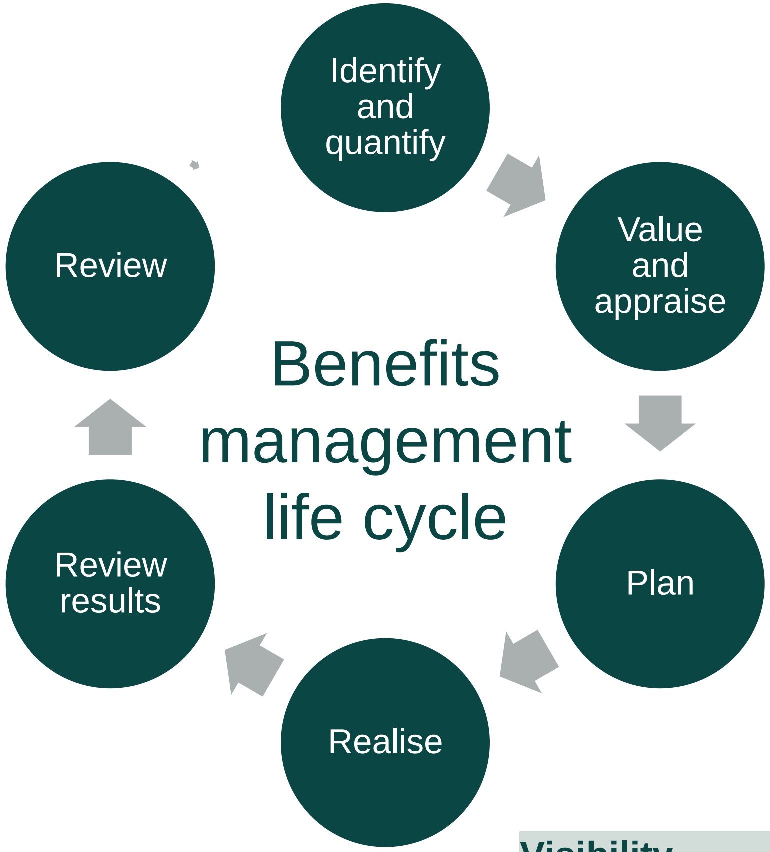
A transformative impact

The introduction of benefits into an organisation can be regarded as a transformational change in its own right. To be effective, the culture and behaviours of an organisation must support internal and external collaboration. Open and frank discussion around benefits and solutions must be accommodated at all levels of the organisation.

All stakeholders have a role to play in achieving successful change outcomes. Benefits management activities help bring stakeholders closer to the reasons for change, options for change and the development of the required solutions

Ownership

Having sponsorship for benefits management at board level helps with adoption across the business. It's essential that the sponsor of the programme understands that they are accountable for realising the benefits in the business case. It is also important to get the affected parts of the business involved in ensuring the necessary changes happen.



Benefits framework (optional)

This is a portfolio level document used to support the management of benefits across multiple projects and programmes. It sets the standards for benefits definition and management including:

- benefits mapping
- measures rules and guidelines
- valuation methods
- evaluation
- categorisation

Benefits management capability

Benefits management is best implemented and developed as an organisational capability. A capability improvement approach might include:

- benchmarking against other similar organisations
- benefits capability assessments (for example, P3MS maturity assessment)
- benefits capability improvement planning
- networking and knowledge sharing facilities
- training and benefits tool support

Visibility

Introducing benefits management to an organisation increases the visibility of failures as well as successes, so there can be an initial reluctance to embrace it across the organisation. However, the benefits of having a shared vision, shared objectives and everyone understanding their role in achieving them outweighs the risk.

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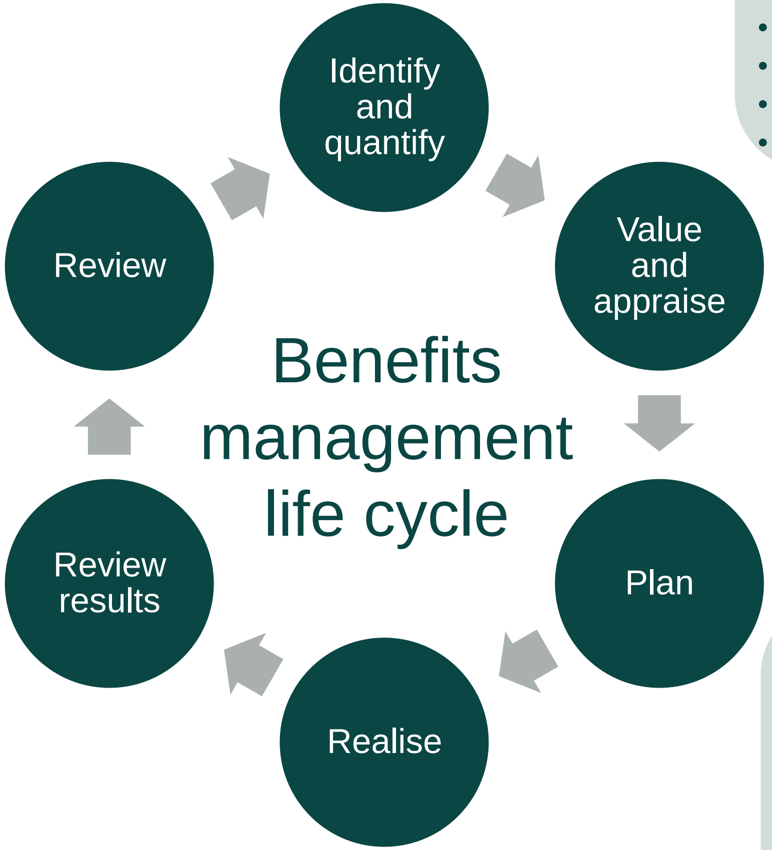
Figure 'The Benefit Realisation Cycle' based upon APMG International's Managing Benefits by Stephen Jenner (TSO, 2012). Reproduced with permission of APMG International. All rights reserved; ©APMG International, 2012

Benefits management life cycle: tools

Post-investment report
An initiative level report that details the benefits realisation performance:
- benefit realised - actual versus forecast
- effectiveness of selected measures
- business case review
- baseline management
- lessons learned



Benefits management strategy
An organisational level document that sets out the policy, approach, methods and processes for benefits management. It can include benefit:
- identification schema
- type definitions
- categories
- mapping techniques
- responsibilities
- definitions
- processes

Benefits profiles
Each benefit and disbenefit is characterised in a benefits profile. For each, the data includes:
- unique identifier
- title
- description
- owner
- realisation timeline
- measures
- risk

Benefits dashboard
A dashboard established for reporting benefit realisation performance. It can include:
- unique identifier
- benefit title
- risk RAG status
- forecast value
- actual value of benefits realised
- enabling change KPIs

Benefits realisation plan
A document that details the benefits and the arrangements made to evidence their realisation. It can include:
- benefits realisation schedule
- resource management
- benefit risk management
- reporting
- the enabling changes on which the benefits depend
- the benefits measures along with their target and forecast values

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Figure 'The Benefit Realisation Cycle' based upon APMG International's Managing Benefits by Stephen Jenner (TSO, 2012). Reproduced with permission of APMG International. All rights reserved; ©APMG International, 2012

9b) Understand the importance of aligning benefits with strategic objectives and ways in which the benefits of a project can be communicated to stakeholders.

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Align benefits with strategic objectives

Business case

- Benefits should be defined within the business case for each project or programme, detailing how the contribution of benefits will aim to achieve the operational, organisational or business strategy.
- The accountability of benefits realisation sits with the sponsor but will not be known until after transition into to the ‘business-as-usual’ phase.

Benefits mapping

- Benefits should be strategically aligned to investment decisions taken.
- All projects are designed to bring benefits to an investing organisation.
- Benefits have a tangible value usually expressed in monetary terms to justify the investment.

Benefits identification and measurement

- Benefits management involves identifying and agreeing the benefits and how they will be measured, monitored and managed over the course of the project and into ‘business-as-usual’ phase.
- This helps to assess and track the impact of programme or project benefits on business performance across the organisation.
- End benefits should equate to achieving a given strategic objective; intermediate benefits are steps towards achieving these.

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Communicate the benefits of a project to stakeholders

Benefits are quantifiable and measurable improvements resulting from the completion of deliverables that are accepted, utilised and perceived as positive by stakeholders.

End benefits will be known following transition into use. Therefore, these need to be communicated to stakeholders by the project sponsor.

It is important that benefits and ways in which these will be measured, monitored and managed throughout the project are agreed with stakeholders early on.

Stakeholders should be consulted to understand how benefits should be attributed within the investment appraisal and business case, and how they should be mapped to organisational strategic drivers.

Reporting on the achievement of intermediate benefits will demonstrate progress towards the end benefits.

Benefits reviews can be held some time following transition to demonstrate to stakeholders how the benefits identified within the business case have been achieved.

Audits facilitate an independent review of how the project or programme has achieved the intended benefits. This independent assessment of compliance can demonstrate how well the benefits achieved align with the original business case objectives.

Photo credit: Israel Palacio

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 4.1.1 (Success and benefits)

APM ‘Benefits management life cycle’ apm.org.uk/resources/find-a-resource/benefits-management-lifecycle/

APM (2019) ’Benefits management – an opportunity not just a process’ apm.org.uk/blog/benefits-management-an-opportunity-not-just-a-process/

APM (2019) ‘What is benefits management?’ https://youtu.be/4qfaNFYtKa8

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Sample Exam Questions

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10 minutes

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Stakeholder engagement and communication management

Learning objective:

10. Understand stakeholder engagement and communication management as the ability to work with people internally and externally to achieve intended outcomes.

10a) Understand the relationship between stakeholder analysis, influence and engagement.

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Stakeholder analysis

- Enables project managers to identify a project’s key stakeholders, their interests in the project and the ways in which those interests affect project riskiness and viability.
- Supports stakeholder influence and engagement by identifying the views of stakeholders towards the project.
- Provides an understanding of who the project stakeholders are, their relationship to the project and their objectives which may impact the project.
- Enables the development of an appropriate engagement strategy appropriate for each stakeholder group.

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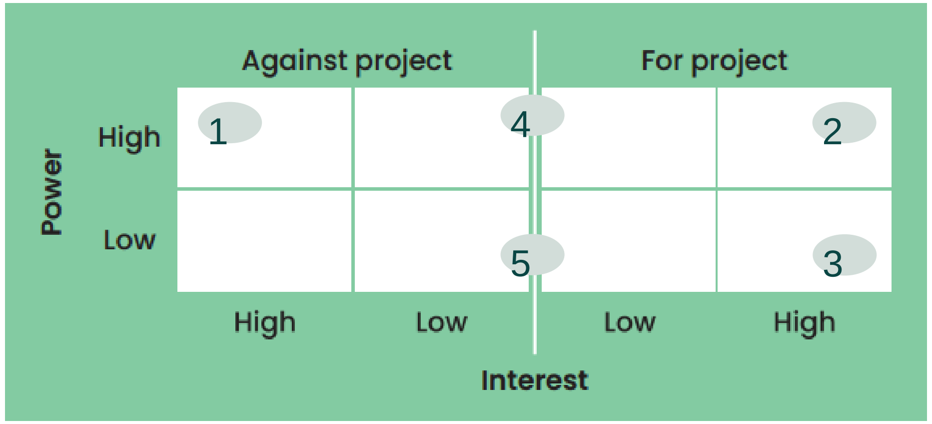
Stakeholder analysis and communications

A stakeholder identified as:

- **High power, high interest and negative attitude** to the project may need intensive engagement from the project manager, and possibly the project sponsor or the senior management team, to avoid any risk they might pose.
- **High power, high interest and positive attitude** to the project may need regular engagement to maintain their support and commitment to the project so that they can continue to champion the project and support the removal of any obstacles or risks.
- **Low power, high interest and positive attitude** to the project may need regular engagement with sufficient information to keep them on side. Miscommunication or lack of communication can cause concerns within stakeholder groups which could lead to negativity towards the project, which could then cause challenges.
- **High power, low interest and a neutral attitude** to the project will likely need to have regular engagement of specific information, common with regulatory bodies such as the Health and Safety Executive, where providing specific information is a regulatory requirement and keeps the stakeholder group neutral towards the project.
- **Low power, low interest and a neutral attitude** to the project are likely to need little engagement. Provision of some information will be needed but this can be just a newsletter or similar. This will free up resources to focus on engagement with others with more power and interest.



Source: *APM Body of Knowledge*, 7th edition.

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10b) Understand the relationship between stakeholder analysis and an effective communication management plan.

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Stakeholder interests and their potential impact

- Stakeholders will have conflicting interests. Therefore, it is impossible to please everyone, but the aim should be to ensure the highest possible satisfaction among stakeholders is achieved.
- This can be done by assessing stakeholders' level of power or influence on the project and by understanding their level of interest in the project. A more in-depth assessment could also understand their attitude towards the project.
- By understanding these two / three parameters, the project manager and team can then decide how they should engage with the stakeholder and document this within a communication management plan.
- The frequency of communications should be tailored to the stakeholder group, as well as the type of media used, and the level of information and detail provided.

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Stakeholder communication plan and engagement strategy

The stakeholder communication plan and engagement strategy should address:

- The particular message(s) to be communicated.
- The person best placed in the project organisation to carry out the communication.
- The most appropriate form of message, media or method that is likely to motivate / engage a stakeholder the most.
- The frequency of communications.
- The level of feedback that should be solicited or expected.
- The barriers that can be proactively identified and acted upon prior to communication taking place.
- The stakeholders which should / should not communicate with each other.

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10c) Understand the benefits to a project of a communication plan.

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Benefits of a communication plan

- Enhances stakeholder engagement.
- Identifies target audience.
- Tailors communications according to individual preferences, and power/interest/attitude.
- Ensures relevant information is provided to stakeholders.

- Records the best method of communication for stakeholders, e.g. use of appropriate technology, opportunities for in person meetings, etc.
- Avoids mis-understandings / conflict.

- Ensures a consistent approach to communication.
- Assigns responsibilities for communications.
- Provides a structured and systematic approach to communications.
- Ensures project team understands requirements of communication.

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10d) Understand the importance of managing stakeholder expectations to the success of the project.

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Managing stakeholder expectations

Carrying out an in-depth analysis of stakeholders consumes resources. To justify this consumption, the project manager must acknowledge the importance of managing stakeholder expectations and how that will influence overall success of the project, particularly in the following areas.

Enabling more effective risk management

- Identifying negative stakeholders as risk.
- Identifying stakeholders who can support risk identification.
- Establishing risk mitigation actions.

Improved communication planning

- Defining necessary and relevant communications.
- Understanding of stakeholder key interests and requirements:
  - time
  - cost
  - quality
  - scope
  - benefits

Enabling effective engagement actions

- Ensuring a productive team is formed.
- Using and sustaining positive interest.
- Minimising and reducing negative interest.
- Building relationships, partnering with stakeholders.

Increased likelihood of project being accepted

- Matching the agreed scope with the outputs to ensure stakeholder acceptance and satisfaction.
- Fulfilling stakeholder needs by ensuring these have been fully identified and understood as early as possible.
- Reducing risk of mis-understanding through regular reporting and communicating of the intended project outputs.

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10e) Know the range of communication methods available and understand the importance of tailoring messaging to meet stakeholder requirements.

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Communication methods and tailoring

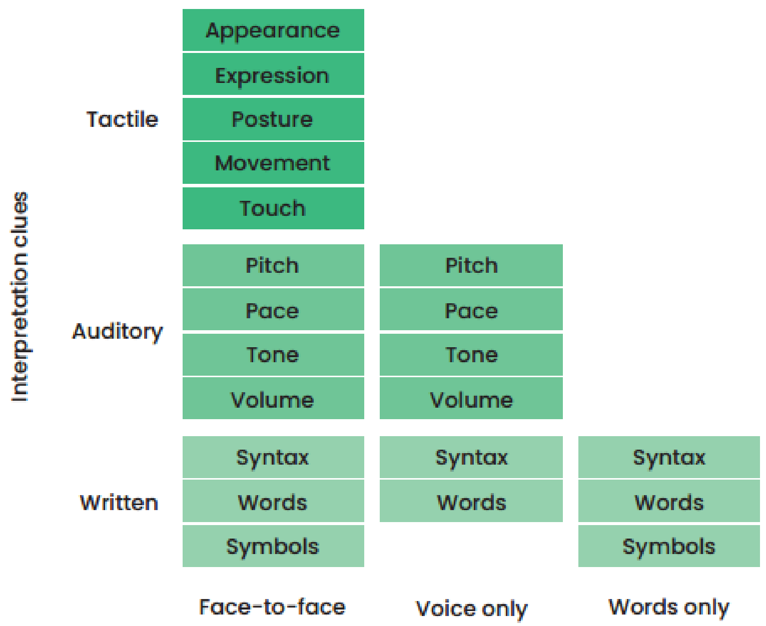
Communication methods

- Written using words / symbols.
- Electronic formats: be careful not to rely overly on one method, such as emails; the medium chosen must be appropriate.
- Verbal and face-to-face: consider the impact of body language (non-verbal signals).

The importance of tailoring messaging

Effective communication needs to consider not only the message, but the medium used for communicating:

- It is important to understand who the target audience is, the intended impact and the risks / potential unintended impact of the approach.
- Cultural influences and language used should be considered.
- Organisations will have communication norms which should be considered.
- Communications should support timely governance procedures and approvals.



Source: *APM Body of Knowledge*, 7th edition.

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10f) Know the factors which can positively or negatively affect communication.

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Factors that affect communication

In groups identify 3 factors which could have a NEGATIVE impact on communication

State how you would resolve each of these?

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Factors that affect communication

Physical location / time zones	• Project team members can be in different offices and locations around the world. The project manager can put systems in place to overcome the issues related to distance, such as virtual meetings.
Cultural diversity and language	• The impact of people’s background culture and the organisational communication culture needs to be planned for. Project team members may not speak the same language or have the same cultural references, for example.
Skill and technical capabilities	• Having a variety of skills and capabilities is inevitable. Support mechanisms should be in place to assist those with less experience and knowledge. For example, technical jargon used on a project could be accompanied by information such as technical summaries and glossaries.
Physical environment	• A suitable, safe and comfortable environment with necessary facilities will enhance the team’s experience. Distractions and noise can lead to a loss of focus and messages not being received correctly.
Body language	• Non-verbal signals can affect communication and sometimes can have a greater impact than the words themselves. It is important to be able to manage non-verbal signals so that these align with the message.
Personal circumstances and personality traits	• A person’s interest and attitude, which may be related to their levels of workload or stress, can affect communication. People’s preferences in terms of communication also come into play – whether they prefer written, virtual or in person communications, or through a detailed or a more strategic level approach.

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 3.1.1 (Understanding who needs to be engaged and influenced)

Elizabeth Harrin (2020) *Engaging Stakeholders on Projects: How to harness people power.* apm.org.uk/book-shop/engaging-stakeholders-on-projects-how-to-harness-people-power/

APM ‘What is stakeholder engagement?’ youtube.com/watch?v=ZzqvF9uJ1hA

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Sample Exam Questions

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Breakout group to answer 10.7

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Conflict resolution

Learning objective:

11. Understand conflict resolution as the ability to identify and address differences between individuals and/or interest groups.

11a) Know the sources of conflict within a project.

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Sources of conflict within a project

- There are many sources of conflict between stakeholders that can occur at various points in the project life cycle.
- Conflict is when two or more parties have a differing opinion or attitude towards something.
- Conflict can arise in a project when these differing opinions and/or opposing attitudes between stakeholders are not easily reconciled.

Concept phase

Securing funding, business case approval or ongoing support for the project.

Balancing stakeholder influences and agreeing high-level scope for the project.

Definition phase

Alignment of scope to project budget and priorities.

Agreeing resource plan and securing resources for the project.

Establishing the various components of the project management plan.

Deployment phase

Maintaining resource availability and commitment.

Deadlines not being met, or progress not as expected.

Agreed plan, processes or methods not being followed.

Transition phase

Acceptance criteria not met.

End-user expectations not achieved.

Adoption phase

Outputs not adopted as intended by business as usual.

Benefits realisation phase

Agreement of measurement and realisation.

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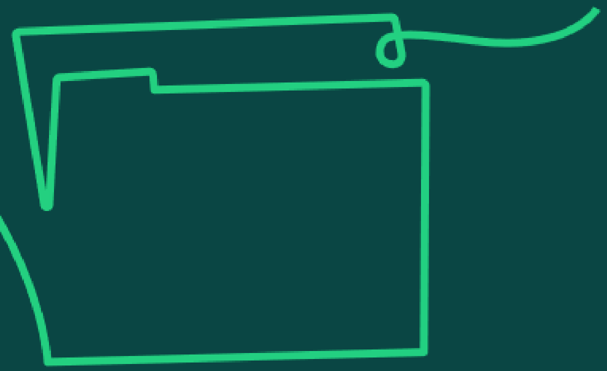

11b) Understand that conflict can have both positive and negative impacts within a project.

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Conflict: positive and negative

Conflict resolution is the process of identifying and addressing differences that, if left unresolved, could affect objectives.

Conflict should be seen as an opportunity to add value as it:

- Creates open conversations to debate and discuss points.
- Shows feelings and agendas towards the point in question.
- Enables direction to be sought.

Ignoring the conflict and the people involved is not an acceptable or productive way of safeguarding the success of the project.

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11c) Know ways in which conflict can be addressed in different situations (such as Thomas Kilmann Conflict Mode Instrument).

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Thomas Kilmann Conflict Mode Instrument

- The Thomas Kilmann model encourages people to think about conflict using two dimensions:
  - The desire to achieve one's own objectives.
  - The desire to help others achieve their objectives.
- Sometimes other parties are required to resolve a conflict to prevent the conflict escalating.
- Clear protocols within a project are needed for when conflicts must be escalated for a resolution or required decision for project delivery and governance.

High

Desire to meet personal need

Compete

Quick solution or decision is necessary.

Emergencies occur or unpopular decisions need to be made.

Issues with a right and wrong solution.

Collaborate

Time required to explore concerns and find a consensus.

Focused on developing strong relationships.

Win-win approach through sharing of ideas and information to consider insights from different perspectives.

Compromise

When goals are important, but solutions are required under time pressure.

Through negotiation, parties find a mutually agreeable solution.

No one wins or loses; concessions are made to reach a middle ground.

Issue is small or more important issues are pressing.

Potential disruption outweighs the benefit of resolving.

Want to avoid an overt disagreement.

Avoid

Relationship is more important than the situation.

Maintain stability and cooperation.

To show reasonableness and to learn and let others learn through trying.

Accommodate

Low — High

Desire to meet the needs of others

Figure adapted/modified from the TKI Conflict Model, copyright © Kilmann Diagnostics LLC, 2009-2023. All rights reserved. Original figure is available at: [http://www.kilmanndiagnostics.com/overview-thomas-kilmann-conflict-mode-instrument-tki;](http://www.kilmanndiagnostics.com/overview-thomas-kilmann-conflict-mode-instrument-tki) Reproduced with permission. © Kilmann Diagnostics LLC, 2009.

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When to Use Each Mode

Mode	When to use
Competing	When quick, decisive action is vital. Important issues when you know you’re right. To protect yourself against people who take advantage of non-competitive behaviour.
Accommodating	When you know that you’re wrong. When the issue is much more important to the other person. To build up social credits for later issues.
Avoiding	When an issue is trivial. Temporarily used while gathering information. When the potential damage of confronting a conflict outweighs the benefits of its resolution. When others can resolve the conflict more effectively.
Collaborating	To gain commitment by incorporating other’s concerns into a consensual decision, To work through hard feelings which have been interfering with an interpersonal relationship.
Compromising	When goals are moderately important, but not worth the effort or potential disruption of more assertive modes. To achieve temporary settlements to complex issues.

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Resolving conflict

Some important considerations for conflict resolution include:

Anticipate different people being involved with different perceptions, agendas and outcomes to be achieved.

Acknowledge very rarely does it just go away.

Resolve and avoid placing blame. Work through solutions that will support achieving the project's aims.

Encourage the sharing of perspectives. Open and honest dialogue will enable better solutions.

Facilitate collaboration to explore potential resolutions.

Neutral – Keep things neutral, using neutral tones, language and body language to avoid defensiveness.

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 3.1.5 (Conflict resolution)

APM Learning module, Managing and resolving conflicts apm.org.uk/resources/apm-learning/

APM ‘A project professional’s guide to conflict management and resolution’ apm.org.uk/v2/media/1yxg4klg/a-project-professional-s-guide-to-conflict-management-and-resolution.pdf

APM (2022) ‘Why conflict might not (always) be a bad thing for your project team’ apm.org.uk/blog/why-conflict-might-not-always-be-a-bad-thing-for-your-project-team/

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Sample Exam Questions

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Leadership

Learning objective:

12. Understand leadership as ways to empower and inspire others to deliver successful projects.

12a) Understand how leadership impacts on team performance and motivation (using models such as Maslow, Herzberg and McGregor).

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Maslow, Herzberg and McGregor

Maslow's hierarchy of needs

- A leader can support team performance by identifying which level of need the team member is seeking to address.
- Each need must be satisfied in turn and once satisfied then ceases to continue to motivate.
- Motivation will derive from individuals looking to move to the level above.

Herzberg's two-factor theory

- Leaders should address 'hygiene factors' for individuals first to remove potential job dissatisfaction and then focus on 'motivators'.
- Hygiene factors to consider are working conditions such as pay and policies.
- Motivators should increase job satisfaction by providing opportunities for tangible results and recognising individuals when they achieve their goals.

McGregor's theory X & Y

- This theory considers that effective leadership comes from adapting the leadership style to the individuals in the team.
- The leader determines whether team members are:

  1. Theory X workers, where they dislike work and are naturally unmotivated to work.
  2. Theory Y workers, who are self-motivated, creative, enjoy work and take pride in what they do.

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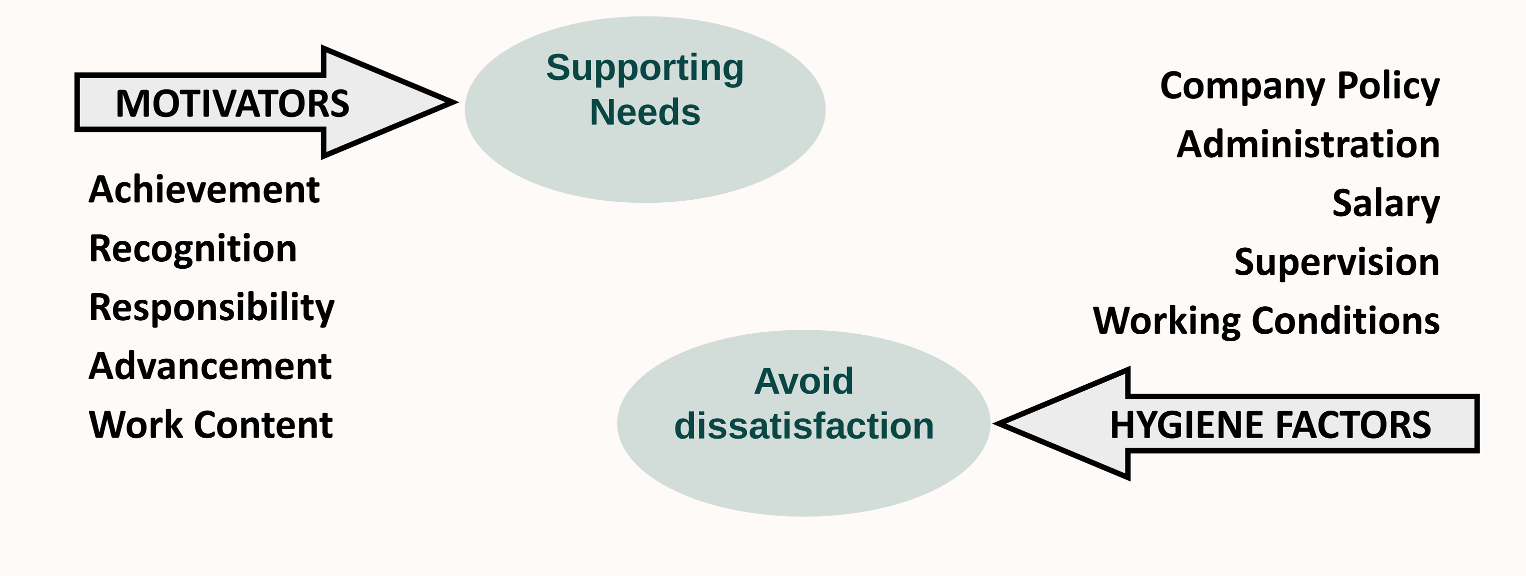
Maslow's Hierarchy of Needs

Project Hierarchy of Needs

Herzberg's Hygiene Theory

- Motivating Factors lead to job satisfaction, supporting people's needs.
- The absence of Hygiene Factors leads to job dissatisfaction, frustrating people's needs.



Mc Gregor’s Theory X and Theory Y

Theory X

The Theory X manager:

- Management style **assumes** that the typical worker has little ambition, avoids responsibility and is individual-goal oriented, and interacts with them under this belief

- Manager believes in the importance of heightened supervision, financial reward as the only real motivation, and the use of penalties.

- May be appropriate IF people are of this type, but if not they run the risk of demotivating workers who are confident, work well together and enjoy the work

Mc Gregor’s Theory X and Theory Y

Theory Y

The Theory Y Manager:

- Management style gravitates towards relating to the worker on a more personal level and developing a warmer relationship with their team. Thus creating a healthier atmosphere in the workplace

- Incorporates a type of democratic environment to the workforce ,encompasses creativity and discussion. Manager focusses on an empowering role and encourages workers to approach tasks without direct supervision.

- Works well with workers of a similar type, but workers that conform more to Theory X, will not be productive and take advantage of a Theory Y Manager

12b) Understand why it may be necessary to change leadership styles depending upon the situation.

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Different Leadership Styles

Leaders need to adapt their style and approach to the needs of the team and the work that needs to be accomplished.

Autocratic	• Strong one-dimensional leadership style, where one leader has full authority. • Appropriate for addressing an issue that threatens the achievement of objectives, where issues become critical or where team members are new and lacking motivation.
Participative / Democratic	• Leader involves team members in decision-making process. • Suitable when there is time to focus on development of the team, to build team relationships and collaboration is valued. Some decision making will be passed on to suitably skilled team members.
Delegative / Laissez-faire	• Self-rule, where employees are empowered to make decisions. • Suitable when the team is established and working well, and has the required knowledge to work independently without direction.

Figure 'Different leadership styles' by Kurt Lewin, Ronald Lippitt and Ralph. K White from "Patterns of Aggressive Behavior in Experimentally Created "Social Climates"", Journal of Social Psychology Vol.10: 271–301, 1939. Reproduced with permission of Informa UK Limited, trading as Taylor & Francis Group, www.tandfonline.com

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12c) Understand the importance of a coaching and mentoring style in leadership, and the role of emotional intelligence.

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Coaching and mentoring style

Understanding individual strengths and weaknesses

- Adapting leadership style to an individual’s needs.
- Providing a greater amount of direction for new team members with an appetite for learning and advancing within the project.
- Coaching when team members become more embedded and experienced.
- Engaging and delegating responsibilities where skill is fully established, and motivation is high.

Understanding the situation

- Where there are critical, urgent or changing situations, a directive style might be more appropriate.
- Where change is needed, guidance from the leader might be needed.
- When the change is embedded, the leader can then move to a supportive, coaching and engaging style.

Dealing with uncertainty

- The team may need the leader to create a supportive environment through coaching and engaging in order to feel safe when working.
- By being supportive and less directive, the leader demonstrates that they trust the team to cope with this project circumstance. This will make the team members feel valued and empowered.

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Leadership and emotional intelligence

Emotional intelligence is critical to successful leadership. It is the ability to understand and manage one's own emotions and those of the people around you. Leaders with high levels of emotional intelligence will understand their own emotions and know how these can impact their teams. There are five important elements within emotional intelligence:

Self-awareness	• Knowing how you feel and how your emotions and actions can impact on the people around you. This means understanding your strengths and weaknesses.
Self-regulation	• Avoiding making rushed or emotional decisions, remaining calm and staying in control. • Understanding your own values and not being willing to compromise on those.
Motivation	• Self-motivated leaders consistently work towards their goals and have extremely high standards for the quality of their work.
Empathy	• Empathetic leaders can put themselves in someone else's position. They develop the team, challenge those who act unfairly, give constructive feedback and listen to those who need it.
Social skills	• Leaders with strong social skills are great communicators, open to hearing good and bad news, and experts at getting their team to support them and be excited about a new project. • They are good at managing change and resolving conflicts diplomatically; they set an example with their own behaviour.

Figure 'Intra & Interpersonal intelligence' adapted from Emotional Intelligence: Why it can matter more than IQ by Daniel Goleman, Bloomsbury, 1996, copyright © Daniel Goleman, 1995. Reproduced with permission from Bloomsbury Publishing Plc; and Bantam Books, an imprint of Random House, a division of Penguin Random House LLC. All rights reserved

Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 3.2.4 (Leadership)

APM (2021) 'Cultivate different leadership styles to create collaborative teams' apm.org.uk/blog/cultivate-different-leadership-styles-to-create-collaborative-teams/

APM (2018) 'How leadership style affects team ownership' apm.org.uk/blog/how-leadership-style-affects-team-ownership/

APM (2023) 'How does emotional intelligence contribute to project management success?' apm.org.uk/blog/how-does-emotional-intelligence-contribute-to-project-management-success/

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Sample Exam Questions

Learner Study Guide

Sample Exam Questions

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Team management

Learning objective:

13. Understand team management as the ability to work with team members to create and sustain teams.

13a) Know the different stages in the development of a team, including creation, development, maintenance and leadership, and understand the factors which influence these stages and knowledge of the models that are used to understand team development (such as Belbin, Margerison McCann, Myers-Briggs, Hackman, Tuckman, Katzenbach and Smith).

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Models of team development

Belbin

- Model shows a number of roles that are optimal for an effective team.
- Individuals tend to have roles that they are most comfortable with.
- If a leader understands how each individual in the team tends to behave and contribute, and where their strengths lie, they can develop the team by utilising these strengths and behaviours.

Margerison McCann

- Model similar to Belbin and works on understanding an individual's preferred role within a team.
- Explores how individuals relate to others, how they gather and use information, how they make decisions and how they organise themselves and others.
- If the leader understands these aspects of each team member, then they can select the correct mix of complementary skills to achieve a high performing team.

The Team Management Profile | Team Management Systems

Myers-Briggs

- This model recognises that how individuals perceive a situation, along with their communication preferences, will influence how they react to the situation.
- The leader can use this to influence perception or manage their reaction.
- Allows project team members to be integrated more easily into the project environment, aligning to values and therefore improves performance.

MBTI | The Myers-Briggs Company

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Models of team development (continued)

Hackman

This model suggests that five conditions need to be put in place and maintained by the leader for the team to be effective:

1. Team must be a real team, rather than a team in name only.
2. Direction of work must be clear.
3. Structure must support teamworking.
4. Organisation must be supportive.
5. Coaching should be available within the team.

Hackman model of team effectiveness, explained | Five factors for great teamwork — BiteSize Learning

Tuckman

- This model recognises how a project team may take time to mature into a performing team.
- The leader should facilitate progression through the following stages of team development:

1. Forming
2. Storming
3. Norming
4. Performing

Katzenbach and Smith

- This model recognises that the more effective the group are as a team, the better the performance.
- The leader can support this by developing the level of commitment, accountability and the skills within the team.
- Actions should be in place to ensure team members are clear on specific goals, that the right mix of skills is in place and accountabilities are clear.

Katzenbach and Smith Model of Team Effectiveness

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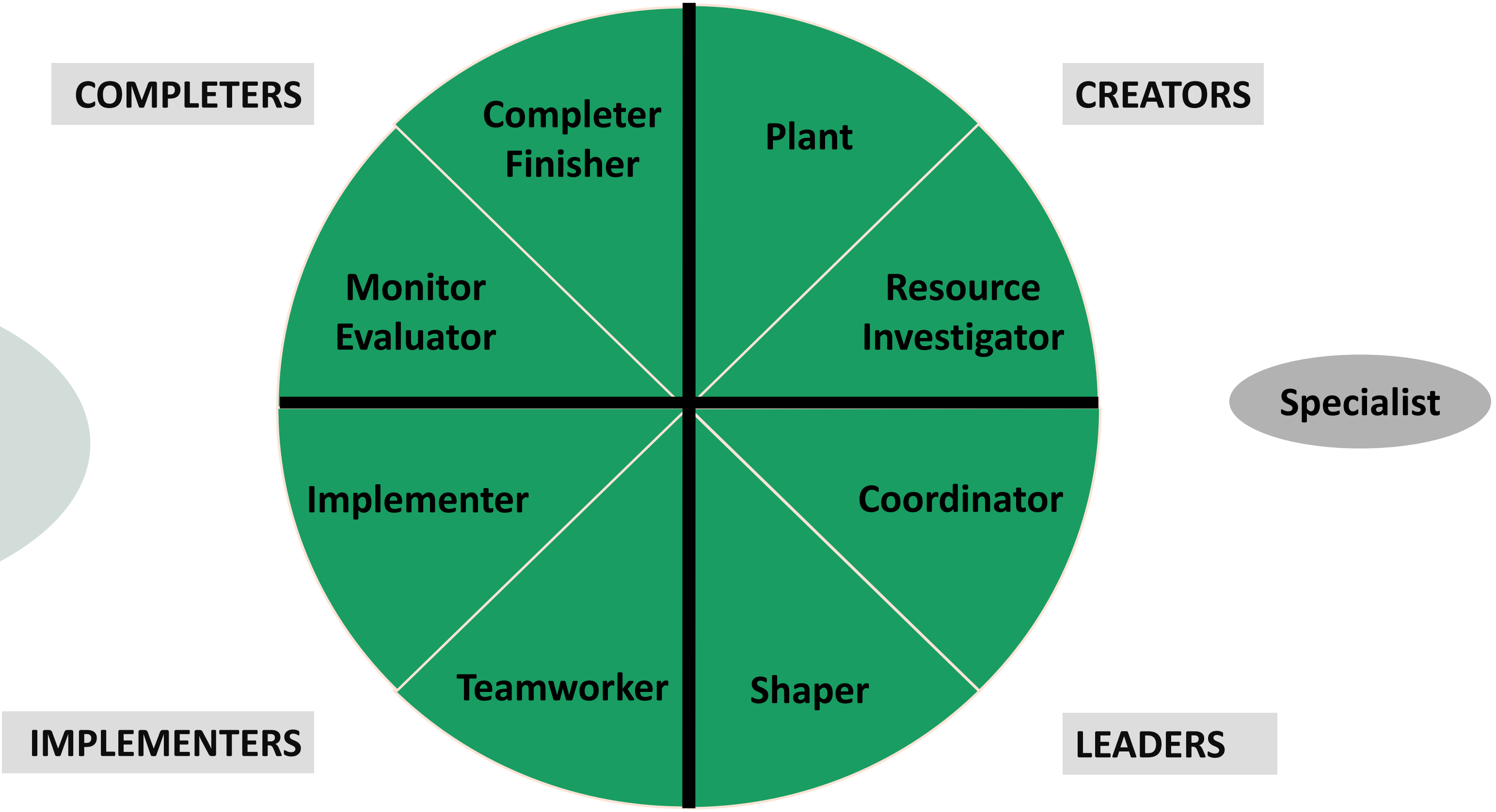
Models of team development (continued)

Belbin's Team Roles

Look at the descriptions on the next two slides, which Belbin role(s) sum you up best?



Models of team development (continued)

Belbin’s Team Roles

Role	Characteristics	Strengths	Allowable weakness
Plant.	Individualistic, serious-minded, unorthodox.	Genius, imagination, intellect, knowledge.	Inclined to disregard practical details or protocol.
Resource-Investigator.	Extroverted, enthusiastic, curious, communicative.	Capacity for contacting people. Responds to challenges.	Liable to lose interest quickly. Over optimistic.
Co-coordinator.	Calm, self-confident, controlled.	Promotes decision-making, clarifies goals, strong sense of objectives.	Can be seen as manipulative. Delegates personal work.
Shaper.	Highly strung, outgoing, dynamic.	Thrives on pressure, drive, challenging.	Prone to provocation, and impatience. Tendency to offend others.
Monitor-Evaluator.	Sober, unemotional, prudent.	Judgment, discretion, hard-headedness.	Lacks inspiration, lacks ability to motivate others.

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Models of team development (continued)

Belbin's Team Roles

Role	Characteristics	Strengths	Allowable weakness
Team Worker.	Socially orientated, perceptive, diplomatic.	Promotes team spirit, listens and averts friction.	Indecisive in crisis, avoids pressure situations.
Implementer.	Conservative, dutiful, predictable.	Organising, common sense, self-disciplined.	Lack of flexibility, resistance to unproven ideas.
Completer-Finisher.	Painstaking, orderly, conscientious, anxious.	Will follow through, perfectionist, delivers on time.	Reluctance to 'let go' or delegate, tendency to worry unduly.
Specialist.	Professional, self-starting, dedicated.	Provides knowledge or technical skills in rare supply.	Contributes only on a narrow front, dwells on technicalities, overlooks the 'big picture'.

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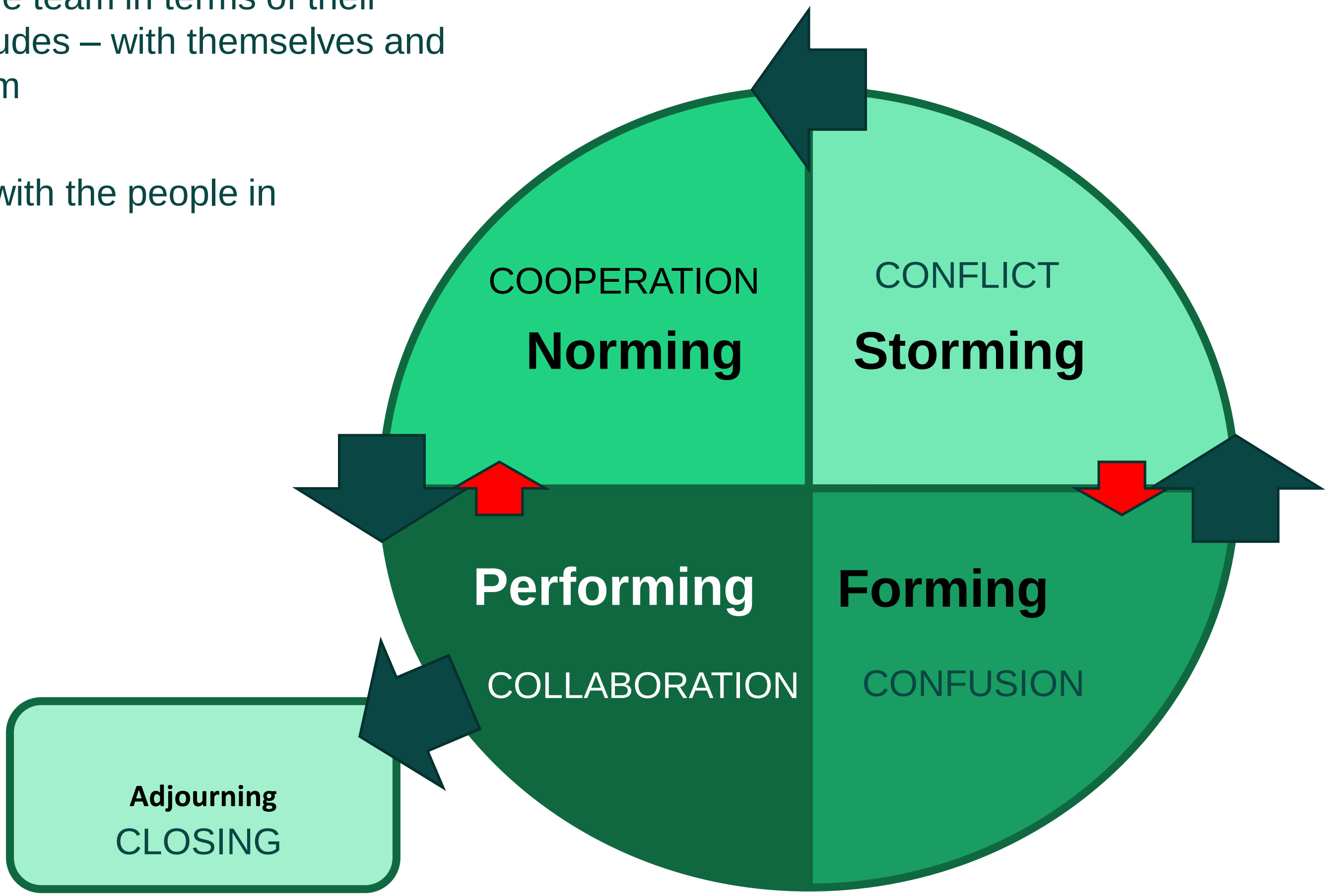
Models of team development (continued)

Tuckman's Stages of Group Development:

Focus 1 – looks at the people in the team in terms of their behaviours, chararacteristics, attitudes – with themselves and with the other members of the team

Focus 2 - How the leaders deals with the people in the stage and from stage to stage



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Models of team development (continued)

Tuckman's Stages of Group Development:

Stage	Behaviours
Forming.	Team members are positive and polite. Some are anxious, not fully understanding what work the team will do. team members' roles and responsibilities are not clear. Others are simply excited. The leader plays a dominant role at this stage
Storming.	This is the stage where many teams fail. Starts where there is conflict between team members' natural working styles. Storming can also happen in other situations where team members may challenge the Project Managers authority.
Norming.	Team starts to resolve their differences, appreciate colleagues' strengths, and respect the authority of the Project Manager as their leader.
Performing.	The performing stage - hard work leads, without friction, to the achievement of the team's goal.
Adjourning.	Project teams exist for only a fixed period, and even permanent teams may be disbanded through organisational restructuring - Some team members may find this stage difficult, particularly if their future now looks uncertain.

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13b) Know the characteristics and benefits of effective teams and teamwork.

Characteristics of effective teams and teamwork

In your group, identify 5 characteristics of effective teams

10 minutes

Characteristics of effective teams and teamwork

- Clear roles and responsibilities should be aligned to skill and experience so that team members know what is expected of them.
- Open, clear and honest communication will build trust and confidence between the leader and the team, as well as providing a platform for innovative discussion to take place.
- A common goal and sense of community led by the project manager will develop the common focus to deliver the outputs to achieve the goal.

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Benefits of effective teams and teamwork

- Improved motivation through a sense of belonging and shared sense of accomplishment.
- Good problem solving through sharing of ideas, knowledge and experience to overcome challenges.
- Increased team member confidence to try new things, to be creative and innovative as a team as there is a sense that the risk is shared.
- Increased productivity through sharing the workload and allocating tasks to match skill and experience.

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13c) Understand why different leadership approaches are needed to support virtual and hybrid teams.

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Leadership and virtual and hybrid teams

How leaders can enable trust to be established and avoid hidden conflict when teams are disparate	How leaders can overcome cultural differences	How leaders can overcome challenges to open and synchronous communication	How leaders can facilitate contribution of all team members
• Coordinate activities to help develop trust. • Hold regular team meetings allowing everyone to contribute. • Create opportunities for team dialogue. • Mixture of formal and informal meetings. • Collaborative online workspaces, such as Microsoft Teams.	• Promote the right behaviours within the team and demonstrate an appreciation of cultural nuances and expectations. • Encourage listening to and the sharing of each other’s perspectives. • Give clear messages and promote respect. • Build a clear project team identity to help establish team norms of behaviour and common values, across cultures.	• Agree norms of communication so that the same message can be understood by all the members of the team. • Acknowledge time zone differences and accommodate adaptations to messages to achieve a level of consistency. • Use technology to overcome communication barriers.	• Ensure team has the right training and tools to use the technology platforms that promote collaboration. • Encourage equal participation in virtual spaces, taking individual needs into account. • Provide opportunities to collaborate outside meetings, share information ahead of time and allow questions via email.

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 3.2.3 (Team development)

APM (2014) ‘Team dimensions’ apm.org.uk/blog/team-dimensions/

APM (2018) ‘How leadership style affects team ownership?’ apm.org.uk/blog/how-leadership-style-affects-team-ownership/

APM (2018) ‘What Makes a High Performing Team?’ apm.org.uk/news/what-makes-a-high-performing-team/

Belbin: The Nine Belbin Team Roles, belbin.com/about/belbin-team-roles

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Sample Exam Questions

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Diversity and inclusion

Learning objective:

14. Diversity and inclusion: Understand diversity and inclusion as the ability to build and maintain an inclusive environment that embraces a diverse culture.

14a) Knowledge of diversity, including characteristics which may cause a person to be treated less favourably, such as: age, disability, marriage and civil partnership, pregnancy and maternity, race, religion or belief, sex, and sexual orientation.

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Diversity

- Diversity leads to higher performance through diverse skill sets and perspectives.
- This is of great value to projects and helps to enable successful delivery of outcomes.
- Project-based working is made up of teams with different backgrounds, experiences and capabilities which should be embraced.
- National cultures are developed over long periods of time and are influenced by different drivers, which in turn influence people’s values and beliefs, and these shape people’s behaviours.
- As part of stakeholder analysis, project professionals can research different cultures to understand typical behaviours.
- Value is created from difference.

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Diversity

- In many countries, it is illegal to discriminate against people based on protected characteristics. It is important that project professionals, if unsure, seek advice on how to treat people fairly, particularly as protected characteristics can vary by country.
- Age,
- Disability,
- Marriage and civil partnership,
- Pregnancy and maternity,
- Race,
- Religion or belief,
- Sex, and sexual orientation,
- Gender.

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14b) Understand why incorporating diversity into all parts of a project, from team members to customers, is a factor in creating a positive working environment and understand the importance of embracing diverse thinking in teams as a means of generating innovative solutions.

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Diversity in project teams

- In project-based working, temporary teams come together and need to perform quickly.
- These teams can be based in the same office, in different countries or in different continents.
- Leadership and communication styles are enabling factors in maximising opportunities to create an open, honest and psychologically safe environment.
- The workplace environment needs to consider different individuals' perspectives, abilities and any reasonable adjustments required to form ways of working that enable a positive working environment and ensure everyone has a fair chance.

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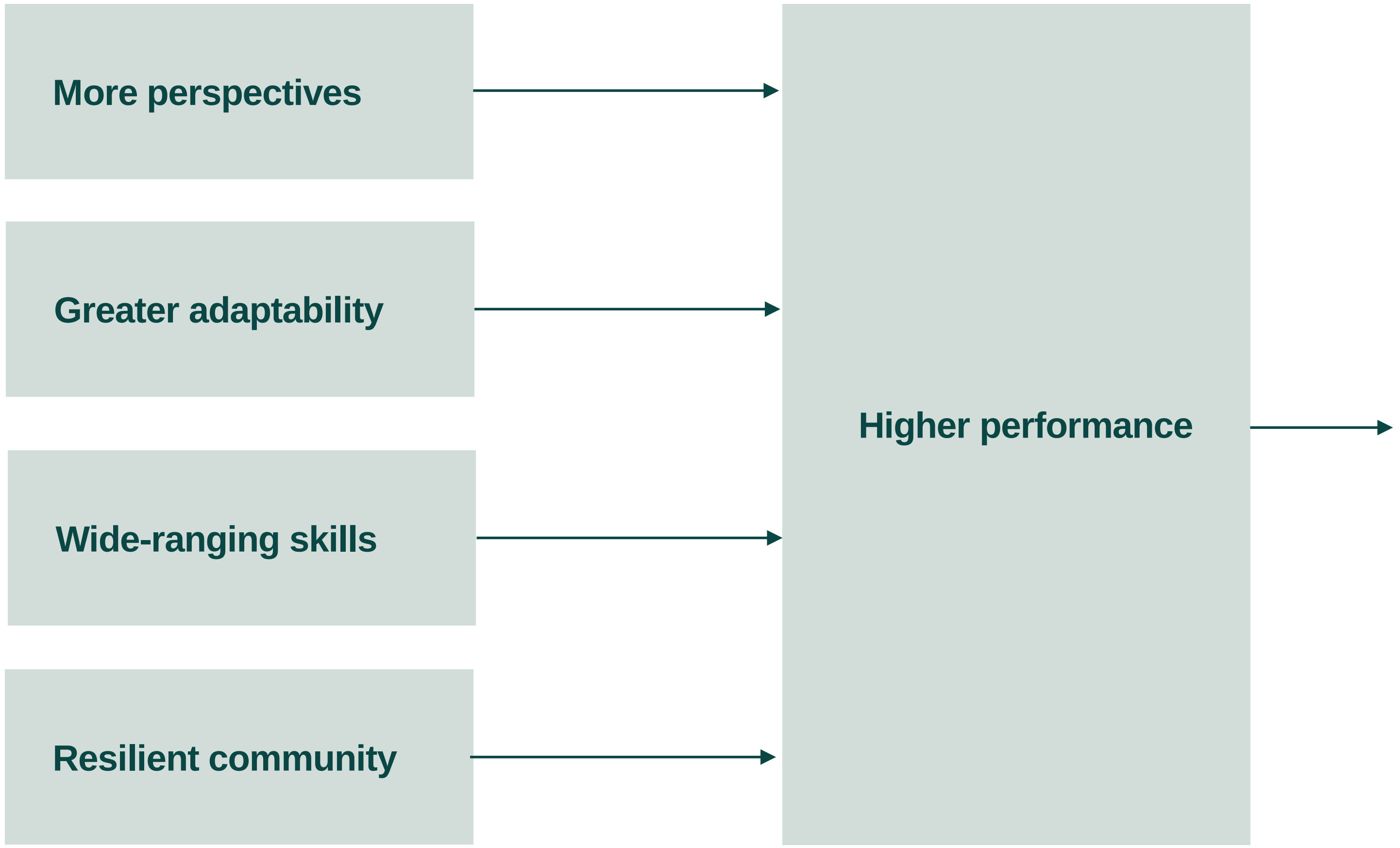
Diversity leads to higher performance

Consider:

- Do people feel they can be themselves at work?
- Are people encouraged to share ideas, and challenge respectfully?
- Do people understand and welcome differences?
- If we can answer yes, to the above, then we will benefit in the following ways:



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14c) Understand how conscious and unconscious bias can affect actions and know how to treat people fairly, including adapting behaviours to support individuals’ needs and facilitate their contributions.

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Conscious and unconscious bias

- Human nature is to favour the norm ‘like us’ rather than ‘different from’ us: this leads to unconscious and conscious bias.
- Conscious and unconscious biases are ingrained preferences or prejudices that influence our perceptions and decision making.
- Conscious bias involves deliberate thoughts and attitudes, while unconscious bias operates involuntarily, often stemming from societal stereotypes.
- These biases can affect our understanding, actions and decisions in an implicit manner.
- A strong sense of inclusion and community is fostered when people with different backgrounds come together for a common purpose. Project professionals can improve outcomes by harnessing the diversity of people involved.

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Improving performance

If individuals feel engaged and valued, there will be an improvement in team performance and solutions.

- Create and sustain a positive, inclusive working environment and address any potential barriers.
- Make reasonable adjustments to ensure that everyone has a fair chance.

- Manage differences to achieve a common purpose.
- Consider the contributions that can be made by individual team members.

- Think about ways to communicate in a meaningful way.
- Remember that multiple dimensions of culture exist.

- Balance priorities and actively accept the need to treat people fairly.
- Explore and respond to signs of bias.

- Lead by example: demonstrate appropriate behaviours and communication.
- Adapt to facilitate contributions from all team members.
- Recognise signs of resistance, either individually or organisationally, to inclusion and diversity.

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 3.2.6 (Diversity and inclusion)

APM Learning module, Creating an environment of inclusion apm.org.uk/resources/apm-learning/

APM ‘What is diversity and inclusion in project management?’ apm.org.uk/resources/what-is-project-management/what-is-diversity-and-inclusion-in-project-management/

APM (2022) ‘Mythbusting diversity and inclusion in project management’ apm.org.uk/blog/mythbusting-diversity-and-inclusion-in-project-management/

Binna Kandola (2009) *The Value of Difference: Eliminating Bias in the Workplace.*

The implicit bias test https://implicit.harvard.edu/implicit/

UK protected characteristics gov.uk/discrimination-your-rights

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Sample Exam Questions

Learner Study Guide

Sample Exam Questions

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14.5 Breakout Groups to provide answers

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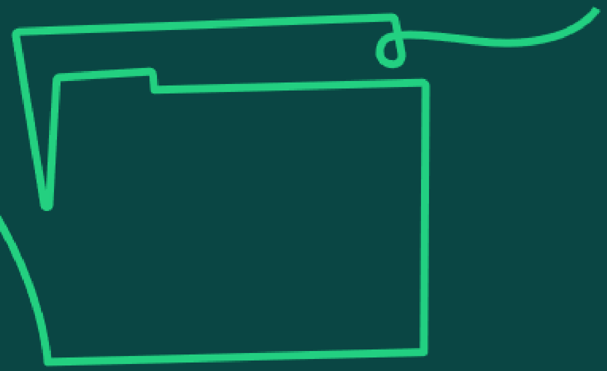
Ethics, compliance and professionalism

Learning objective:

15. Understand ethics, compliance and professionalism as the ability to work consistently in a moral, legal and socially responsible manner.



15a) Understand the importance of continuing professional development, which should cover both knowledge, skills and behaviours, and the individual's role in identifying and addressing their own competence gaps.

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Considerations for project professionals

Consider:

- Taking responsibility for identifying own competence gaps and pursuing learning opportunities to bridge gaps with the support from your employer.
- Different approaches to learning; they can vary from informal on-the-job learning and coaching, to formal courses and qualifications.
- Being a ‘reflective practitioner’, which means you take the time to actively reflect on experiences and engage in the process of continuous learning and development from those experiences.
- Using a wide range of resources and a variety of informal and formal approaches to achieve more effective learning.

Informal approaches that are great for sharing and learning include:

- academic journals
- articles in professional magazines
- conferences
- blogs and podcasts

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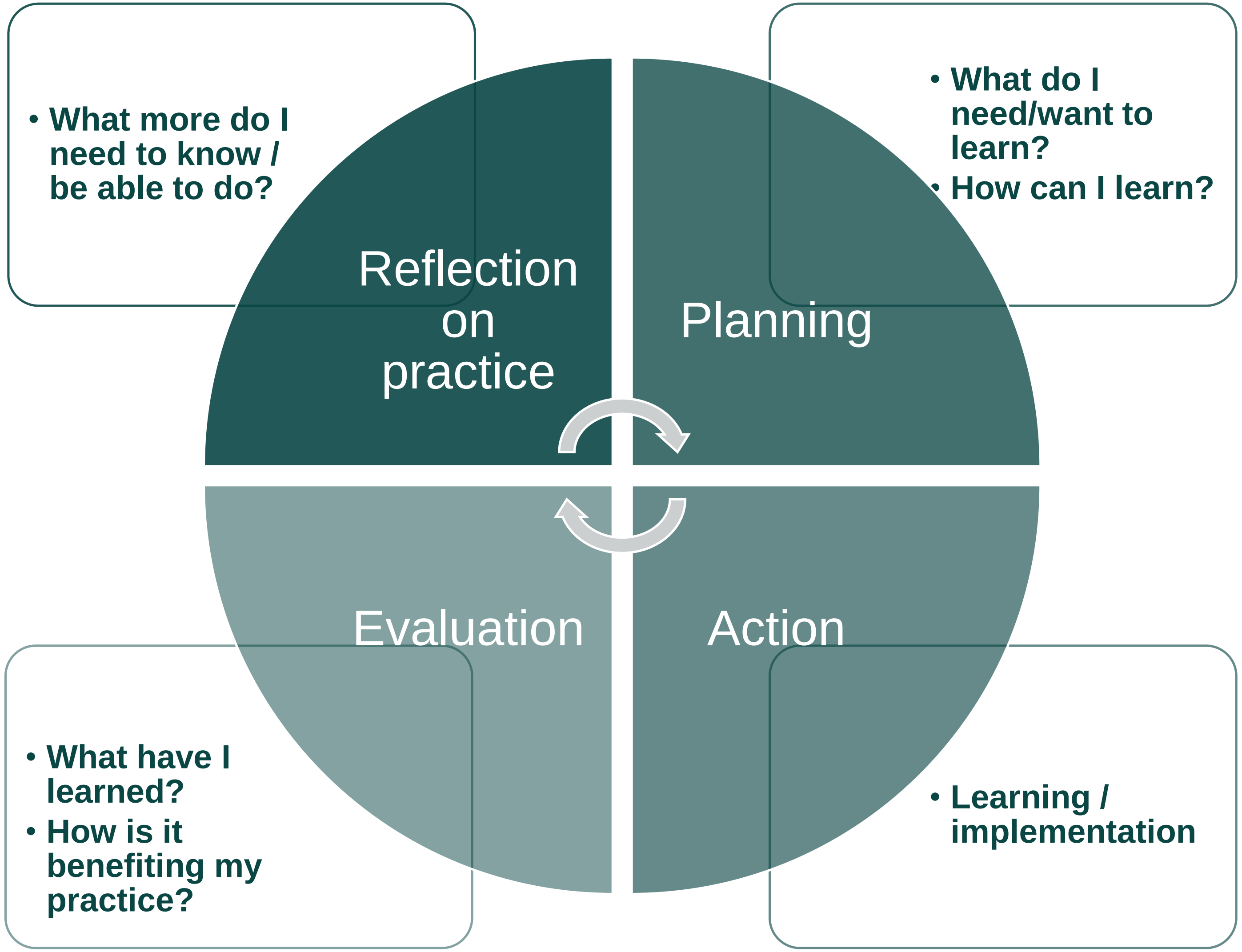
CPD cycle



Source: *APM Body of Knowledge,* 7th edition
Figure 'CPD cycle' adapted from *Continuing Professional Development* by Andrew L. Friedman, Routledge, 2012, figure 1.2, copyright © Andrew L. Friedman, 2012. Reproduced with permission of Routledge via Copyright Clearance Center.

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Importance of CPD

The project profession is navigated by a set of standards that articulates a defined level of technical knowledge, professional practice and ethical behaviour.

Ongoing continuing professional development (CPD) of your own competence enables you to demonstrate capability of those standards and the desire to further develop them promotes and supports a trusted profession.

In order to make sure that you're behaving in a consistent ethical way, it's important to consider:

- The impact ethics has on change activity.
- How to make sure that you're acting within the limits of your own competence and authority.
- Professional codes of conduct.
- When to seek specialist advice to understand legal and contextual obligations.
- The cultures, values and policies that the project is operating within.
- The impact regulation might have on project governance and processes.

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15b) Knowledge of the sources of specialist advice and standards that need to be adhered to.

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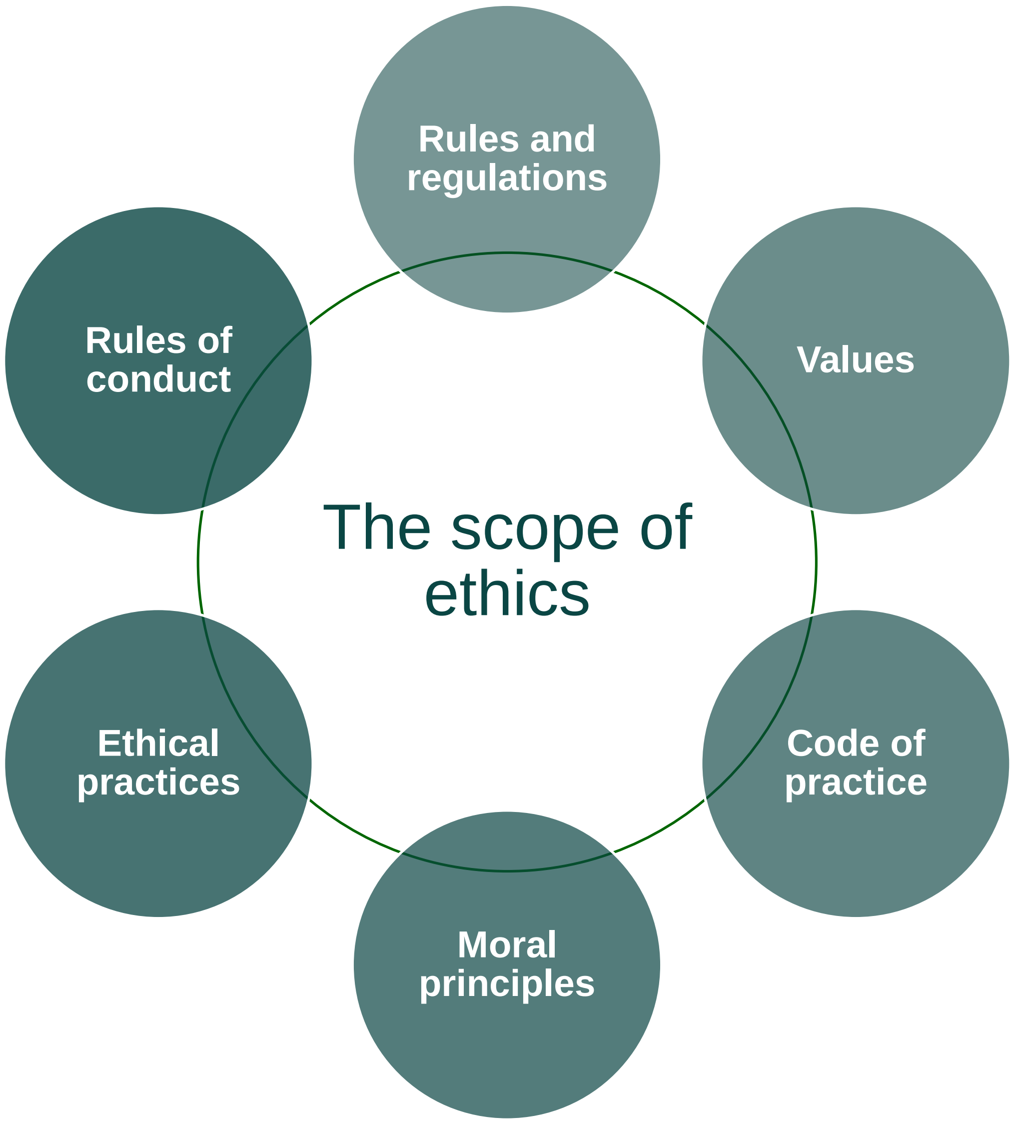
Sources of specialist advice and standards

- Trust and respect are vital to anyone who wants to be regarded as a professional.
- Trust is gained through consistently working in a moral, legal and socially responsible manner in accordance with organisational and professional codes of conduct.
- Professionalism is a demonstratable awareness and application of competences and qualities, including knowledge and appropriate skills.
- The APM FIVE Dimensions of Professionalism provides a framework to set standards and guide project professionals.
- Where standards for ethical conduct are not being met, there is a duty as part of professional standards to report any wrongdoing.
- Everybody has a responsibility to ‘speak up’; organisations are required to have a policy and process to enable individuals to do so.



Source: *APM Body of Knowledge*, 7th edition

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APM FIVE dimensions of professionalism

- **Breadth** — The *APM Body of Knowledge* defines the knowledge needed to manage any kind of project.
- **Depth** — The *APM Competence Framework* maps levels of knowledge and experience to help progress skills and abilities.
- **Achievement** — APM offers recognised qualifications.
- **Commitment** — Continuing professional development (CPD) helps develop project management practice.
- **Accountability** — The APM Code of Professional Conduct outlines ethical practice expected of a professional.

15c) Understand the impact of the legal and regulatory landscape on projects (such as the impact on working conditions, risk management, governance and sustainability).

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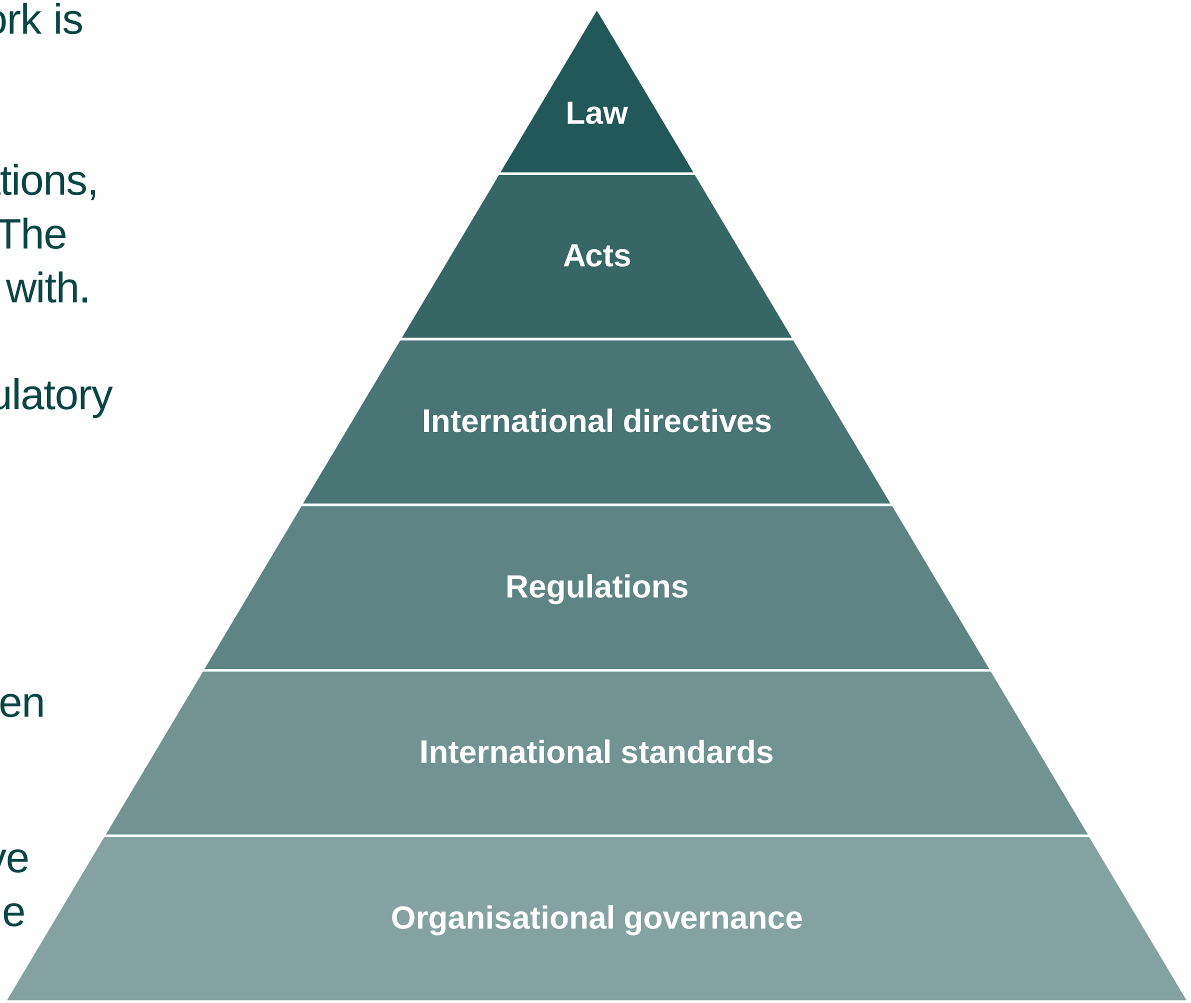
Legal and regulatory landscape

- Projects operate within a legal and regulatory framework. This framework depends on the industry and country in which the work is being delivered.
- Where the work is being delivered in different geographical locations, there may be conflicting regulations and standards to consider. The relevant governance structure decides which standards to align with.
- Project professionals are responsible for understanding the regulatory environment they are operating in; if in doubt, seek advice.
- Regulations generally fall into two categories:
  - Those with specific instructions that must be taken.
  - Those that help employers set procedures where they can then decide how they control risks identified.
- Failure to comply with regulations can result in advised corrective action through to custodial sentences for both individuals and the organisation.



Source: *APM Body of Knowledge*, 7th edition

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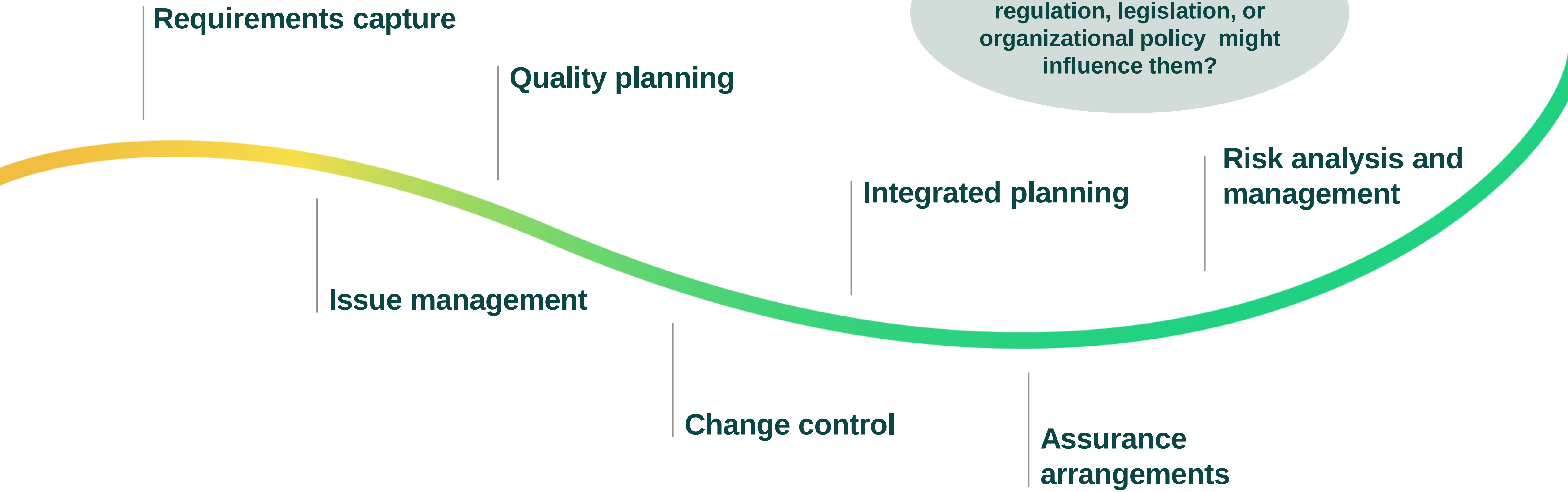
Examples of ‘Laws & Regs’ relevant to Projects

Also, for example – Contract Law. Environmental laws/regs, Specific Industry regulations, also Procurement policies, finance rules etc within organisations

Law	Explanation
Employment Contracts	Written statement setting forth the terms and conditions of the employment.
Minimum wage and working hours	Varies per age group up to 21. Adult employees may not be required to work in excess of 48 hours per week. Employees entitled to 5.6 weeks paid holiday time per year. Part-time workers - pro-rata amount of holiday pay.
Discrimination Laws	Prohibiting discrimination and harassment based upon characteristics such as age, disability, gender, etc.
Maternity and paternity leave	Both sexes entitled to paid or unpaid leave.
Terminations	Covers conditions of termination by mutual agreement or by retirement or redundancy.
Redundancies	Long-term employees entitled to compensation from their employers in the event of redundancy.
Data Protection (GDPR)	Personal data obtained only for specified lawful purposes.
Freedom of Information	Providing public access to information held by public authorities.
Health & Safety	Providing a safe place of work for those at work and impacted by it.

Several project processes are influenced by the regulatory environment

Select 3 of these processes and determine what sort of regulation, legislation, or organizational policy might influence them?



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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Sections 3.3.4 (Regulatory environment), 3.3.5 (Ethics and standards), 3.3.6 (Continuing professional development)

APM Learning module, Ethics and professionalism apm.org.uk/resources/apm-learning/

APM (2017) *The importance of ethics in professional life* apm.org.uk/resources/find-a-resource/thought-leadership/road-to-chartered-series/the-importance-of-ethics-in-professional-life/

APM 'Ethics in project management' apm.org.uk/resources/find-a-resource/ethics-in-project-management/

APM 'FIVE dimensions of professionalism' apm.org.uk/about-us/apm-5-dimensions-of-professionalism/

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15.5 Breakout Groups to provide answers

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Requirements management

Learning objective:

16. Understand requirements management as the ability to capture and monitor the requirements of a project.

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16a) Understand how to establish scope through requirements management processes (such as gather, analysis, justifying requirements and baseline needs).

Requirements management processes

Requirements management is an ongoing process throughout the project life cycle and should be commenced in the initial phase of the project. It facilitates the development of an agreed scope management. It is important to go through these processes:

Gather

- After having identified key stakeholders, information needs to be gathered from these to ensure that all possible requirements are gathered, recorded and documented.
- Where some stakeholders are missed from these early discussions, this can lead to important information being missed or to conflict further down the line.

Analysis

- Function analysis or function cost analysis will provide detail on the value that the requirements will contribute to the organisation.
- Analysis should enable any gaps to be identified or conflicting requirements resolved. This can also provide information on alternative solutions.
- The outcomes should be fed back to stakeholders and consensus reached

Justify requirements

- Functional requirements should be prioritised, linking directly back to the benefits, using techniques such as MoSCoW (M – must have; S – should have; C – could have; W – won’t have).
- Reporting on the outcomes to stakeholders is then required to mitigate any misunderstandings or dispute later.

Baseline needs

- Once the functional requirements have been set, these form the baseline requirements for the project.
- From these, the project manager and project team can determine the project deliverables, manage change and mitigate scope creep.

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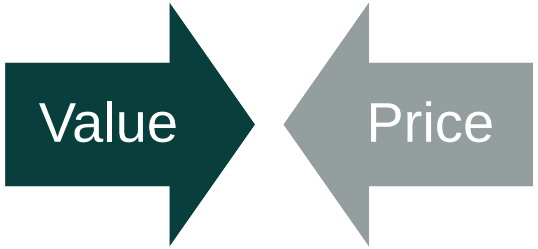
Function analysis or function cost analysis

- We should only accept benefits at a cost that makes them worthwhile to the organisation, thus creating value. Value must be created at the beginning of the project through the development of the business case and in the selection of the option to be delivered, and then be maintained throughout the project.

- Function analysis and function cost analysis are value-based techniques which allow the analysis of the requirements and their business drivers by combining information from functions such as schedule management and investment appraisal. Analysing the time and cost required to deliver the requirements will result in thorough understanding of requirements and the value they will contribute to the objectives.



- The results of the analysis are communicated through individual consultation or group workshops. The product owner has a vital role in ensuring that requirements are always relevant to business needs and advises on the acceptance criteria necessary to achieve this.

- Value represents the amount of benefit that will accrue from the requirement and can be used to justify its priority – how important this requirement is compared to others.

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Prioritisation techniques

- Sometimes, more requirements are requested than it is feasible to deliver, so a prioritisation exercise is needed to highlight the most essential requirements and justify why the chosen requirements should be the ones that are baselined. A common prioritisation technique is the MoSCoW approach:
    - (M) Must have.
    - (S) Should have.
    - (C) Could have.
    - (W) Won't have.
- If using an agile methodology, the 'could have' and 'should have' requirements could be sacrificed if at any time the project was predicted to go over budget or be late. The 'must haves' would only be reduced as a very last resort.
- The main benefit of the process is to ensure that essential requirements are understood as an input to the work to select the optimal solution and then define the detailed scope of work to be delivered with acceptance criteria.
- This level of rigour mitigates the risk that, later in the life cycle, there will be dispute between the project and the organisation about completion and transition of the deliverables into use.

Photo credit: Brett Jordan

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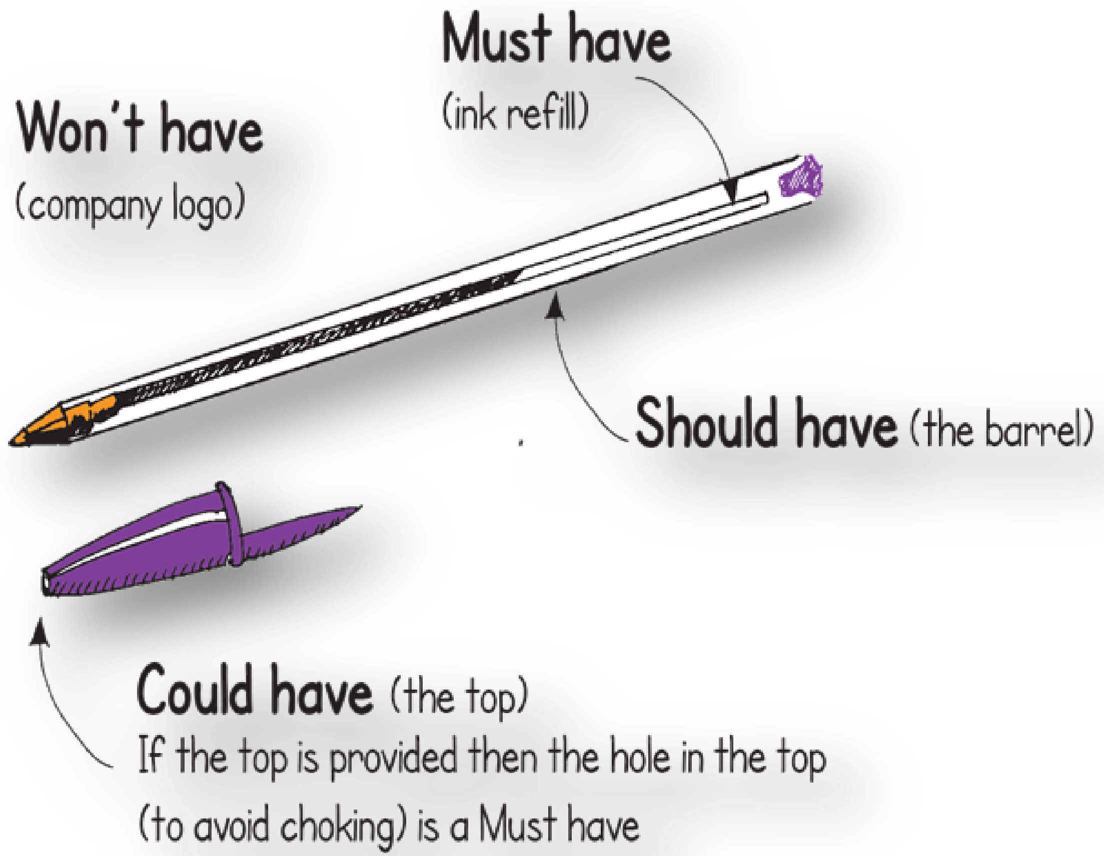
Prioritisation techniques

Must have some flexibility
What can be sacrificed if time and cost are threatened?

MUST
O
SHOULD
COULD
O
WON'T



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16b) Understand how to manage scope through configuration management processes (such as planning, identification, control, status accounting and verification audit).

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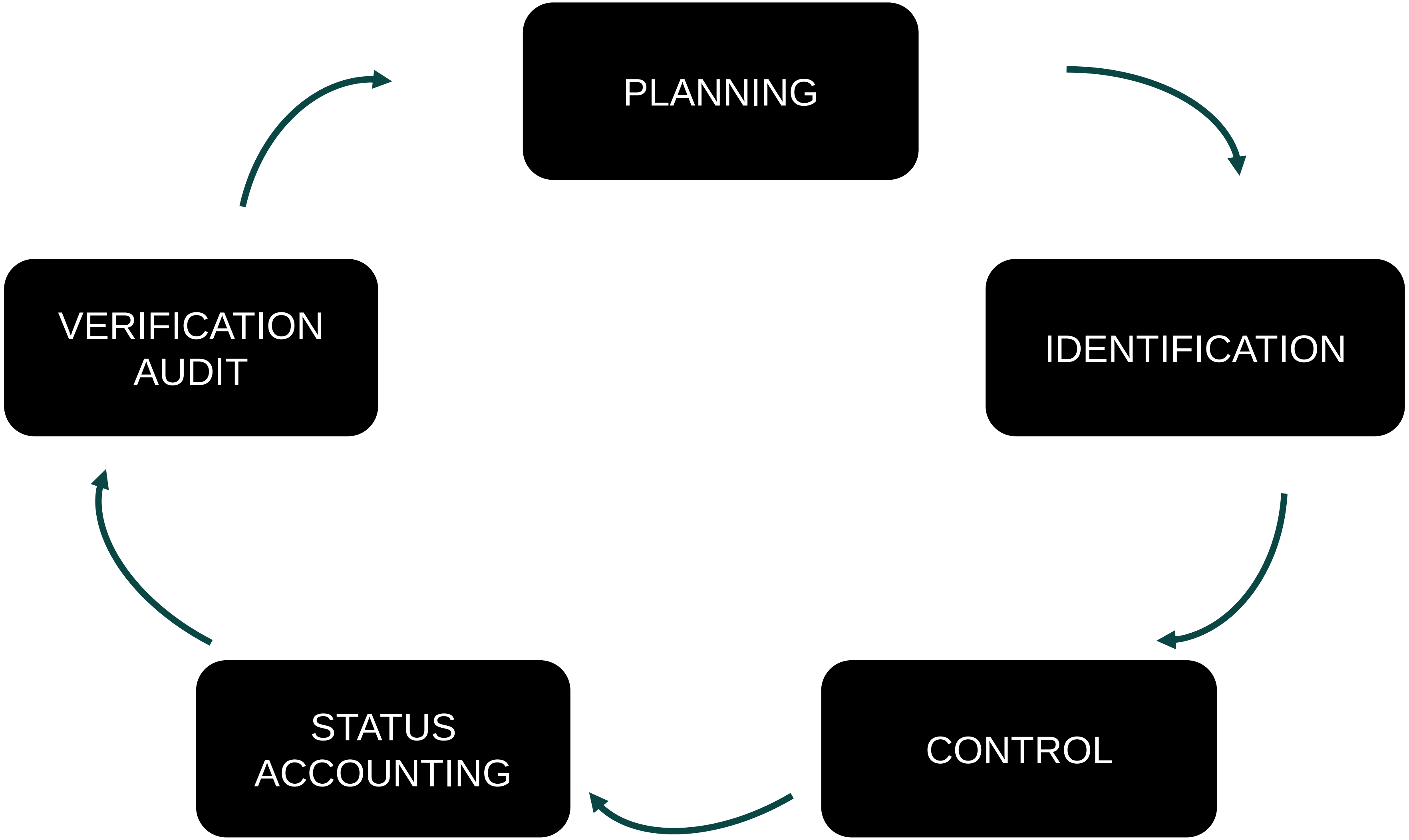
Configuration management process



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Configuration management process

Scope management is closely linked to configuration management by planning, protecting and controlling changes to requirements. The activities of configuration management contribute to scope management as follows:

Planning

- Defines process and procedures, along with setting clear roles and responsibilities to communicate requirements.

Identification

- Divides the project into configurable items, each with a unique reference so that the project manager and team can identify the deliverables of the project.

Control

- Seeks to ensure that changes to these items are controlled and interrelationships between the items are understood to fully understand the impact of the change.

Status accounting

- Ensures traceability of configuration items throughout their development. This supports record keeping and document control for the project, thus maintaining a robust audit trail of configuration records over the course of the project.

Verification audit

- This ensures that the deliverable conforms to its requirements and configuration information, and that completed deliverables match those documented. Audits should be undertaken throughout the life cycle to maintain full end-to-end configuration management of the project.

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 4.1.2 (Objectives and requirements)

APM (2020) 'Understanding requirements, the basis of delivering quality' apm.org.uk/blog/understanding-requirements-the-basis-of-delivering-quality/

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Sample Exam Questions

Learner Study Guide

Sample Exam Questions

16.1

16.2

16.3 (Just list the steps)

16.4

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Solutions development

Learning objective:

17. Understand solutions development as the ability to determine the optimal solution to satisfy agreed requirements.

17a) Understand how to evaluate and prioritise requirements in order to deliver the optimal solution and understand the different approaches for different life cycle models, e.g. the use of MVP and MMP in iterative life cycles.

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Evaluating and prioritising requirements

- The business case should document the options considered, including an option to ‘do nothing’.
- Solutions development requires clarity about the problem to be solved and then exploring different options until a preferred solution is identified and subsequently maintained and refined.

Things to consider:

- Priority of objectives and requirements.
- Acceptance criteria necessary to be achieved.
- Priorities of the project end users.
- The life cycle approach for project delivery.
- Value of alternative solutions.
- Potential for stakeholder pressure to adopt a ‘preferred solution’ which may not be the optimal choice.

The Russian Space Pen!

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Defining the options

Depending on the type of project, there can be many criteria to think through when defining different options.

- **Technical:** process, material, software, future.
- **Social:** people, interactions, needs, relationships.
- **Procurement:** lease, buy, make, service, partnership.
- **Management:** life cycle, governance.
- **Transition:** phased, all at once ‘big bang’.

Photo credit: Glenn Carstens-Peters

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Prioritising

- Requirements may need to be prioritised if the project is unable to deliver them all.
- A common technique for this is the MoSCoW approach:
  - **M:** Must have.
  - **S:** Should have.
  - **C:** Could have.
  - **W:** Won't have.
- This looks at how important a requirement is compared to the others.
- The main benefit of the process is that essential requirements are understood, and can then form an input to the work, enabling the selection of the optimal solution.

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Solutions development in different life cycle models

Projects using different life cycles take different approaches to developing solutions.

- In a linear life cycle, analysis of options is done early on for the selection of a preferred optimum solution to be agreed with scope. This is then confirmed, and any changes thereafter are managed through change control.
- For an iterative life cycle, there is the opportunity to explore options for longer, allowing different design options to be investigated while the needs are being investigated in another timebox.
- Iterative development provides an environment whereby the solution can be delivered in increments providing opportunity for 'Must have' requirements to be delivered as early as possible, in the form of a minimum viable product (MVP).
- The project can then gain progressive user feedback and the gradual delivery of requirements in the next steps of product development to achieve a minimum marketable product (MMP) through to future versions of the product eventually delivering all requirements.

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 4.1.3 (Options and solutions)

APM Learning module, Developing effective approaches and solutions apm.org.uk/resources/apm-learning/

APM ‘Time for Speed’, *Project* https://www.apm.org.uk/news/time-for-speed/

Edward de Bono (2009) *Lateral Thinking: A Textbook of Creativity.*

UK Government (2022) *The Green Book* gov.uk/government/publications/the-green-book-appraisal-and-evaluation-in-central-government

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Sample Exam Questions

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Sample Exam Questions

17.1

17.2

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Quality management

Learning objective:

18. Understand quality management as the ability to ensure that outputs are delivered in accordance with requirements.

18a) Understand what is meant by quality planning, including an understanding of quality indicators and how these relate to the business case.

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Quality planning

- Quality planning is the process for establishing quality for a project. It involves taking the defined scope and listing the criteria to be used to validate that the outputs are fit for purpose, and the methods that will be used to do this. This then forms the quality plan, which is approved by the sponsor and wider governance board.

- The quality plan is a key part of the project management plan (PMP) and sets out the work in scope, aligned to the business case, and how it will be assessed. It includes regulations, standards, specifications and the agreed acceptance criteria. This then provides guidance to the project team as it answers the question what does good look like?

- The quality plan guides the planning of quality control and assurance activities that also confirm that outputs meet requirements. All these activities require resources and will influence tasks on the project that need to be estimated and planned into the schedule and cost.

- Finally, getting stakeholder agreement for the quality plan supports a successful handover on project completion. It ensures that what is being delivered by the project team achieves the outcomes of the business case and to the required standards.

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Quality plan

- The contents of the Quality Plan could include:
  - Project Deliverables and the Acceptance Criteria (Pass/Fail metrics)
  - Quality Standards, Policies, Systems of the host organization and other bodies
  - Relevant Quality Tools and techniques
  - Quality Control Processes and Procedures
  - Quality Assurance Processes and Procedures
  - Organisation, Roles & Responsibilities
  - Reporting requirements

Photo credit: Alejandro Escamilla

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18b) Knowledge of how quality control techniques are used to determine whether success criteria are met.

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Quality management

- Quality control is verifying compliance of the output to requirements.
- Quality assurance is providing confidence that the right processes and methods have been used to deliver it.
- When using an iterative life cycle, acceptance criteria will be defined for each time period, and then will also be verified for the final product, once the project is delivered, to confirm the acceptance criteria have been met for the overarching outputs.
- If a project is part of a programme, some quality requirements may be planned and controlled at programme level.

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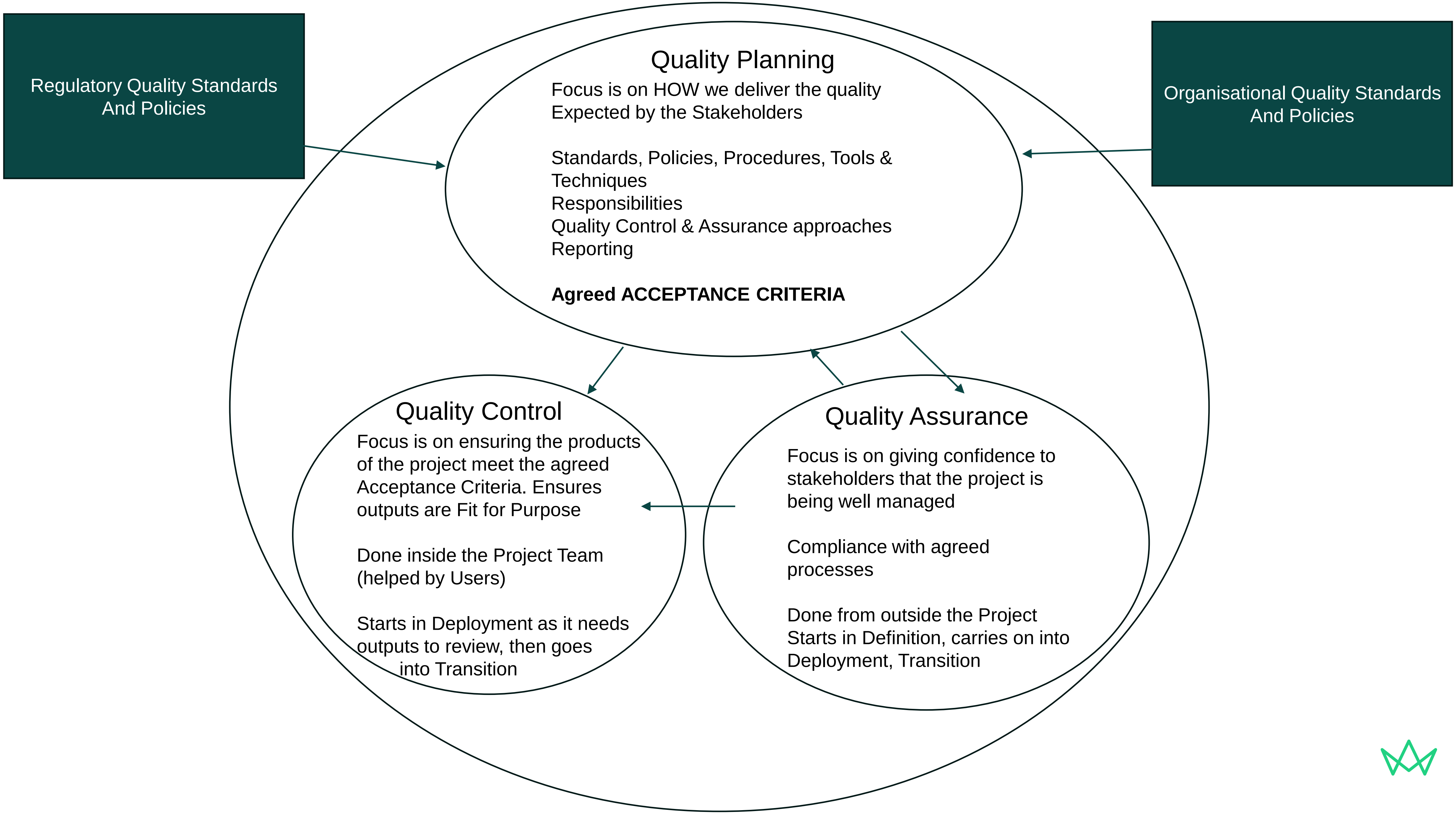
Regulatory Quality Standards And Policies
Organisational Quality Standards And Policies
Quality Planning
Focus is on HOW we deliver the quality Expected by the Stakeholders
Standards, Policies, Procedures, Tools & Techniques
Responsibilities
Quality Control & Assurance approaches
Reporting
Agreed ACCEPTANCE CRITERIA
Quality Control
Focus is on ensuring the products of the project meet the agreed Acceptance Criteria. Ensures outputs are Fit for Purpose
Done inside the Project Team (helped by Users)
Starts in Deployment as it needs outputs to review, then goes into Transition
Quality Assurance
Focus is on giving confidence to stakeholders that the project is being well managed
Compliance with agreed processes
Done from outside the Project
Starts in Definition, carries on into Deployment, Transition

Quality control versus quality assurance

Quality control

- Consists of inspection, measurement and testing to confirm project outputs meet acceptance criteria defined during quality planning.
- Focused on finding problems to stop them from being passed onto the end customer.
- Can be performed by a member of the project team.
- To be effective, the test plans agreed during quality planning need to be managed through change control, so any amends are agreed and communicated.
- Test plans include aspects such as:
  - sample size of tests
  - test protocols including resources required
  - independent performance or witnessing of tests
- To start when physical outputs are delivered to be tested against.
- Outputs are checked against degree of conformance to the requirements and acceptance criteria. If non-conformance, decisions need to be made on what action is needed.
- Quality control as a process will provide a pass or fail result; there is nothing in-between.

Quality assurance

- Looks to build in quality using standard processes and procedures supported by training and feedback.
- Looks to maintain stakeholder confidence that a project has been delivered in the right way.
- Is performed by people independent to the project (eg Programme, Internal Audit, Corporate Quality Team, external bodies).
- To be effective, a framework with supporting documentation on how to do things and methods to use needs to be in place as standard practice.
- To be applied as soon as the project commences management activity (Definition onwards).
- In the event a deviation to organisationally set standards is identified, a decision will need to be made on what action is needed, by the relevant corporate governance board.
- Quality assurance as a process may not be followed fully, yet still deliver an acceptable output, but having an independent QA function will increase the chance of success.

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Sections 4.1.5 (Quality planning), 4.3.8 (Quality control)

APM Learning module, Quality management apm.org.uk/resources/apm-learning/

APM ‘What is quality management and control?’ apm.org.uk/resources/what-is-project-management/what-is-quality-management-and-control/

APM Contracts and Procurement Specific Interest Group (2017) *Guide to Contracts and Procurement* apm.org.uk/book-shop/apm-guide-to-contracts-and-procurement/

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Sample Exam Questions

Learner Study Guide

Sample Exam Questions

18.1

18.2

18.3 Breakout Groups to provide answers

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Integrated planning

Learning objective:

19. Understand integrated planning as the ability to incorporate multiple plans and processes into an integrated project management plan.

19a) Know the format for an effective integrated project management plan and its typical contents.

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An effective integrated project management plan

The intention of the project management plan (PMP) is to capture the fundamental components to deliver project outcomes successfully.

- It involves collating a suite of plans and processes to support a project to create an integrated plan (the PMP).
- The size and complexity of a project will influence the amount of content in the PMP.
- Format may also vary based on complexity, ways of working and who's involved in the project.
- Once the information is assembled in a self-contained document, it pulls together all the essential management processes needed to show the project teams' intentions of how the project will be managed.

Photo credit: Brands&People

Typical contents of a project management plan

The PMP details the why, what, when, who, how much, where and how of the project.

Benefits – why?

Referencing the business case, looking at the value the project benefits will deliver the business once achieved.

Scope – what?

Looking at descriptions of the scope involved, considering the product breakdown structure (PBS) and work breakdown structure (WBS) and detailing the acceptance criteria.

Time – when?

Showing the timeline for project activities and deliverables aligned to the project life cycle approach.

Resources – who?

Showing the roles and responsibilities and reporting lines – the organisational breakdown structure (OBS).

Cost – how much?

Showing how the budget has been allocated to the work for the project duration – the cost breakdown structure (CBS).

Delivery logistics – where?

Detailing project location, delivery requirements, any environment constraints, technology to support virtual locations, or security protocols.

Management – how?

The most wide-ranging part of the document outlining the management aspects for deployment covering areas such as quality, risk and issues, communication and project controls.

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19b) Understand the importance of producing an integrated project management plan.

Why is an integrated project management plan important?

- It communicates the details of approved project plans to stakeholders.
- Key stakeholders involved in the deployment of the project have been engaged with its production.
- It's a reference source for all stakeholders.
- It provides one source of the truth to all resources involved and can be used as induction material for new members, providing continuity to the project.
- It's referred to as the contract between the project manager and project sponsor, providing clarity on what is expected of the project manager.
- It guides the project team and is the baseline for measurement and analysis of progress to take place.

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 4.2 (Integrated planning)

APM Learning module, Introduction to project planning

APM Learning module, Project management plan

APM project management plan template

apm.org.uk/resources/apm-learning/

APM (2015) *Planning, Scheduling, Monitoring and Control: The Practical Project Management of Time, Cost and Risk*
apm.org.uk/book-shop/planning-scheduling-monitoring-and-control-the-practical-project-management-of-time-cost-and-risk/

APM ‘What is project planning?’ apm.org.uk/resources/what-is-project-management/what-is-planning/

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Sample Exam Questions

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Sample Exam Questions

19.1

19.2

19.3

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2	June 2024	Diagram updated for clarity	15

Schedule management

Learning objective:

20. Understand schedule management as the ability to undertake time-based planning with an emphasis on activities and resource.

What is schedule management?

Time scheduling is used to develop schedules for when work is planned to be completed in projects.

It's time-based planning with an emphasis on activities and resource.

It's the process for developing and maintaining schedules.

It shows when work for a particular project is planned to be done.

It considers dependencies and can be for internal and/or external resources and activities.

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20a) Know how to define scope in terms of outputs, outcomes and benefits (including use of product, work and cost breakdown structures).

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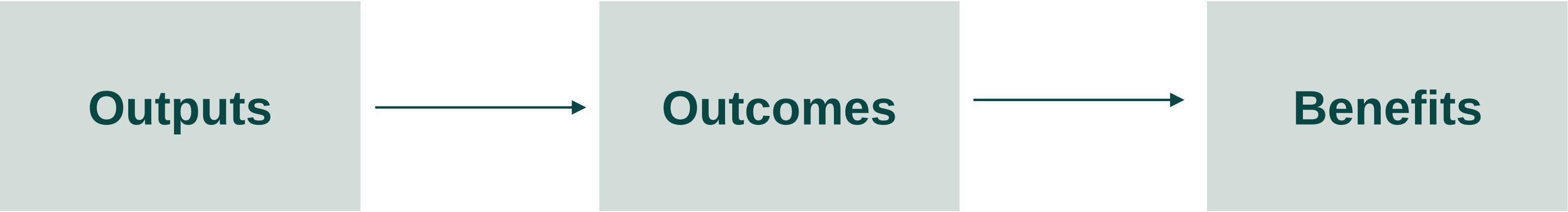
Defining scope

- Scope is defined by the outputs (and the work needed to create them), outcomes and benefits being delivered. Typically, a high-level scope will be recorded in the business case.

- A project’s outputs are the tangible or intangible deliverables or products being created which produce the outcomes of changed circumstances, behaviours or practices, to create the benefits in line with the defined acceptance criteria.



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Defining scope in detail

- It’s important to be clear about the boundaries and interfaces with adjacent projects to avoid duplication, conflicts or omission of work within projects or business-as-usual work.
- The detailed scope of work emerges from the breakdown of the chosen option to meet the sponsor’s requirements.
- Once a solution has been identified to meet these requirements, breakdown structures can be used to illustrate what products and activities are needed to deliver the project.
- For projects using a linear life cycle, breakdown structures provide a hierarchical view of what is needed to deliver the project. Iterative lifecycle projects will often use a Product Backlog (essentially a prioritized list of what is needed).
- Content created at each level supports the creation of the next level of the hierarchy until the project deliverables are identified. The same can then be done for the detailed ‘workpackages needed to create the products.
- This enables decisions to be made about who in the project will take responsibility for carrying out aspects of work, supervising activity and reporting on progress.

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Breakdown structures

Illustrating what needs to be done

Product breakdown structure (PBS)

A PBS identifies the scope of a project to be delivered.

The end output of the project is placed at the top level, defining all components moving down levels until reaching the individual products to be delivered.

Each product has defined acceptance criteria and quality control methods.

This can be used to secure stakeholder agreement of the deliverables and agree what is in and out of scope.

Work breakdown structure (WBS)

A WBS defines a baseline scope of work capturing the activities or work to be completed to deliver the project requirements.

The lowest level of detail identified is a work package.

The work required and skill set needed to complete can be reviewed, and work requiring a similar skill set can then be grouped. This supports allocation of resources, estimating and scheduling of the work packages.

Cost breakdown structure (CBS)/ Organisational breakdown structure (OBS)

A CBS provides a financial view by breaking down the project into cost elements.

It can be used to show either the costs assigned to work packages, using the WBS, such as people, equipment, materials or other required resource, or the costs assigned to functional areas and third parties using the organisational breakdown structure (OBS).

This can be used to track project spend to compare against the estimates produced.

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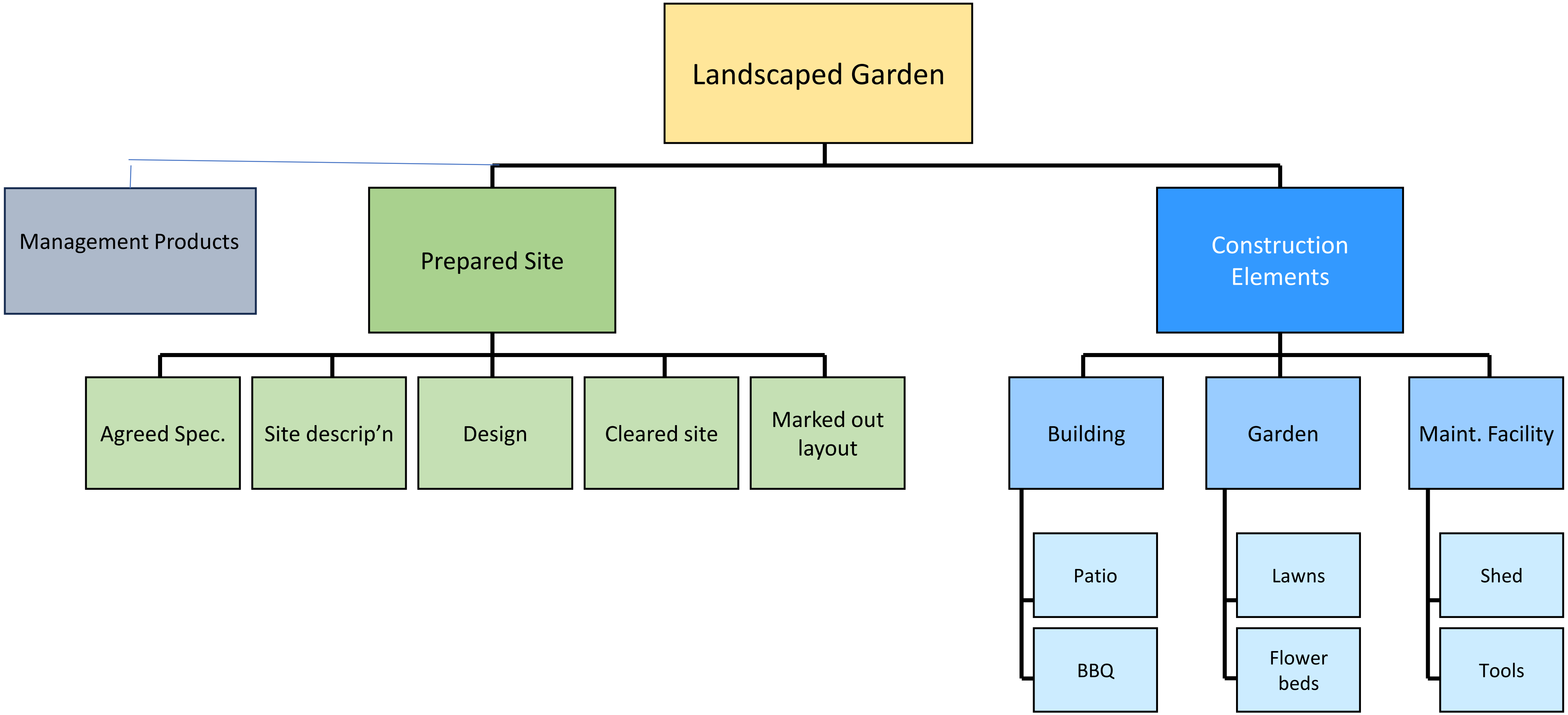
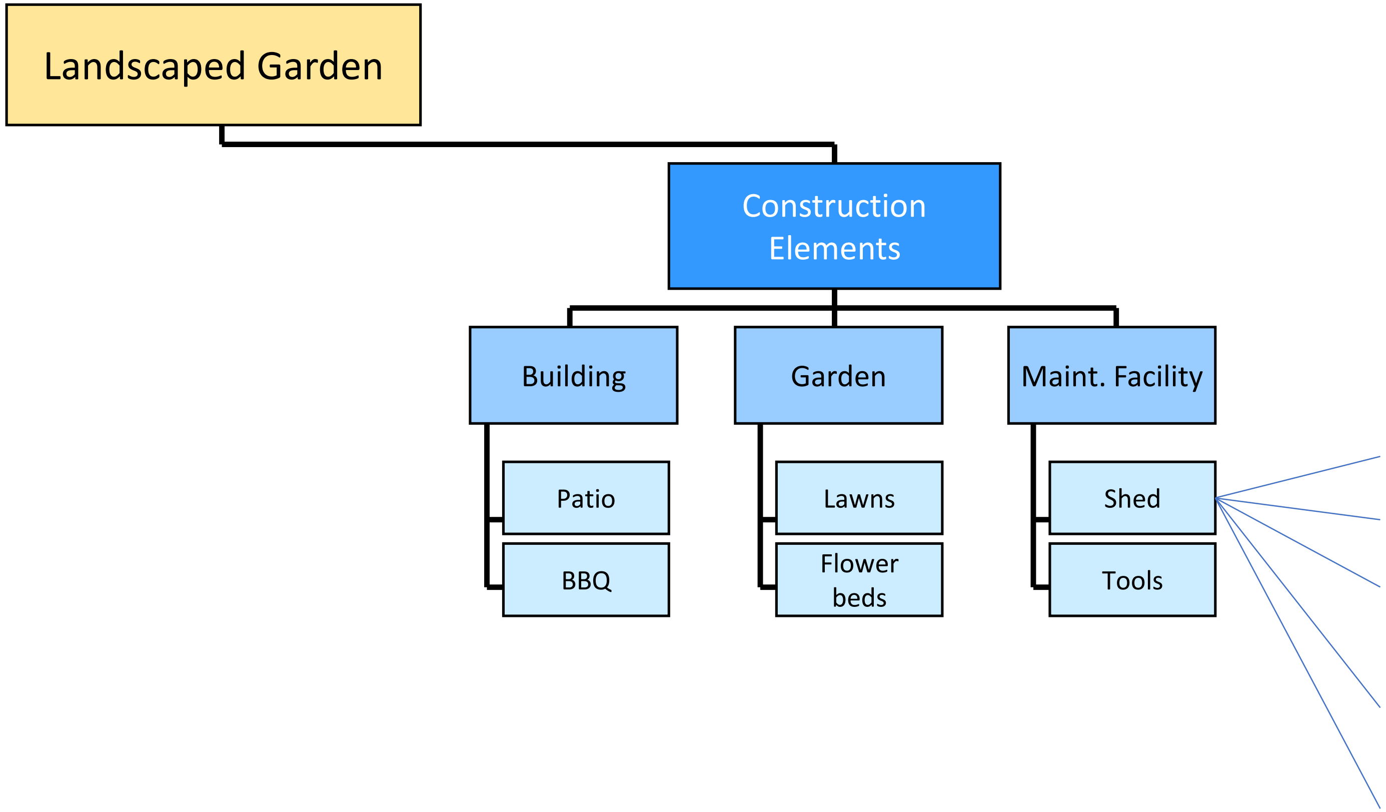
Product Breakdown Structure



Work Breakdown Structure



Workpackages for “Shed” Output

Digging out for pad

Pour concrete for pad

Build shed shell

Fit out shelves

Fit windows

20b) Understand there are links and dependencies between activities within a project and business-as-usual activities, for example business-as-usual activities, costs, quality, risks and scope can all impact the schedule for the project.

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Links and dependencies

All the following can have an impact on a project and should be considered as part of schedule management.

Relationship between project tasks

The links between activities and their implications.

Scope definition

Prevent risk of delays through misunderstanding of what is in and out of scope for the project.

Dependencies

Between different project activities and business as usual (BAU).

Stakeholders

From a technical and BAU perspective to inform on potential impacts to time / resource.

Business change/ Adoption

Activities that are resourced by BAU not the project.

Schedules Clashes

Advise of activities and any potential conflicts requiring resolution.

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20c) Understand the reasons for and benefits of re-estimating, and schedule optimisation, throughout the project life cycle.

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Why re-estimate and optimise?

An estimate is just that – an approximation of time to complete tasks. The activity of estimating tasks should not be a one-off but should be ongoing. As more information is known, re-estimating should happen. This will increase confidence in the estimates being used to create the schedule.

Greater involvement of project team

- Apply technical expertise to clarify what is needed to achieve project scope.
- Minimise the effect of any previous or ongoing estimating errors.

Opportunity to incorporate lessons learnt and reduce contingency reserves

- Validate assumptions as scope is clarified to reduce uncertainty.
- Include variances and trends from previous work to remaining estimates to adjust them for ongoing activity to incorporate findings.

Realisation of an accepted risk

- Understand the impact of the realised risk on the project.

Increased likelihood of adhering to overall estimate

- Provide visibility of variances to inform stakeholders and take any necessary actions.
- Apply productivity data for better understanding of final costs and timescales.

To fully optimise scheduling techniques, the concepts that feed into the activity must be understood. The WBS, estimating, dependencies, resource allocation, alignment of success criteria and baselining are the foundations for scheduling and for the optimisation of the schedule for a successful project.

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The benefits of re-estimating

Greater involvement of project team

- Increases the accuracy of estimates through technical expertise providing increased confidence in the project's ability to deliver against the success criteria.

Opportunity to incorporate lessons learnt and reduce contingency reserves

- Removes reserves that are no longer needed and can be used elsewhere on other projects as forecasts become more realistic to circumstances.

Realisation of an accepted risk

- Increases awareness of the situation and any potential impact with key stakeholders and, if necessary, requests to draw down contingency for the project to continue.

Increased likelihood of adhering to overall estimate

- Increases confidence in the project's performance.
- Provides information that supports any required escalations for further funding or time for the project to deliver against the success criteria.

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Critical path and Critical Chain

- Two of the main methods used to support scheduling optimisation are critical path and critical chain:
  - The **critical path** method places emphasis on the time required to complete all the individual activities in a logical order. Identifying the relationship between tasks, and the time estimate for each is essential in identifying the critical path. Resources can be used flexibly.
  - The **critical chain** method places emphasis on **resources**, attempting to keep them at a constant utilisation. It uses an overall Project Buffer, rather than flexibility in individual activities.
- The critical chain method, alternatively called the ‘resource critical path’, places the emphasis on the resources (labour and non-labour) in a project, while the **critical path** emphasises the activities & time taken.
- If resources were unlimited and always available, the critical path and critical chain method for scheduling would provide the same result.
- The results of these methods are usually presented as a visualisation, such as a Gantt chart that shows either activities or resources as bars on a timeline.

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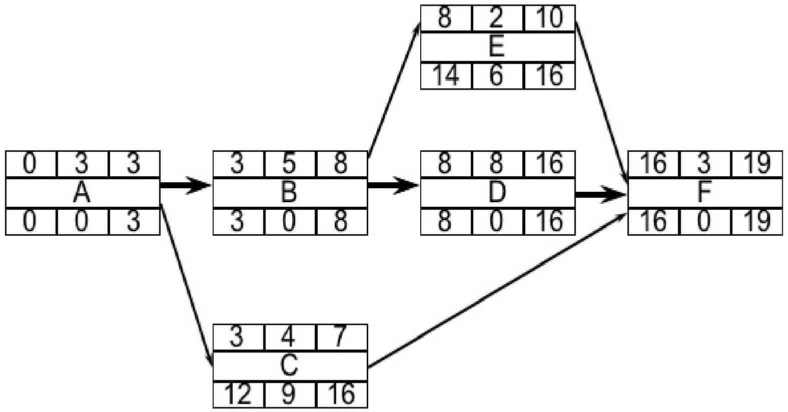
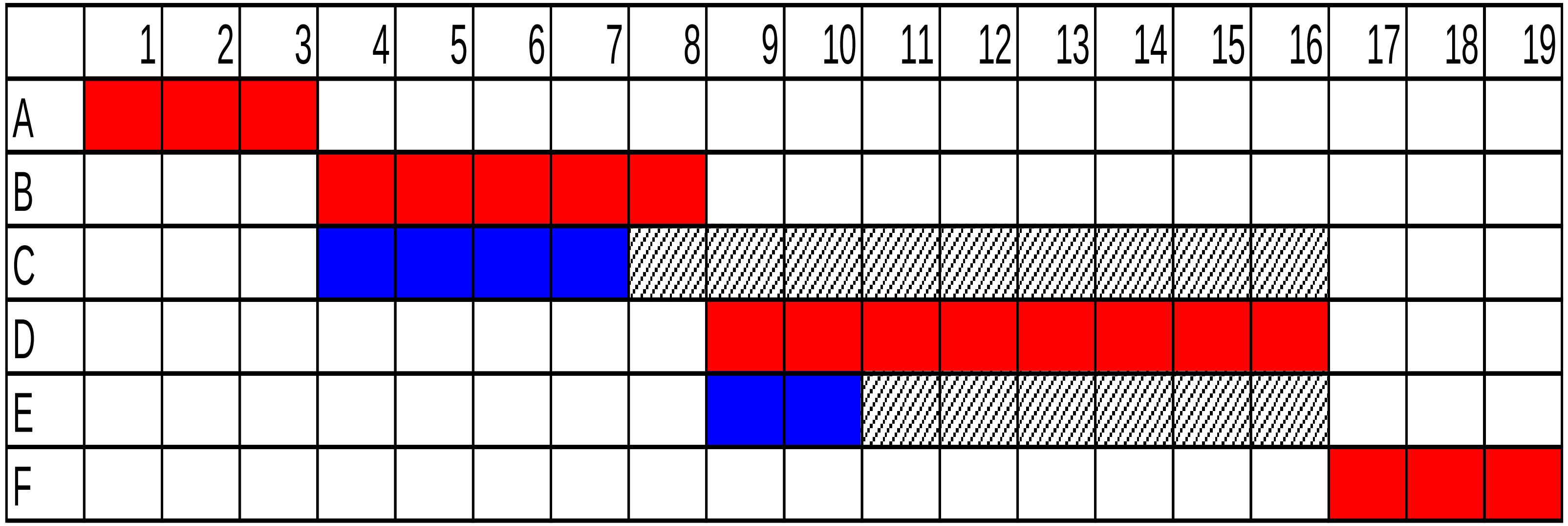
Critical Path

- Identifies the activities or tasks in the plan that if delayed will affect the end date.
- These critical activities, have no ‘float’ and create the longest pathway of activity on the project and therefore form the Critical Path.
- We can manage a critical path schedule by moving resource from non-critical to critical activities, utilising the available float.
- The start and finish of activities are the focus for critical path reporting.



Activities with ZERO float are critical, the path of these if the CRITICAL PATH



Red = Critical Path; Blue = Non-critical activities; Hashed = Float in non-critical activities

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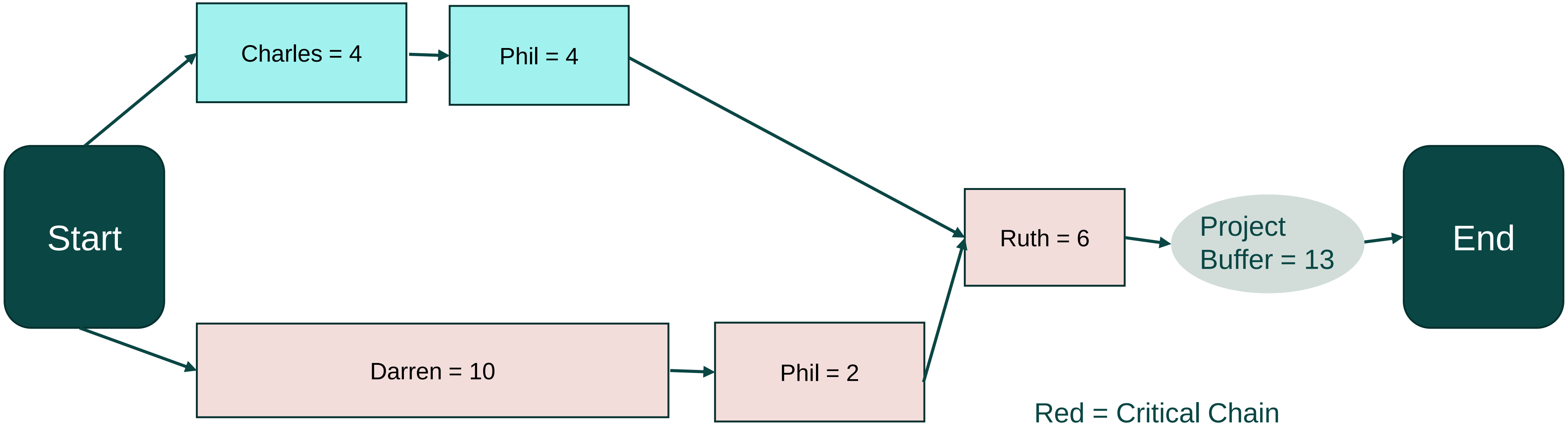
Critical chain

- The critical chain method takes the most likely duration estimates for each activity using a set resource, this resource is for THAT activity and should not be moved to other activities.
- A common approach is to add up these estimates and an amount equal to half of this overall time is held as a project buffer at the end for the chain of activities rather than having a float for individual activities, as in Critical Path.
- The buffer is there to protect the whole of the project and extra time will be taken from it if the 'most likely' times are forecast/are exceeded.



Red = Critical Chain

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Sections 4.2.5 (Scheduling – critical path), 4.2.6 (Scheduling – critical chain)

APM Learning module, ‘Project Controls: Scheduling’

APM Learning module, ‘Planning, Scheduling, Monitoring and Control: The practical project management of time, cost, and risk’

apm.org.uk/resources/apm-learning/

APM ‘What is scheduling in project management?’ apm.org.uk/resources/what-is-project-management/what-is-scheduling-in-project-management/

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Sample Exam Questions

Learner Study Guide

Sample Exam Questions

20.1

20.2

20.4 Breakout Groups to provide answers

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Resource management

Learning objective:

21. Understand resource management as the ability to identify and schedule the required internal and external resources.

21a) Know how to determine the resources required and their availability to deliver activities within a project.

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Resource requirements

Establishing resource requirements for the project requires several considerations:

- life cycle approach
- activities and tasks to be completed
- types of resources needed to support activities
- skill sets and capabilities needed from resources
- an understanding of what the critical tasks for resources are
- timeframe that resources are needed for
- needs of the project versus needs of business as usual (BAU)
- any external resources needed

Resources may:

- Drive project pace by placing constraints on what can be done when.
- Constrain project logistics through space needed to operate and store things.

Photo credit: Markus Winkler

Project managers work with resource or BAU managers to secure resources at the required times for the schedule.

Resource or BAU managers may provide line management to the resources in BAU and therefore may be allocated to the project.

If the resource has come from a supplier, an external resource manager may be allocated to manage relations.

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21b) Understand how an organisational breakdown structure is used to create a responsibility assignment matrix (RACI).

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Creating a responsibility assignment matrix

- An organisational breakdown structure (OBS) details the organisation for the project and the resources available.
- Tasks can then be allocated to the resources defined in the OBS, helping to clarify roles for people. An OBS also makes initial escalation routes clear, which helps with any resource problems that may need escalating.
- For each task, the project manager will decide if the team member is responsible, accountable, consulted, or informed (RACI), creating a responsibility assignment matrix (RAM) for the project.
- The RAM is a communication device to ensure that the people involved in doing the work are aware of the work they have to do and their position in the project organisation.
- Resource or BAU managers (these may be line managers) also consult the RAM to understand and agree the work that has been allocated to their people.

In summary, the RAM provides an understanding of the tasks or deliverables, the specific responsibilities defined within the project procedures, and the level of accountability or contribution expected from named roles or individuals within the project.

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RACI

A common coding structure applied to a RAM is called the RACI matrix:

R — **Responsible**
Does the work and owns any problems.

A — **Accountable**
Approves the works and is held accountable for the deliveries.

C — **Consulted**
Included in decision making and provides a key supportive role.

I — **Informed**
Kept up-to-date with progress and results.

It is best practice for one person to be responsible and one person to be accountable.

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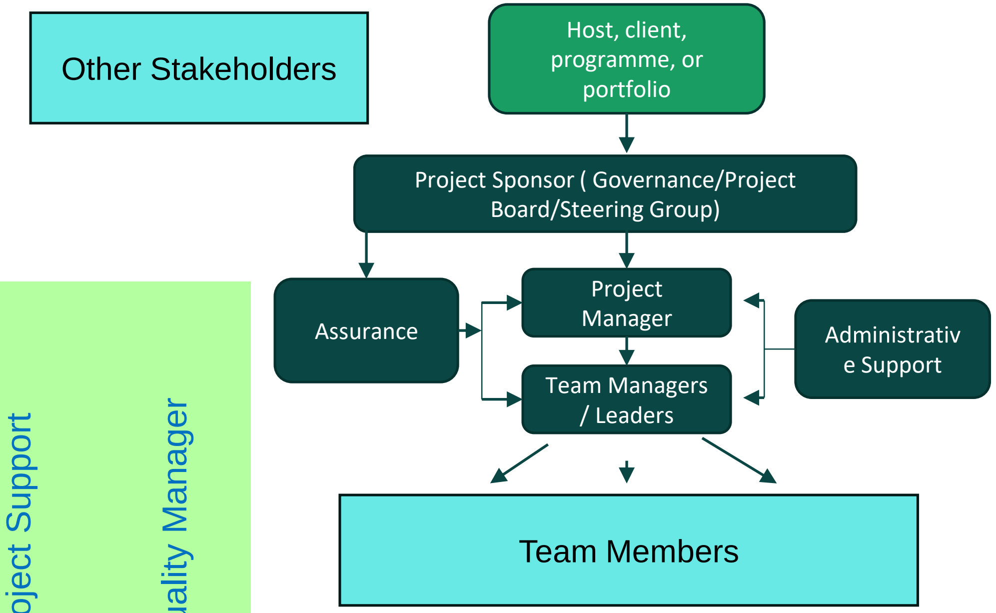
Responsibility Assignment Matrix (RAM)



From OBS

From WBS

	Sponsor	Project Manager	Team Manager	Team Member 1	Team Member 2	Team Member 3	Project Support	Quality Manager
WP A	A	R	C				C	C
WP B		A	R	C	C		I	
WP C	A	R					C	C
WP D		A	R		C	C		
WP E	I	A	R	C		C	I	I

Key:

R – Responsible
A – Accountable
C – Consult
I – Inform

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21c) Understand how resources are categorised and allocated to both linear and iterative life cycle schedules.

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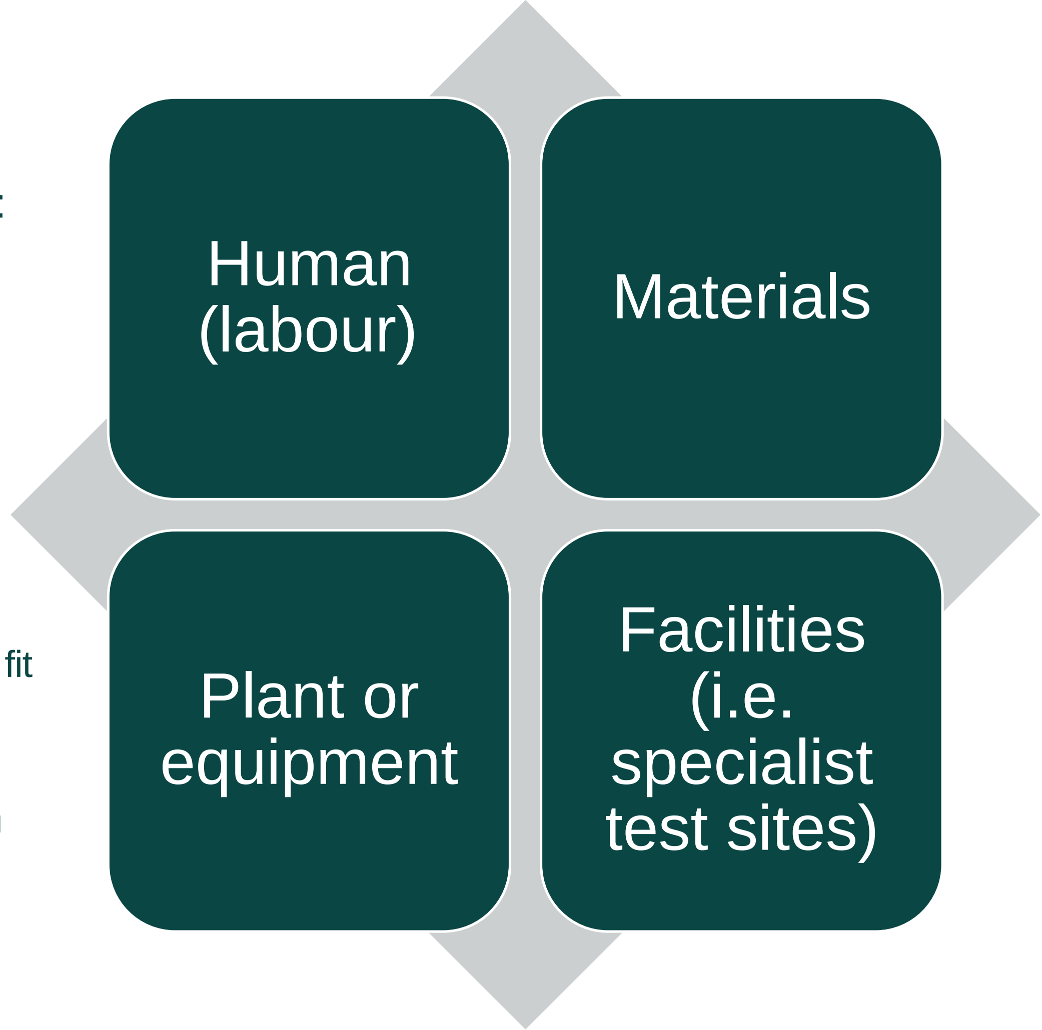
Categorising and allocating resources

Resources are the labour and non-labour items needed to deliver the project objectives successfully to the agreed acceptance criteria.

Resources are categorised into four types: human, materials, plant or equipment, and facilities.

- When looking to allocate resources, the project manager must consider:
  - the allocation of resource type
  - how many resources needed for each type
  - availability to support the schedule
  - cost
- When using a linear life cycle, a planning assumption made when creating the schedule is that all resources will be available when required.
  - If this is not the case, resource management techniques can be applied to try and fit the work in with the resources available as much as possible maintaining the schedule and quality.
- For iterative life cycles, requirements are prioritised and delivered within the timebox using the pre-allocated resources available, varying scope and quality as required.
  - However, if all scope must be achieved to the agreed quality, an extension of time will be needed.



21d) Know the differences between resource smoothing and resource levelling.

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Resource smoothing and resource levelling

Resource management techniques, such as levelling and smoothing, attempt to fit the scope of work into the resource limit, while as much as possible maintaining the schedule and quality objectives.

- **Resource levelling** is used where a fixed amount of resource is available, and the end date of the project can be delayed. It answers the question With the resources available, when will the work be finished?
- Levelling can be achieved through redefining scope and/or specifications, increasing task durations, increasing resources earlier on to bring activities forward, or moving non-critical activities, and even critical path activities, to a different time.
- **Resource smoothing** is used when time is more important than cost, answering the question What resources do I need to deliver the work within the fixed timescale? It looks to protect the end date of the project.
- Smoothing can be achieved through reducing task durations, by adding more resources, avoiding peaks and troughs of resource demand, or changing the order of activities where the logic used originally was optional to run activities in parallel rather than in sequence.
- The overall aim of the techniques is to optimise resources through the flow of work between them, thus preventing the need to change the planned end date of the project. However, if resources are finite (either the people or equipment needed to complete the tasks) then there may be no option but to extend project timelines.

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Resourced Gantt Chart & Histogram (pre-smoothing)

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Activity A																			
Activity B																			
Activity C																			
Activity D																			
Activity E																			
Activity F																			
Activity G																			
Activity H																			
Cumulative Resources	1	2	4	6	8	10	12	15	18	20	22	24	25	26	27	28	29	30	31
Total Resource/Day	1	1	2	2	2	2	2	3	3	2	2	2	1	1	1	1	1	1	1
Resource 3								F	F										
Resource 2			G	G	G	G	G	G	G	G	H	H							
Resource 1	A	A	B	B	B	B	B	C	C	C	C	C	C	D	D	E	E	E	E

Reduce the Peaks & Troughs of the profile to make it easier to manage whilst keeping to the agree timeframe

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Smoothed Gantt Chart

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Activity A																			
Activity B																			
Activity C																			
Activity D																			
Activity E																			
Activity F																			
Activity G																			
Activity H																			
Cumulative Resources	1	2	4	6	8	10	12	14	16	18	20	22	24	26	27	28	29	30	31
Total Resource/Day	1	1	2	2	2	2	2	2	2	2	2	2	2	2	1	1	1	1	1
Resource 3																			
Resource 2			G	G	G	G	G	G	G	G	H	H	F	F					
Resource 1	A	A	B	B	B	B	B	C	C	C	C	C	C	D	D	E	E	E	E

Focus is on maintaining the agreed timescale

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Resourced Gantt Chart & Histogram (pre-levelling)

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Activity A																			
Activity B																			
Activity C																			
Activity D																			
Activity E																			
Activity F																			
Activity G																			
Activity H																			
Cumulative Resources	1	2	4	6	8	10	12	15	18	21	24	26	26	28	29	30	31	32	33
Total Resource/Day	1	1	2	2	2	2	2	3	3	3	3	2	1	1	1	1	1	1	1
Resource 3								F	F	F	F								
Resource 2			G	G	G	G	G	G	G	G	H	H							
Resource 1	A	A	B	B	B	B	B	C	C	C	C	C	C	D	D	E	E	E	E

Current schedule requires a maximum of three resources for the 4 week period (8 to 11)

However, we get informed that our third resource has been moved and we have to manage with two people

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Resource Levelling – Activity F has slipped

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Activity A																				
Activity B																				
Activity C																				
Activity D																				
Activity E																				
Activity F																				
Activity G																				
Activity H																				
Cumulative Resources	1	2	4	6	8	10	12	14	16	18	20	22	24	26	28	29	30	31	32	33
Total Resource/Day	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	1	1	1	1	1
Resource 3																				
Resource 2			G	G	G	G	G	G	G	G	H	H	F	F	F					
Resource 1	A	A	B	B	B	B	B	C	C	C	C	C	C	D	D	F	E	E	E	E

In order to overcome the Resource Constraint we have had to move activity F BEYOND the limit of the Float thus causing the overall time to extend to 20 weeks

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Resource smoothing and resource levelling

Aspect	Resource Smoothing	Resource Levelling
Definition	Adjusting activities without changing the project duration	Adjusting activities based on resource constraints, which may extend the project duration
Primary Focus	Maintaining a consistent level of resource usage	Ensuring resource constraints are not violated
Schedule Impact	Low, as project end date is not changed	Medium to High, as project end date can change
Resource Utilisation	Optimises resource utilisation by minimising peaks and troughs	Balances resource demand with available supply
Critical Path	Not affected, as adjustments are within float limits	May change, as task adjustments can affect the critical path
Flexibility in Task Scheduling	Limited to activities with available float, doesn’t touch critical path activities	Flexible, can adjust any activity including those on the critical path
Usage Context	Used when resource availability is consistent and project deadlines are strict	Used when resource availability varies and meeting resource constraints is critical

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 4.2.7 (Resource optimisation)

APM Learning module, Resource management and capacity planning
APM Information sheet, Resource smoothing and levelling
apm.org.uk/resources/apm-learning/

APM ‘What is resource optimisation?’ apm.org.uk/resources/what-is-project-management/what-is-resource-optimisation/

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Sample Exam Questions

Learner Study Guide

Sample Exam Questions

21.1

21.2

21.3

21.4

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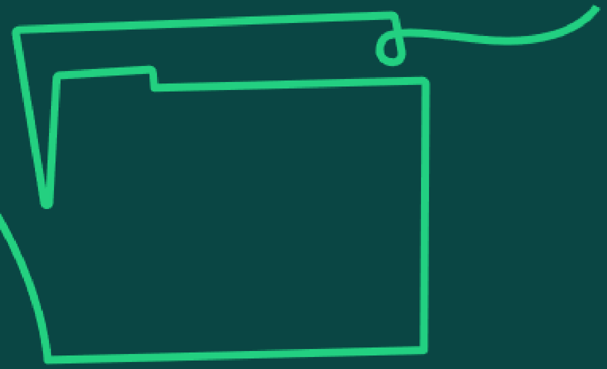
Budgeting and cost control

Learning objective:

22. Understand budgeting and cost control as the ability to estimate costs, develop and agree budgets and monitor actual costs against forecast costs.



22a) Know how to create a budget (including the use of a cost breakdown structure) and the different costs included in a budget (fixed, variable, direct, indirect etc.)

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Creating a budget

In the project profession, estimates are used for multiple purposes, one of which is to support budgeting and cost control. Budgets are created from estimates, looking at the approximation of time and cost to complete the scope of work to the required acceptance criteria.

Estimates are used to:

- Enable budget setting and considerations of affordability.
- Support judgements about value for money for the solution chosen.
- Create a resource schedule which then in turn provides the cost view of the project.

As part of creating a budget, consideration should be given to:

- The different types of costs involved and when they will need to be paid so that they can be budgeted accordingly, i.e. funding drawdowns to cover payment milestones.
- Information required from external sources, such as supplier costs, and how these align to the plan to enable budgeting.
- The assumptions being made on the project and how they underpin the cost estimates so they can be understood, validated and budgeted for.

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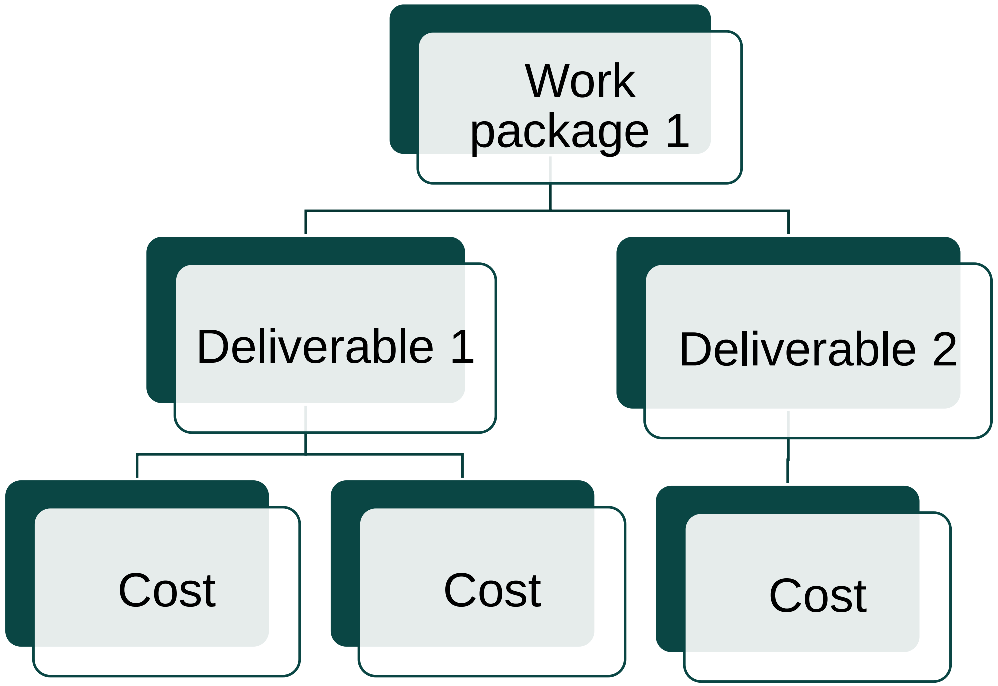
The cost breakdown structure (CBS)

- The work breakdown structure (WBS) is used to create the CBS by taking each of the work packages defined for the project and looking at the estimated costs related to the delivery of activity. Through this process, a budget can be set for the overall project.

- The CBS identifies all activity that has a cost associated with it, both labour and non-labour, in each work package. These costs can then be collected, recorded, monitored and controlled as part of the financial reporting system. Costs can be attributed to:
  - people
  - equipment
  - materials
  - other resources required

- The CBS will reflect the financial coding used in the project for booking and recording. This may be set at an organisational level as part of standard financial reporting procedures.



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The relationship between different costs

- Knowing the overall cost of the project is a good start; however, on its own this is not enough for project control.
- It is key to know, within that overall cost, where the costs fall over time in the schedule.
- It is then possible to manage resources, supplier payments and funding requests effectively.
- The project life cycle determines the frequency with which funds are to be released.
  - In a linear life cycle, funds may only be released at decision gates when the costs spent to date are understood and future forecasts have been approved.
  - In an iterative life cycle, the release of funds may be more frequent due to work being completed in short intervals.
- This cost planning exercise is an essential part of the process to manage cashflow and to ensure the project manager knows when funding drawdowns are needed.

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The relationship between different costs

Cost planning aspects include looking at the type of costs:

- **Are costs fixed?** For example, a contract may set a monthly cost for resources provided by a particular supplier, irrespective of the actual resources used by the supplier over a month.
- **Or are they variable?** For example, paying for materials as and when they are needed or used. The monthly cost may well be different from month to month.
- **Are costs recurring?** For example, software licenses and maintenance reoccur at set periods, happen periodically as an event happens and contribute multiple costs to the project at different times.
- **Or are they non-recurring?** For example, equipment set-up costs are a one-off cost to the project, happening once in a project life and contributing a single cost.

It's important to consider whether costs are direct or indirect, as this makes a difference to how easy it is to identify, track and manage them. Direct costs are directly related to the outputs of the project, whereas indirect costs are associated with the costs of operating the outputs for the business as a whole.

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22b) Know how to forecast and refine budgets using cost control techniques e.g. earned value.

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Cost control techniques

Cost control is necessary to identify variances to project budgets and cost forecasts to inform decision making.

Project managers use cost control techniques to:

- Minimise costs where possible.
- Identify areas of overspend.
- Correct unacceptable levels of overspend.
- Inform future projects by providing insight for lessons learnt.

Forecasts are created and used to determine project costs against scheduled activities.

When changes are made to the agreed baseline, it is important to update the cost information to refine budgets.

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Earned value management

A recognised best practice approach to cost control is earned value management (EVM), which is ideal for identifying where a project is not progressing efficiently.

- Earned value analysis reports on the project spend against the spend curve and the value of the work achieved. If the project is underspent or overspent against the planned spend line, knowing the value of work done at that point helps to refine budgets based on current project status.
- The efficiency of spend is looked at through the cost performance indicator (CPI) and productivity through the schedule performance indicator (SPI).
- A pictorial representation of project progress can be provided by tracking the earned value on an earned value graph.

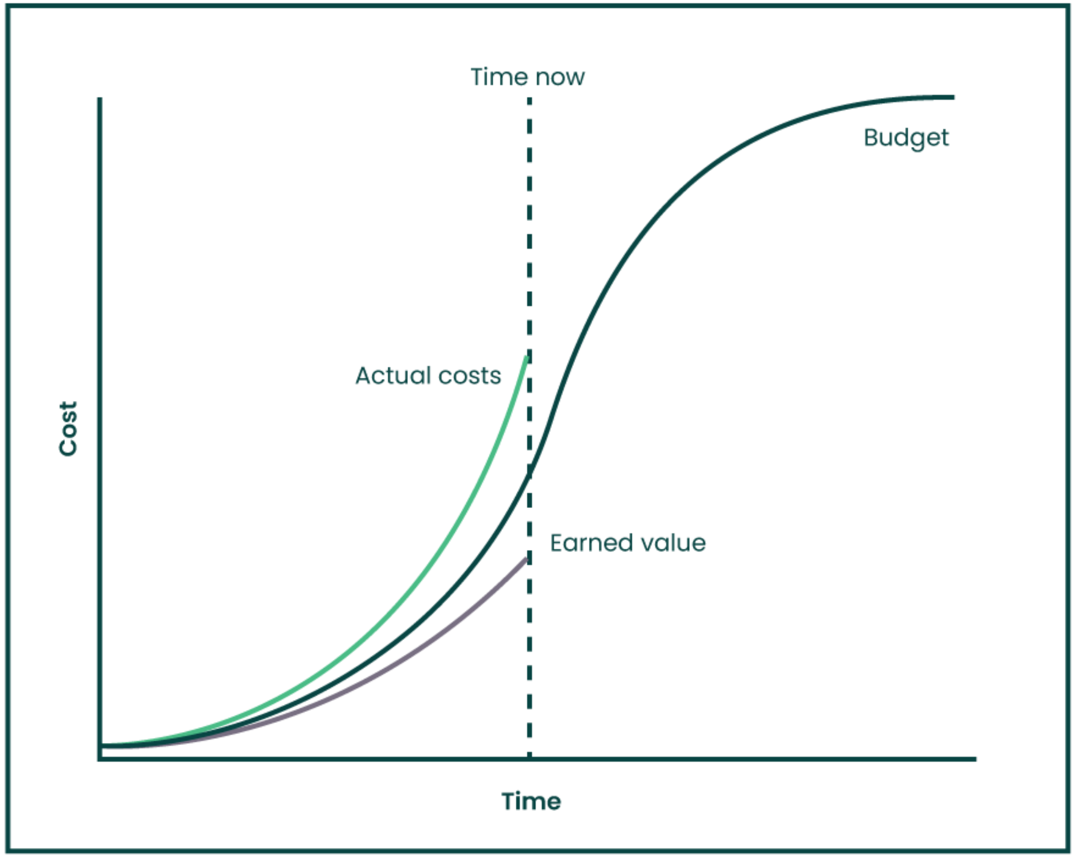

Figure 2: Budget vs actual plus earned value graph
Taken from the Earned Value Management: APM Guidelines (2008)

Earned Value (EV) equates to the value of the work ACTUALLY PERFORMED, and EVM compares this to planned work and the actual cost charged

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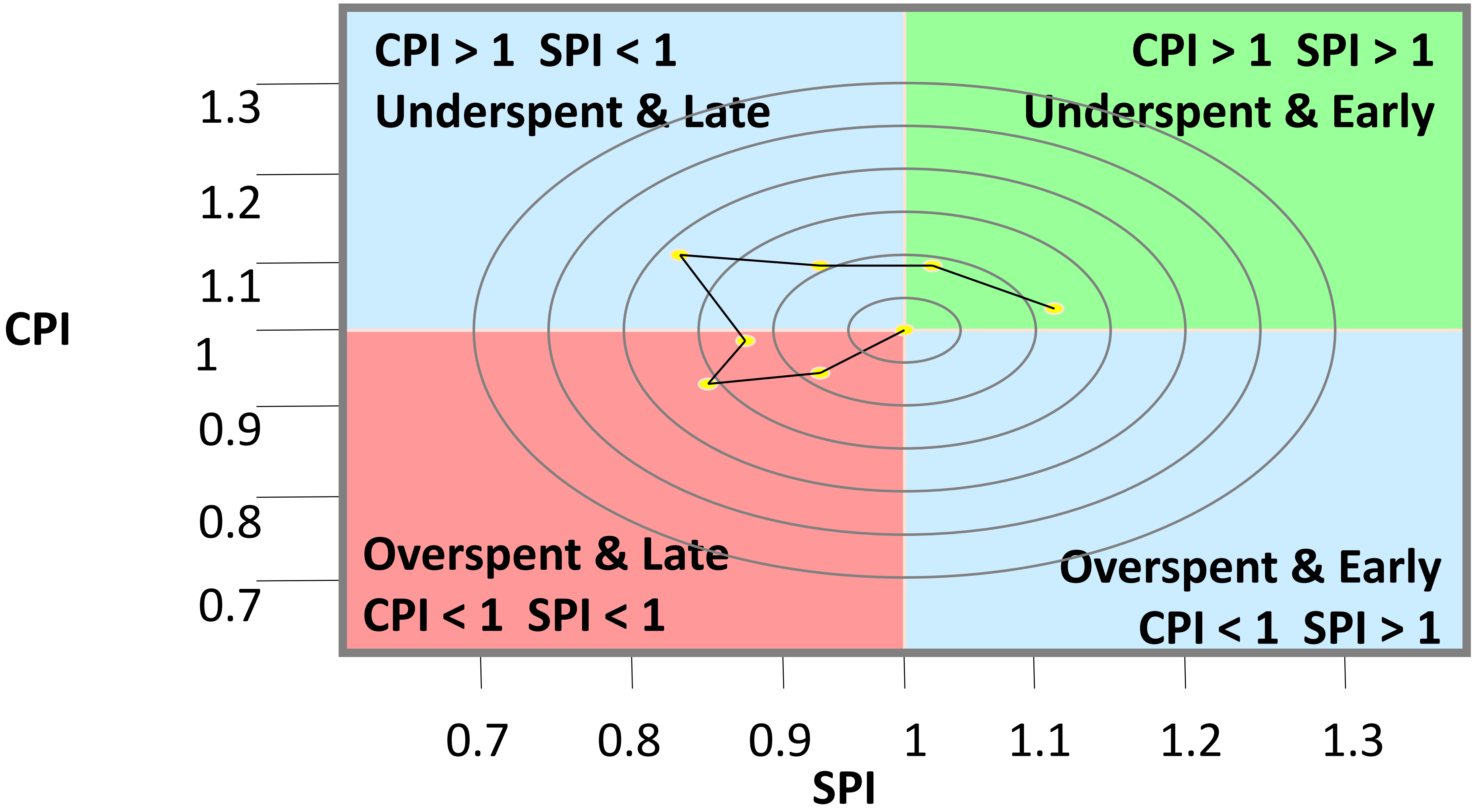
EVM Monitoring



22c) Know how to monitor and report financial performance (including different types of financial reports).

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Monitoring and reporting financial performance

Monitoring takes the baseline that has been agreed and looks at how the project is performing against it. It enables meaningful reports to be shared to improve performance and enable decision making.

- There are different approaches that can be taken to progress monitoring and reporting. The project manager and sponsor agree appropriate methods to monitor financial performance early in the project.
- Monthly monitoring is often appropriate.
- A traffic light approach can be used when reporting to flag the areas that are in or out of the agreed forecast including tolerance levels.
- Financial performance is monitored through different reviews that happen as part of project governance. The following all monitor progress against baselined budgets, timelines and success criteria:
  - decision gates
  - benefit reviews
  - stage reviews
  - audits

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Monitoring and reporting financial performance

Usual reporting could include:

- actual costs versus forecasted costs
- actual spend versus forecasted spend against a set spend curve
- actual spend versus actual work achieved looking at the efficiency of spend (earned value analysis)
- estimated project budget and the forecasted cost overrun based on planned completion dates
- cashflow and drawdown requests
- resource costs
- change request costs
- contingency drawdown requests
- financial benefit realisation

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22d) Know how to close down finances at the end of a project.

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Closing down finances at the end of a project

When the project has completed delivery, as part of governance, the project sponsor will commence close down activities. Typically, this will involve a project closure report, which is then submitted to the final decision gate for endorsement. As well as the report, the following activities need to take place:

Reassign resources

- Move resources back into the business, back to the supplier or out of the organisation.
- Ensure organisation is not charged for resources that are no longer required.

Formally close contracts

- Facilitate closing all contracts in the supply chain.
- Make sure everything has been done, and everything expected to be received has been.
- Agree final accounts.

Accounting and management systems updated

- Close down project on the accounting and management systems.
- Remove delegated authorities and project cost codes.
- No further money can be charged against the project.

Surplus funding returned

- Once all payments have been settled, any budget not spent returns to the business for use elsewhere.

Consider how to close down activities early in the project life cycle so as not to incur any costs that have not been factored into the budget.

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Sections 4.2.4 (Estimation), 4.2.8 (Cost planning), 4.3.1 (Progress monitoring and reporting)

APM Learning module, Budgeting and finance management

APM Learning handbook, Earned value management

apm.org.uk/resources/apm-learning/

APM 'What is project cost planning and control?' apm.org.uk/resources/what-is-project-management/what-is-project-cost-planning-and-control/

APM 'What is investment appraisal and project funding?' apm.org.uk/resources/what-is-project-management/what-is-investment-appraisal-and-project-funding/

APM (2015) *Planning, Scheduling, Monitoring and Control: The Practical Project Management of Time, Cost and Risk* apm.org.uk/book-shop/planning-scheduling-monitoring-and-control-the-practical-project-management-of-time-cost-and-risk/

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Version #	Date	Details of change	Page / slide no.

Risk and issue management

Learning objective:

23. Understand risk and issue management as the ability to identify and monitor risks (threats and opportunities); plan responses to those risks and respond to issues.

23a) Understand the benefits of risk and issue management and the role of contingency planning in projects.

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Risk management

Key aspects

- Risk management allows individual risk events and overall risk to be understood and managed proactively, optimising success by minimising threats and maximising opportunities.
- Risk management is influenced by the risk appetite of the investing organisation that will provide the tolerance levels on what is acceptable or not.
- The risk management process is iterative to reflect the dynamic nature of project work.

Benefits

- Supports more realistic plans, schedules and budgets.
- Increases the likelihood of schedules and budgets being kept to.
- Greater accuracy in allocated contingencies through more meaningful assessment.
- Supports continuous learning and improvement.
- All contributing to increasing stakeholder confidence.

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Issue management

Key aspects

- The process used to manage issues is a simple concept focused on understanding the issue, in order to prioritise and assign owners for a timely resolution.
- Issue resolution requires support from the project sponsor, who may escalate to the governance board.
- The management of issues are tracked from identifying to resolving the issue and may require a re-plan of the baselined project management plan.

Benefit

- Provides the structure and escalation routes required to resolve issues in a timely manner.
- Raises visibility among senior stakeholders who have access to organisational resource to support resolution.
- Provides a consistent way to document issues so they can be assessed against the plan.
- Enables re-prioritisation and decision making.
- Ultimately it enables issues to be resolved efficiently so work can progress preventing further delays.
- Informs continuous learning and improvement.

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Contingency planning

- Contingency is the resource set aside for responding to identified and unidentified risks, also known as the ‘unknown unknowns’.
- Contingency is not hidden extra time or money to deliver planned scope: it is clearly identified and is there to cover the gap between the plan and known risks.
- In addition to the contingency for known risks, some projects will have access to a management reserve for unidentified risks or identified risks that have very low likelihood of materialising but if they did would have a very high impact.
- Contingency is most typically expressed in two ways: budget and time. It’s for dealing with uncertain events that have an impact on either the project costs / financial benefits or the project timescales.
- However, if using an iterative life cycle and timeboxes, the project would consider contingency in terms of scope or quality.
- Where risks are identified, the planned contingency has provision to cover the gap, and the project manager requests to draw down some of the contingency to manage mitigation.

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23b) Understand the purpose and importance of each stage in a risk management process (such as identification, analysis, monitoring, escalation, response and closure) and an issue management process (such as logging and analysis, escalation, and assignment of actions) and understand why these stages are different for linear and iterative life cycles.

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Risk management process

Risk management must be initiated by first establishing the risk management plan for the project. This plan provides the roles and responsibilities, how risks are to be assessed and the impact evaluated, agreed tolerances for risk thresholds, reporting and escalation points. It also serves as an important reference for all stakeholders.

Initiation can be the first step; a typical risk management process will then include the following steps:

Identification

- Identifies the risk events relevant to a project considering risk appetite.
- Ongoing process throughout the project life cycle.
- There are several tools and techniques that can be used to draw out knowable risks.
- Once identified, risks should be documented with the appropriate information in the project risk register and an owner assigned to assess and manage.
- When working in an iterative life cycle, risk identification has a different focus, using a risk-based approach mindset to influence the design of each iteration, while continuing to look at what is risky, why and what can be done about it during deployment.

Analysis

- Assesses the relative impact of the identified risk on the project objectives.
- Undertaken by the project manager, with the support of the project team and subject matter experts, using qualitative and or quantitative techniques.
- Qualitative analysis looks at the impact on objectives, assessing the probability / likelihood of occurrence and the size of impact.
- Quantitative analysis determines the combined effect of risks on objectives and for focus on specific risk-based decisions.
- When working in an iterative life cycle, risks will be analysed for each iteration; the issue of each release could then prompt different risks to be analysed.

Monitoring and Escalation

- Monitoring of risks for occurrence and knowing when to escalate.
- If risk thresholds are reached and considered to have potentially critical consequences to the project objectives, escalation would be required.

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Risk management process (continued)

Response

- Considers the appropriate response to put in place to deal with the risk, whether proactive or reactive.
- Determines whether the risk is deemed a threat or an opportunity.
- Determines whether it makes sense to proactively invest in mitigation activity to bring the risk back within tolerance levels.
- Secondary risks or residual risks may emerge because of the response and should be considered.
- Focus is on putting a risk efficient solution in place, increasing the likelihood of project success.

Closure

- Closes the risk in the risk register when it no longer poses a risk to the project. This could be because it has been removed, been responded to, occurred or passed the point it was likely to occur.
- As the last step in the process, it is important to update the risk register with information relating to the risk closure and lessons learned to avoid unnecessary work of monitoring risks that will no longer occur and to support continuous improvement of project activities and processes.
- Any remaining risks at project closure should be communicated to stakeholders involved in the handover and adoption phase.

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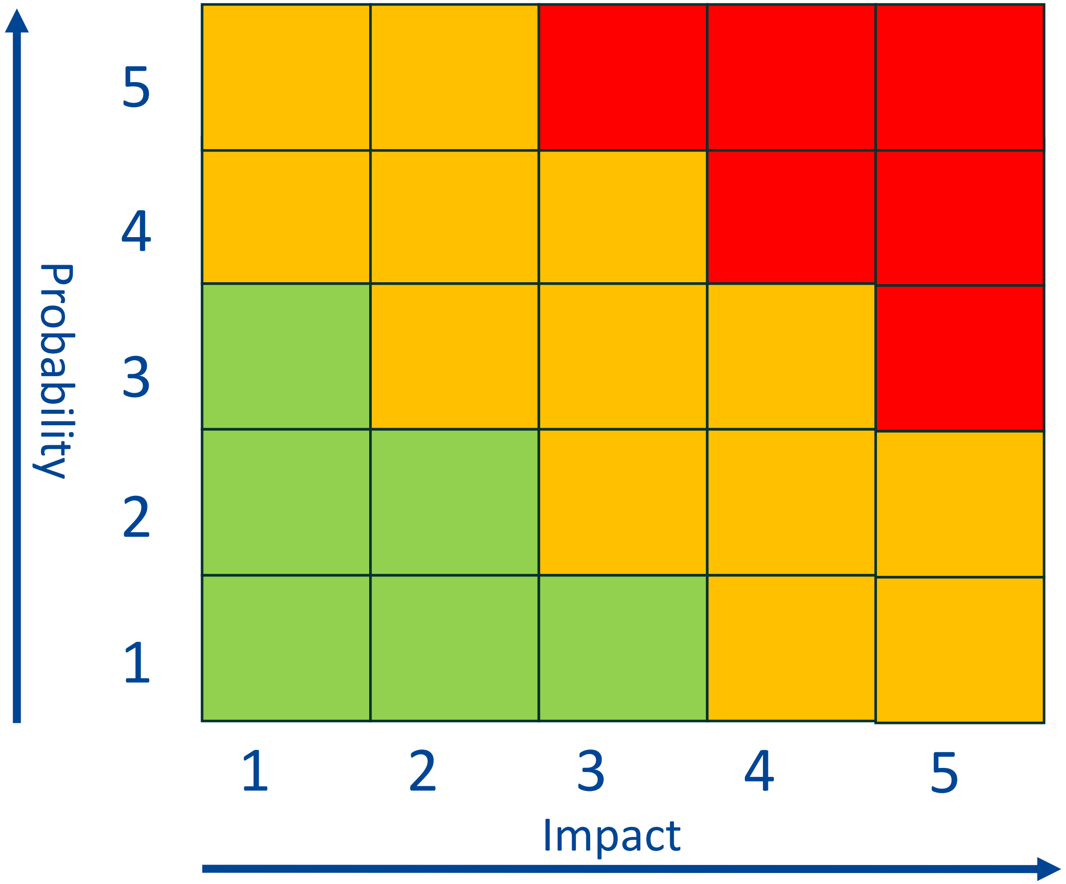
Probability/Impact Grid Example



Key:

Intolerable

Undesirable

Acceptable

Each identified risk is qualitatively assessed and placed on the grid, allowing the risks to be prioritised

Issue management process

Tracking the management of issues happens from identification through to resolution including any change control and replanning activities required to the baselined project management plan.

Log and analyse

- An issue register / log should be maintained and used by the project manager and team to record, analyse, prioritise and track the status until the issue is resolved.
- Prioritisation is based on the success criteria and benefits for the work, considering scope, quality, time, cost and benefits in the business case.
- Issue registers / logs are a source of knowledge for future projects.

Escalate

- The issue will need escalating to the project sponsor, who may then escalate to the governance board.
- The sponsor / governance board will help the project manager define the best course of action for a resolution.

Assign actions

- Actions are assigned to a person or group best placed to resolve.
- Look to implement a resolution in a timely manner.

Apply change control

- Issues that result in a change to scope or the baselined plan will need to be taken through the change control process.
- This ensures all project outputs are updated and checked for no further impact to the project.

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23c) Know proactive and reactive responses to risks (such as avoid, reduce, transfer or accept and exploit, enhance, share and reject).

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Proactive and reactive responses to risks

Risks are categorised as threats or opportunities and there are two main types of responses to them: proactive or reactive.

PROACTIVE A planned and implemented response undertaken to address the likelihood of the risk occurring or the size of the impact if it did occur.

Threats

- **Avoid:** Change objectives or practices to change the cause of the threat so it can no longer occur.
- **Reduce:** Take action to reduce the probability of the threat occurring or the impact on the project.
- **Transfer:** From one party to another, passing on the responsibility for bearing the impact of the threat.

Opportunities

- **Exploit:** Change objectives or practices to implement a response to maximise the opportunity to gain maximum benefit / advantage.
- **Enhance:** The probability of the opportunity eventually being exploited by increasing the likelihood and / or impact.
- **Share:** Collaborating with other organisations to increase the chances of the opportunity being realised.
- **Reject:** Decide not to take advantage of the situation as the opportunity is of little value or the work to gain the result is too great.

REACTIVE A provision for a course of action that will only be implemented if the risk materialises.

Threats

- **Accept:** No acceptable or viable approaches to take to avoid or reduce the threat. Any residual risk must be managed, and contingency plans implemented should the risk materialise.

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23d) Understand why governance is important in risk and issue management.

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Why governance is important

- Governance and risk management work together, providing systems of control for the project and giving confidence to stakeholders.
- Governance informs risk and issue management through processes, procedures, standardised methods, specialist expertise and resources.
- It sets out the arrangements for the organisation and project, the life cycle to be used for project delivery, the route for reporting and escalating, and provides assurance which is targeted where the greatest risks exist.
- Assurance is objective and independent, and governance relies on assurance to support effective decision making when it comes to managing risks and issues.
- Risk analysis provides guidance to assurance on where vulnerabilities lie and in return evidence from assurance improves risk assessment and informs decision making at decision gates.

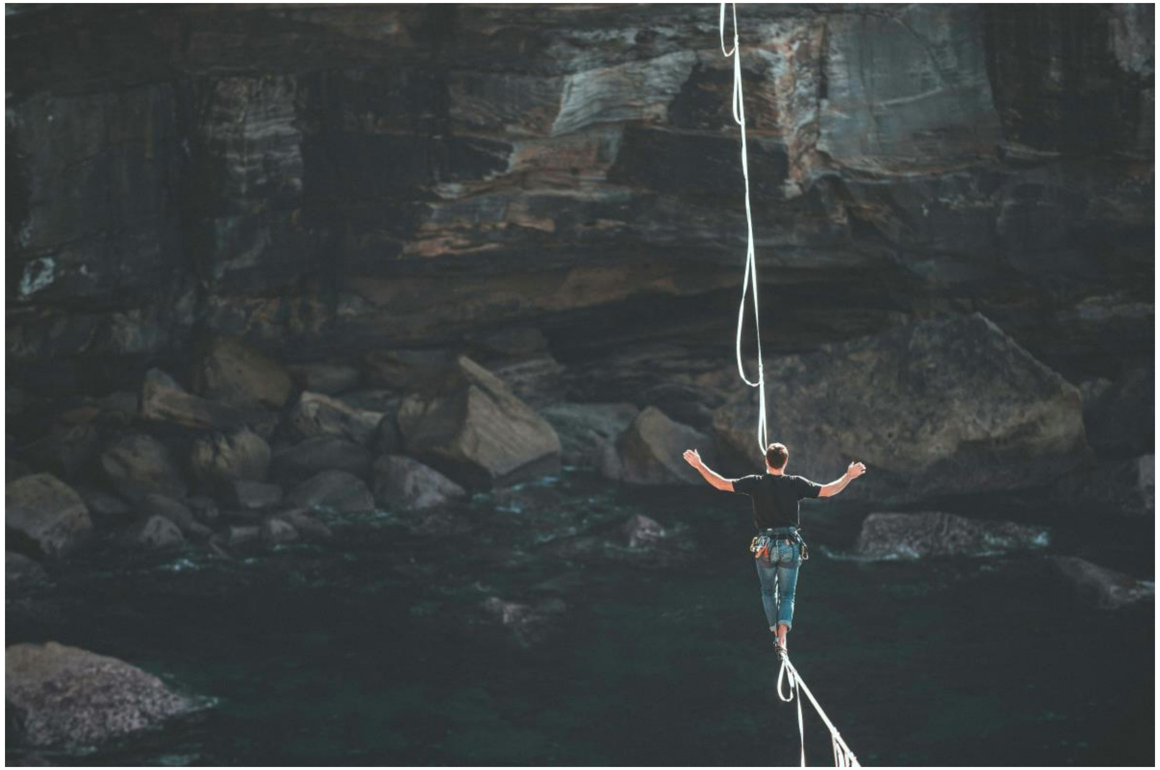

Photo credit: Loic Leray

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Sections 4.3.3 (Risk management), 4.2.2 (Risk identification), 4.2.3 (Risk analysis), 1.3.2 (Assurance principles)

APM Information sheets ‘Risk management’, ‘Issue management’ apm.org.uk/resources/apm-learning/

APM ‘What is risk management?’ apm.org.uk/resources/what-is-project-management/what-is-risk-management/

APM Risk Specific Interest Group (2018) *Project risk analysis and management* apm.org.uk/resources/white-papers/project-risk-analysis-and-management/

APM ‘Risk appetite and risk tolerance’ apm.org.uk/resources/find-a-resource/risk-appetite-and-risk-tolerance/

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Change control

Learning objective:

24. Understand change control as the ability to manage variations and change requests in a controlled way.

24a) Understand the purpose and importance of each stage of a typical change control process (such as request, initial evaluation, detailed evaluation, recommendation, update plans and implement) and understand where these stages are different for linear and iterative life cycles.

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Purpose and importance of change control for linear and iterative life cycles

Change control manages all requests to change the baseline of a project. These requests can come from different sources for different reasons. Once identified, they need to be evaluated and either approved, rejected or deferred.

- Managing change requests in a structured way enables the project sponsor and stakeholders to understand the impact on the outcomes of baselined work and influence the decision on how to respond, considering objectives and risk appetite.
- The consequence of unmanaged change can be far-reaching within the project environment and to business-as-usual activities.
- In some situations, a change freeze may be needed. This prevents any further change be considered in the project, to protect delivery of the objective.

- Projects using an iterative life cycle embed change control as part of the development process, starting each iteration with a planning meeting that clarifies the goals and items to be delivered.
- In iterative life cycles, change is much more normal and expected, rather than needing an exceptional meeting or process to resolve (though change requests can still be raised if required). Changes to existing features or the goals and/or items in that iteration can be considered alongside the existing requirements.

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Stages of a typical change control process

- **Request:** Change requests (CRs) can be made by internal or external stakeholders, documenting relevant information on the nature of the change and reasoning for it. The request should be logged in the change register, given a unique identifier and assigned an owner to evaluate.
- **Initial evaluation:** The owner decides whether the change should be evaluated in detail or rejected at this stage, looking to determine its necessity, feasibility, potential impact on the project and the rationale for raising.
- **Detailed evaluation:** The owner considers the impacts on the project baseline success criteria and on work already completed. Other projects and business as usual should also be considered.
- **Recommendation and decision:** The results of the evaluation are presented to the sponsor and/or wider governance board for a decision. The options are defer (further clarification or information needed), reject (cost or risk of the change exceed the value the change will bring or go beyond agreed thresholds), or approve (the value being created from the change is worth the cost of implementing).
- **Update plans and implement:** The status of the change request is updated in the change log. The sponsor makes sure the decision is communicated. If the change is approved, baselined plans and budgets are updated, including the business case if required, and any new risks raised are added to the risk register. Actions are implemented in line with the revised plan and monitored through to completion. Configuration records are updated to reflect the latest version of project documents incorporating the change.
- **Communicate change:** All updated roles and responsibilities, documentation and plans are communicated to stakeholders.

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24b) Know what should be captured and recorded in change requests.

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A typical change request

Typical information captured in a change request form, to support project evaluations and a decision being made, includes:

- project overview
- description covering the nature of what the change is
- reason for the change
- areas impacted
- source of change
- change request owner
- date raised and unique identifier
- status of the change request
- decision, date made and justification

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24c) Know ways to assess options related to a proposed change and the high-level impact of the proposed change.

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Assessing the options

What's feasible?

- Options are assessed against the impact they have on achieving the success criteria set out in the business case, and whether the skills and resources needed to deliver the change are available.
- A change may impact any of the following:
  - benefits
  - scope
  - schedules
  - costs
  - quality
  - resources
  - risks
  - stakeholders
- Understanding why the change has been raised is also important when assessing options – is it mandatory, a change in regulation, risk profile, customer needs or financial constraints driving the request?
- Another consideration is the resources required to undertake the assessment. Before evaluating, the project should consider whether to assess such a change at all.

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Assessing the options

Assessing High-Level Impact of the Proposed Change

- Understanding the broader implications of a proposed change is crucial for maintaining alignment with the project's goals and the strategic objectives of the organisation.

- ***The following are the steps to assess the high-level impact of the proposed change:***

1. **Project Scope:** Determine whether the proposed change will expand, reduce, or alter the deliverables of the project, potentially affecting project goals. Consider the implications for project complexity and whether additional adjustments will be required in other areas of the project.

2. **Project Schedule:** Evaluate the potential for the proposed change to delay critical milestones or possibly expedite certain phases of the project. Assess the flexibility of the project schedule to accommodate the change, considering the impact on overall project delivery and deadlines.

3. **Resource Allocation:** Examine whether the change will necessitate additional resources or redistribute existing ones, potentially impacting other project areas. Consider the availability of required resources and the feasibility of reallocating them without disrupting other operations.

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Assessing the options

Assessing High-Level Impact of the Proposed Change

4. **Quality of Output:** Assess how the change will affect the overall quality of project deliverables, potentially enhancing or diminishing the value provided to stakeholders. Consider the impact on project standards and compliance requirements, ensuring that quality benchmarks are still met.

5. **Compliance and Regulatory Issues:** Investigate any legal or regulatory impacts due to the change, ensuring continued compliance with relevant standards and laws. Consider the need for additional approvals or oversight, potentially affecting project timelines and costs.

6. **Long-Term Strategic Impact:** Evaluate how the change aligns with the strategic goals of the organisation, considering its contribution to long-term objectives. Consider potential market and competitive impacts, ensuring that the change supports sustained organisational growth and adaptation.

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24d) Understand how to justify recommendations about whether to approve, reject or defer changes.

How to justify recommendations

A recommendation is driven from the evaluation that looks at whether the proposed change could jeopardise the achievement of the project objectives.

Some points to consider when justifying the recommendation include:

- What areas of the project will be affected? How will they be affected? Does the project have the skills and resources needed to handle it?
- Does the project have sufficient budget in the contingency budget to draw down or would further funding be needed?
- If additional resources are needed, does the business have the capacity and the capability to source them? Would external (to the project) help be needed?
- Can the project maintain delivery on time? If not, will the project outcomes still be of use to the business?
- Are the benefits listed in the business case going to be materially impacted in any way? If so, how and what does that mean both for the project and the business?

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24e) Understand the importance of updating plans and schedules to reflect and communicate changes.

Updating plans and schedules

- Poorly managed change is still one of the most common reasons why projects fail. Managing requests for change effectively is a proven success factor for project delivery.
- The project management plan sets out what is expected from the project team. If this is not updated and communicated when changes are made, the team will be working from outdated information, costing the project time and money.
- This can cause unnecessary conflict among the team about what needs to be delivered, impacting team performance and satisfaction levels.
- The solution could end up redundant if what is delivered is not fit for purpose or not delivered in time.
- Stakeholders could disengage through the loss in confidence of the project team's ability to deliver the project's outcomes to the agreed success criteria.
- This could have a knock-on impact on other change activity happening in the business or on business as usual.
- Ultimately, the benefits in the business case may not be realised, impacting the return-on-investment.

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Recommended resources for learners

You can promote independent learning and scaffolding of knowledge by recommending that learners refer to the resources below to gain a broader understanding of the topic area.

APM Body of Knowledge, 7th edition, Section 4.3.6 (Change control)

APM Learning module, Change control: keeping your projects on track apm.org.uk/resources/apm-learning/

APM Information sheet, Change control and change request form apm.org.uk/resources/apm-learning/

APM 'What is change control?' apm.org.uk/resources/what-is-project-management/what-is-change-control/

APM (2015) *Planning, Scheduling, Monitoring and Control: The Practical Project Management of Time, Cost and Risk* apm.org.uk/book-shop/planning-scheduling-monitoring-and-control-the-practical-project-management-of-time-cost-and-risk/

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